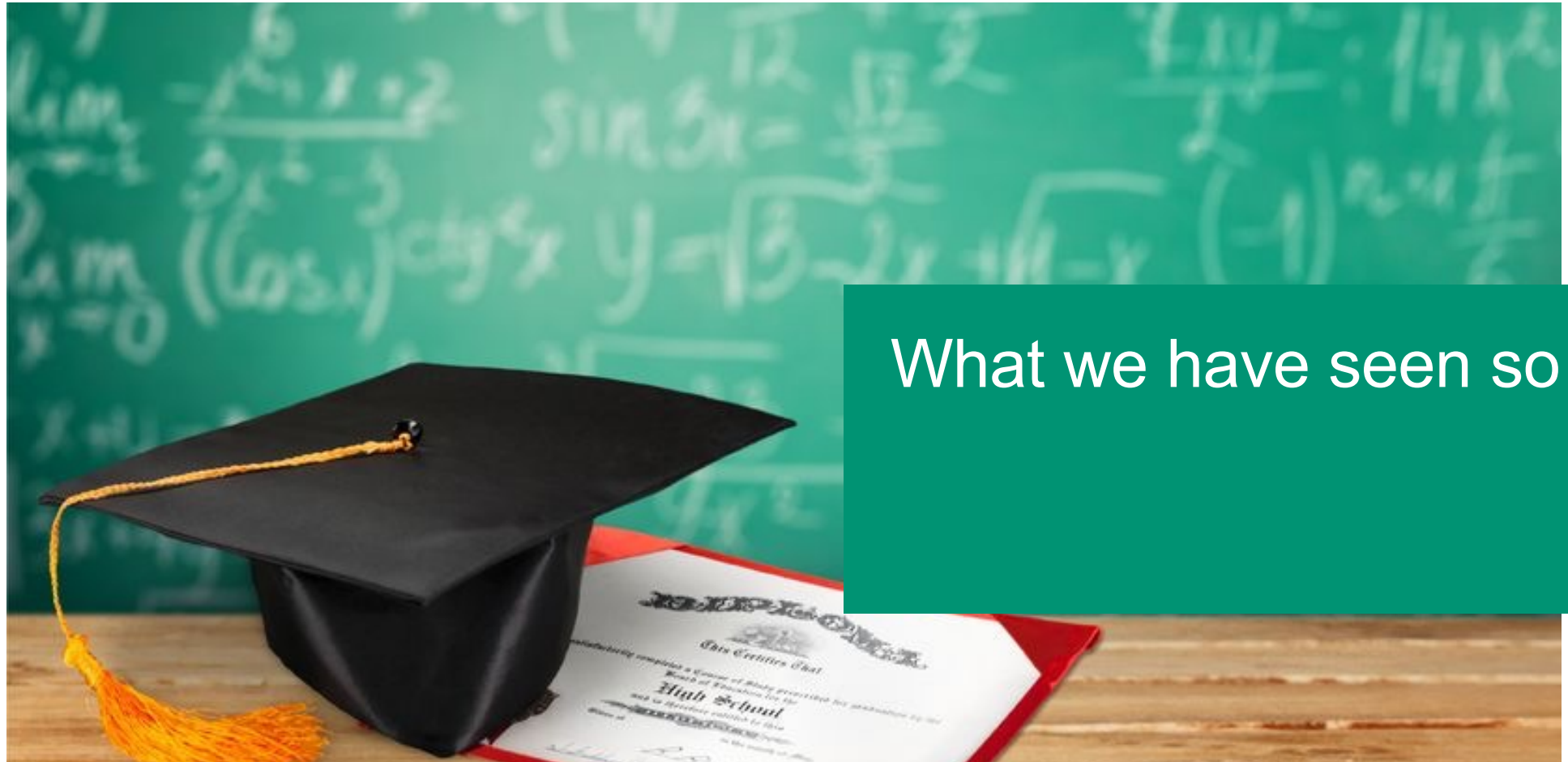


Information Retrieval

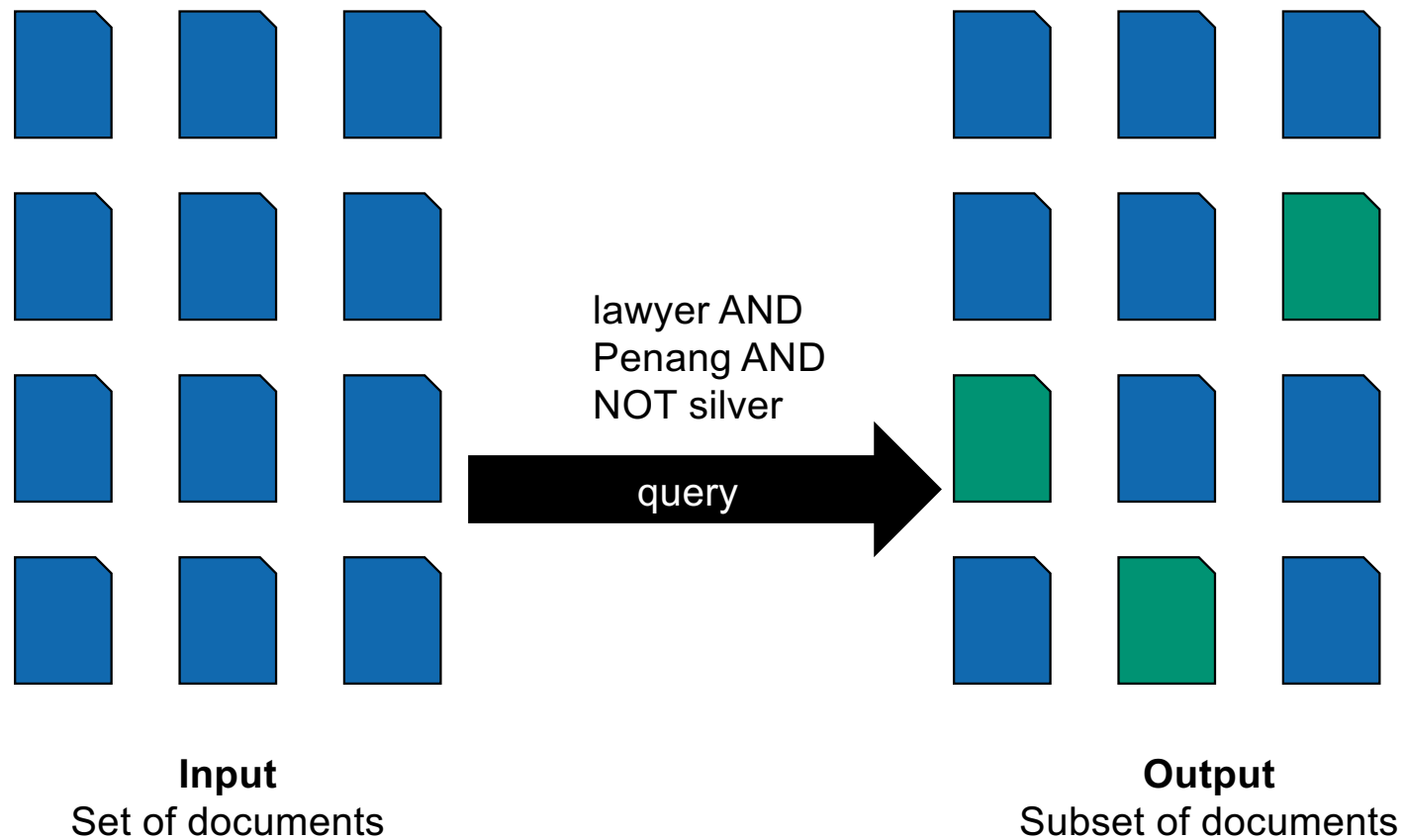
7. Ranked retrieval

Ghislain Fourny
Spring 2022

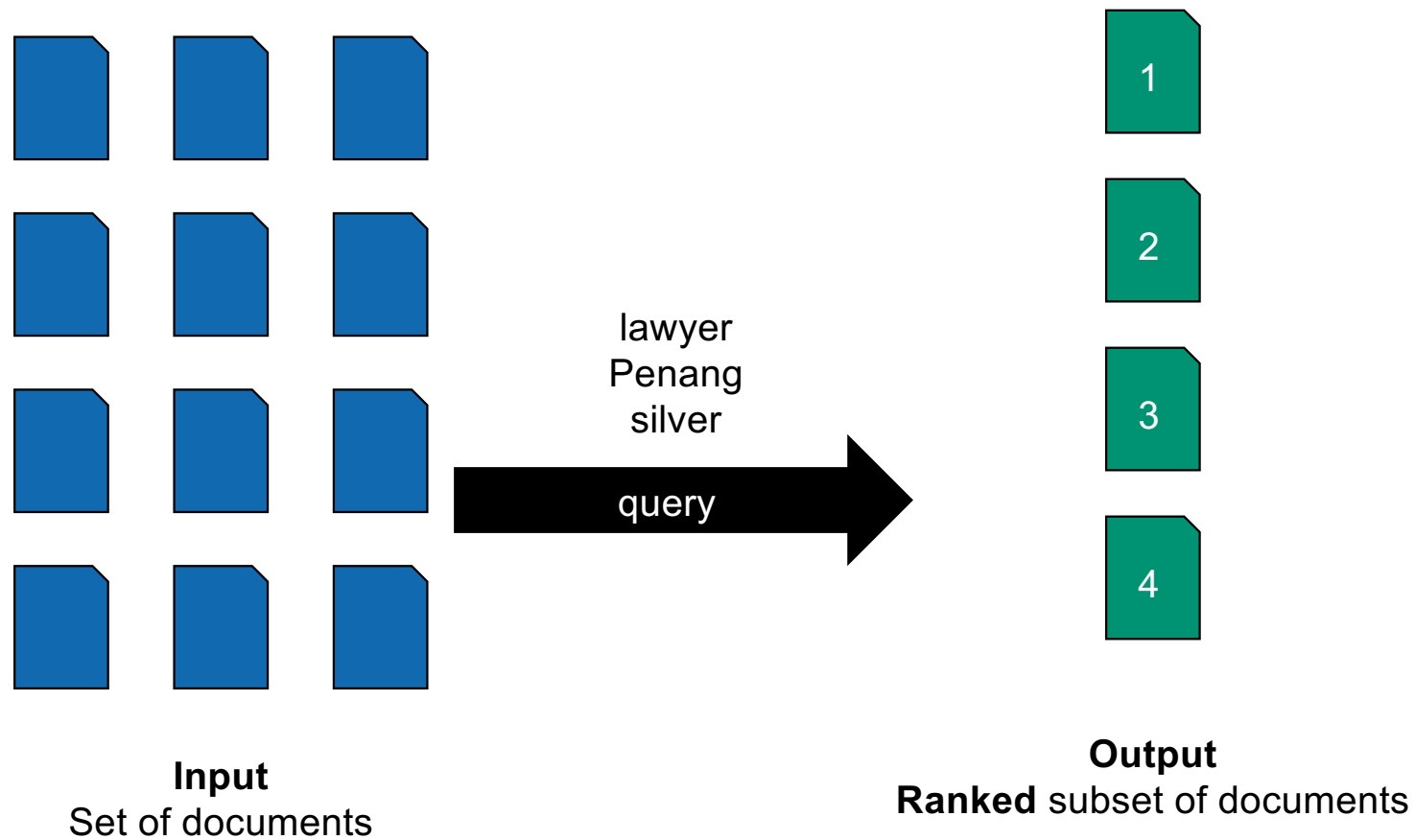


What we have seen so far

Boolean retrieval



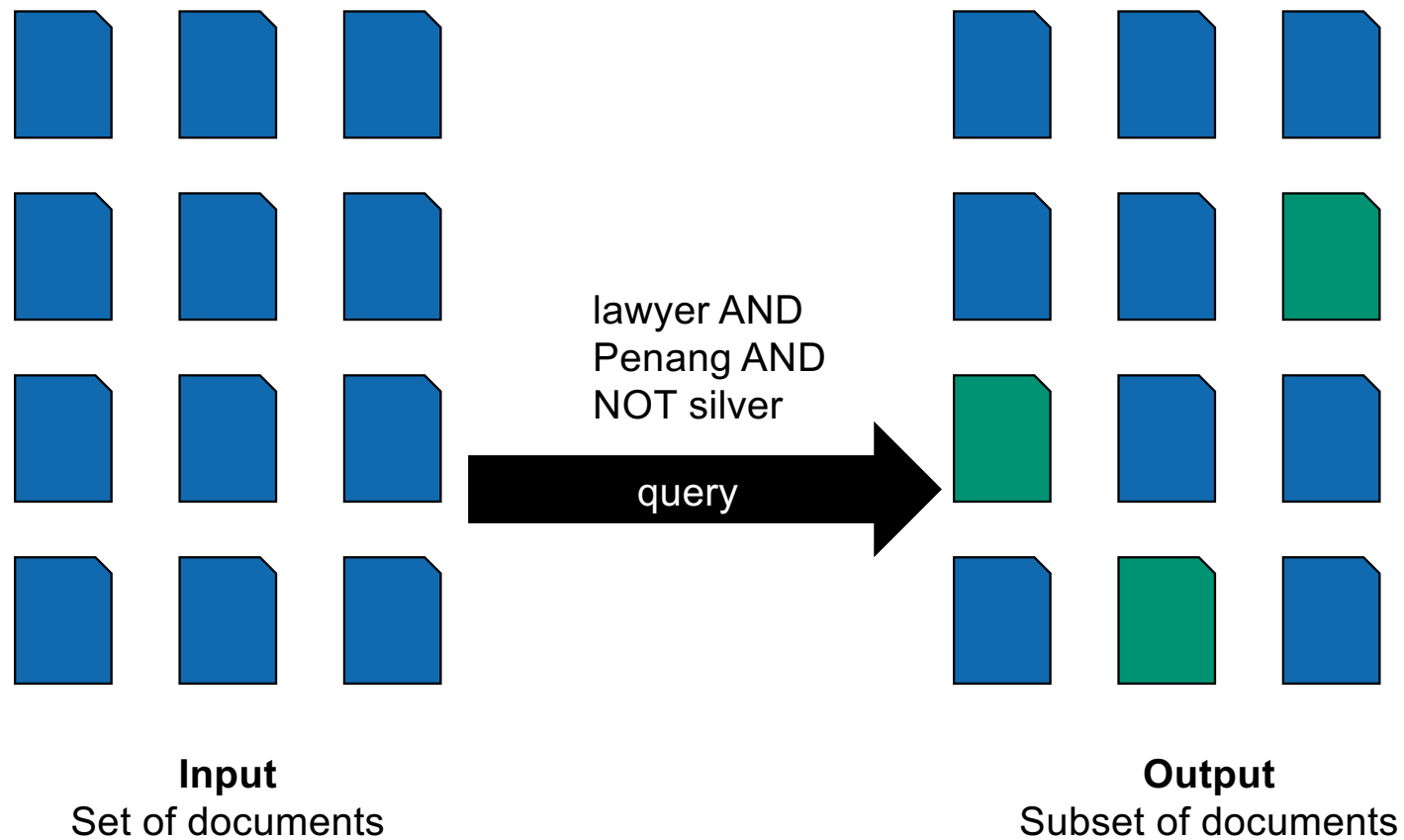
Today: Ranked retrieval



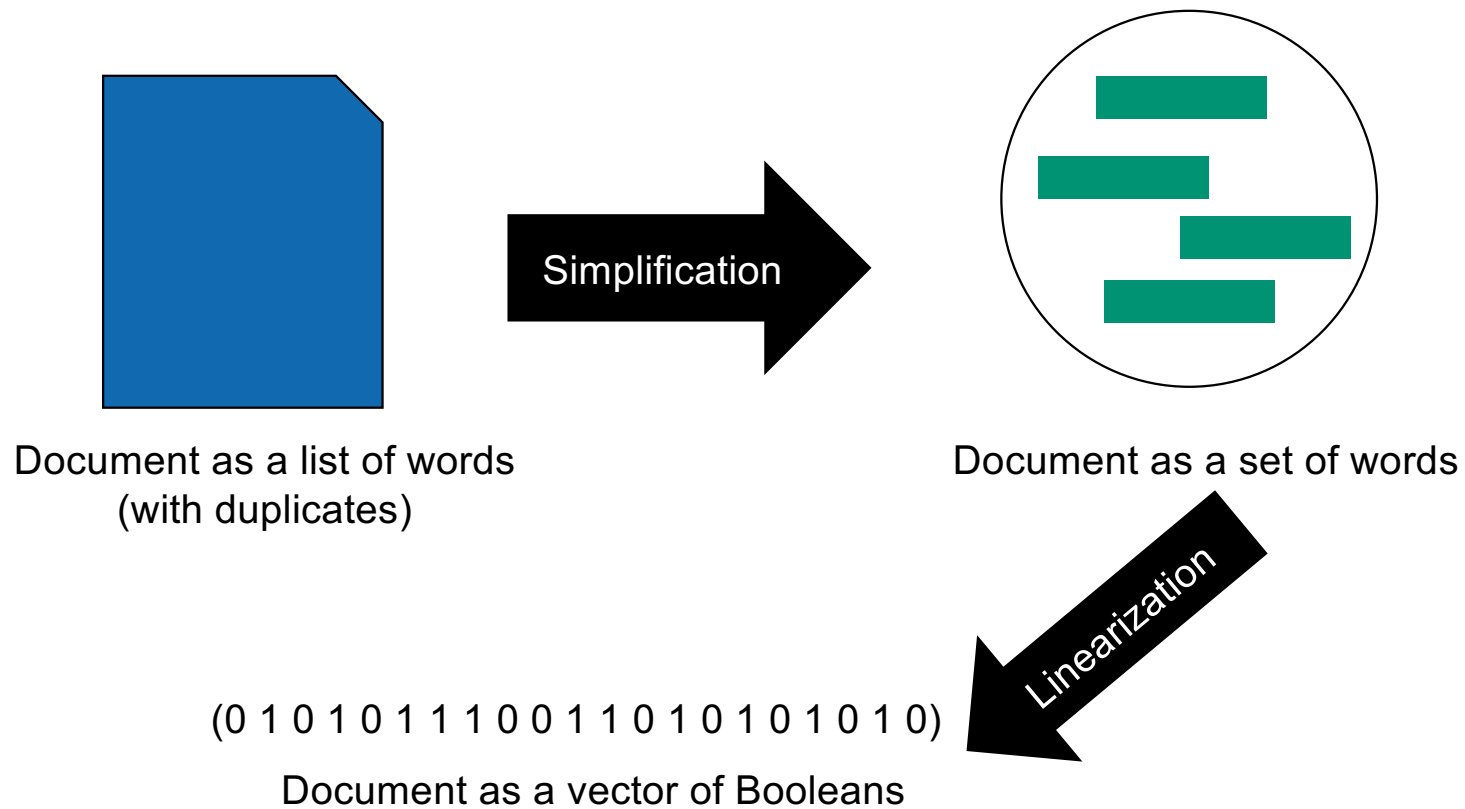


Parametric Search

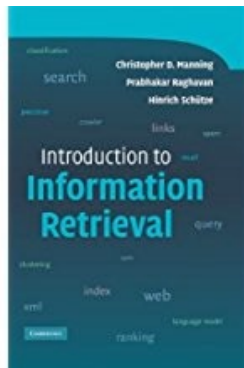
Boolean retrieval



Model and abstraction



Document Metadata



Title: Introduction to Information Retrieval

Authors: Christopher D. Manning and Prabhakar Raghavan and Hinrich Schütze

Hardcover: 506 pages

Publisher: Cambridge University Press; 1 edition (July 7, 2008)

Language: English

ISBN-10: 0521865719

ISBN-13: 978-0521865715

Product Dimensions: 7 x 1.2 x 10 inches

Shipping Weight: 2.2 pounds ([View shipping rates and policies](#))

Average Customer Review: ★★★★★ ▾ 27 customer reviews

Searching Metadata

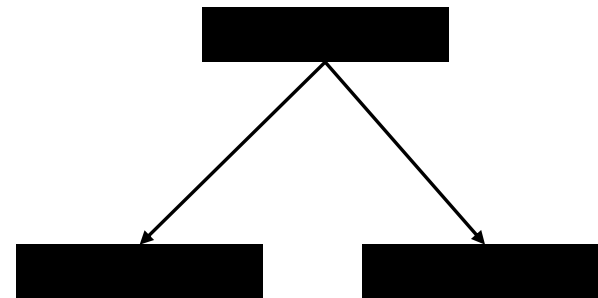
Title	Algorithms		
Author			
Publication Date			
Language			
Country			
Cost		to	
	\$		\$
	Search		

This is a classical database problem!

Title	Author	Publication date	Language	Country	Price
Everything is relative	Einstein	1905	German	Kanton BE	45
Nothing is complete	Gödel	1930	German	Austria	34
Non-halting algorithms	Turing	1950	English	UK	23
Dual-Body theory applied to Suns and Tesla Roadsters	Newton	1687	Latin	UK	12

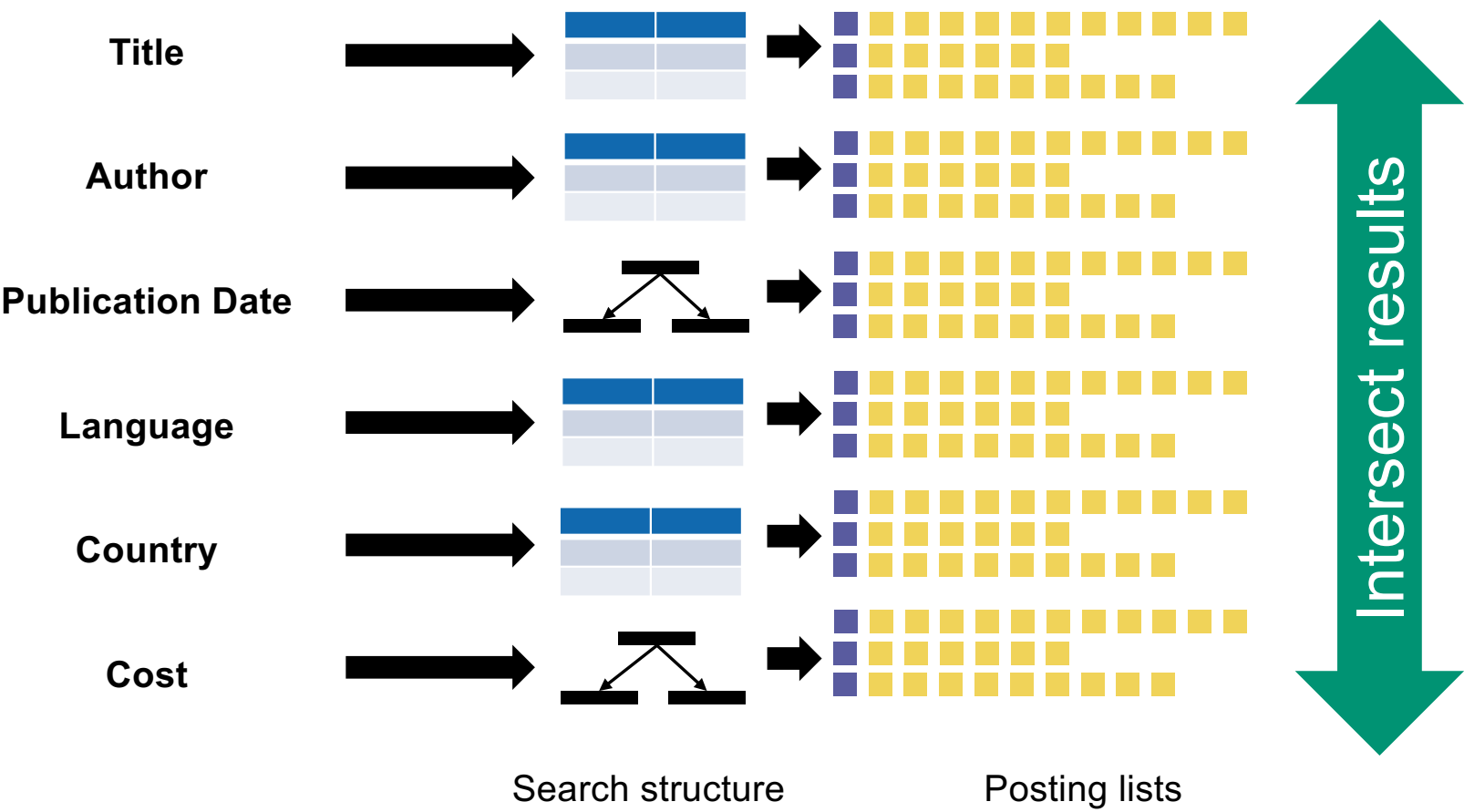
Search structures (Classical databases)

Hash tables

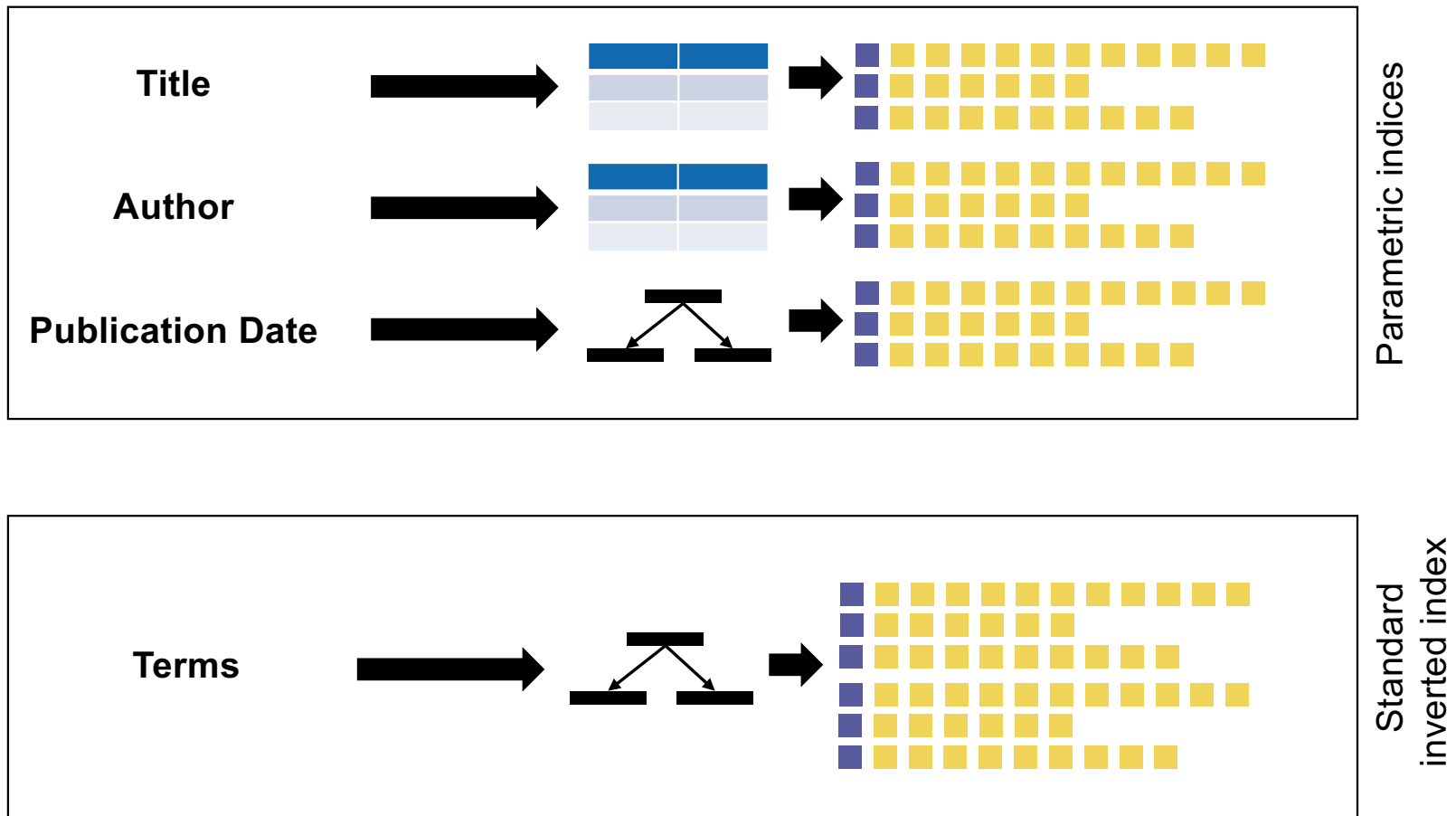


Trees (B, B+)

Parametric indices



The new big picture



A background image of a wooden floor with a grid pattern of square tiles. The wood has a warm, brown tone with visible grain and some minor imperfections.

Zone Search

Searching Metadata

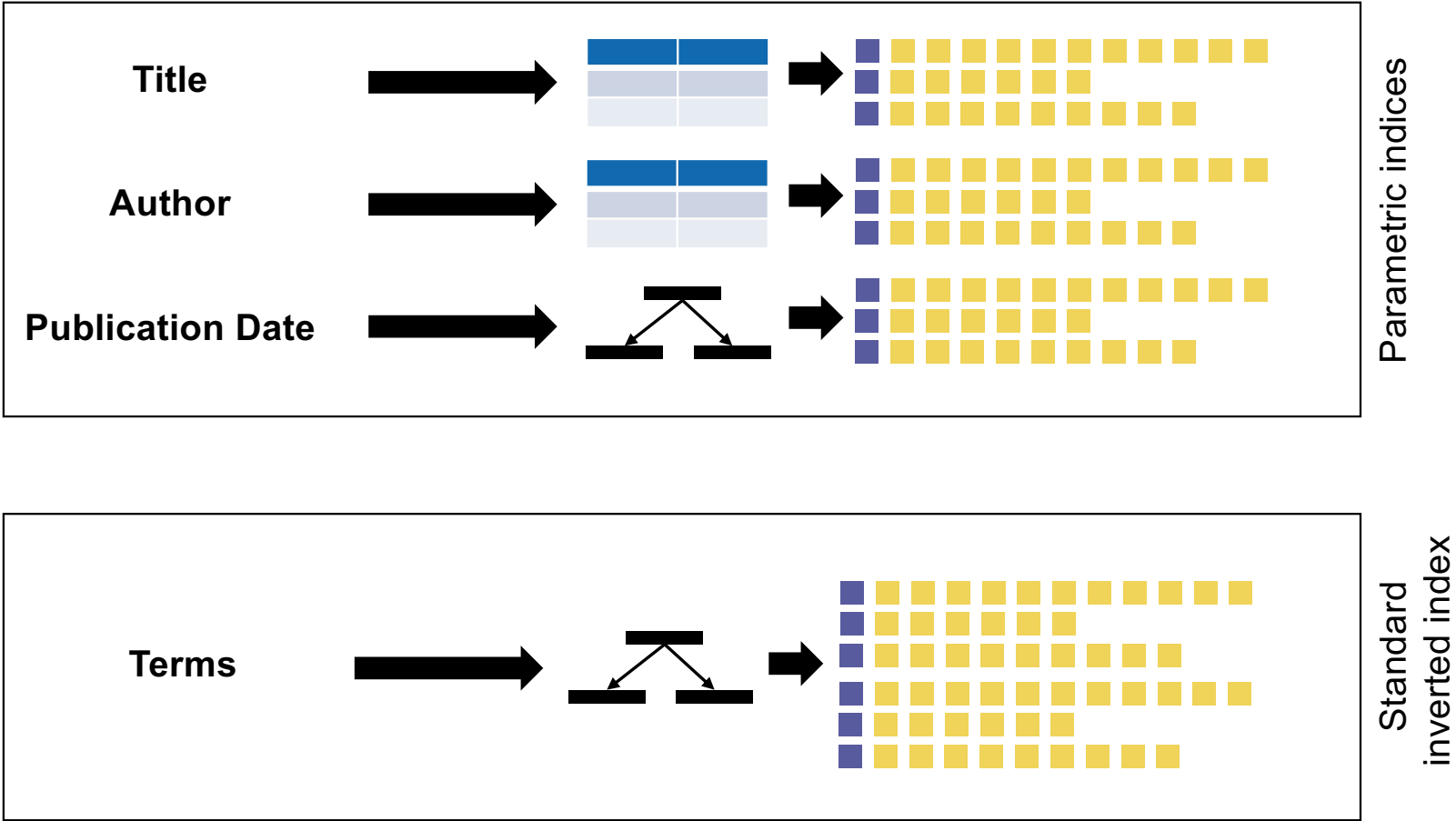
Title	Information retriev		
Author			
Publication Date			
Language	▼		
Country	▼		
Cost	\$	to	\$
Search			

Searching Metadata

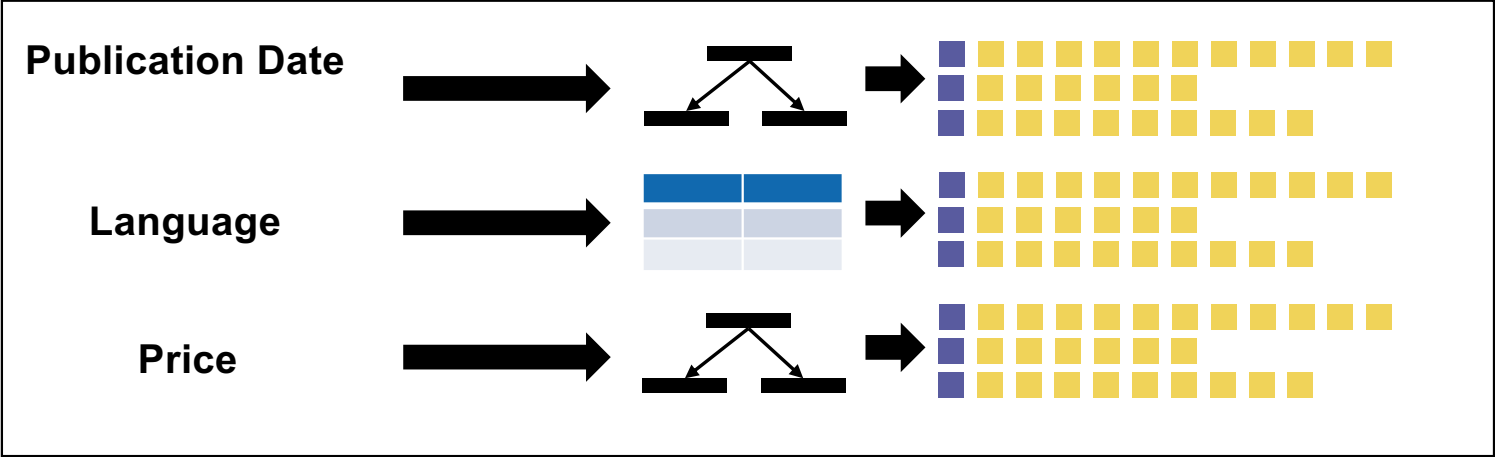
Title	Information retriev
Author	

We could have full-text search here, too!

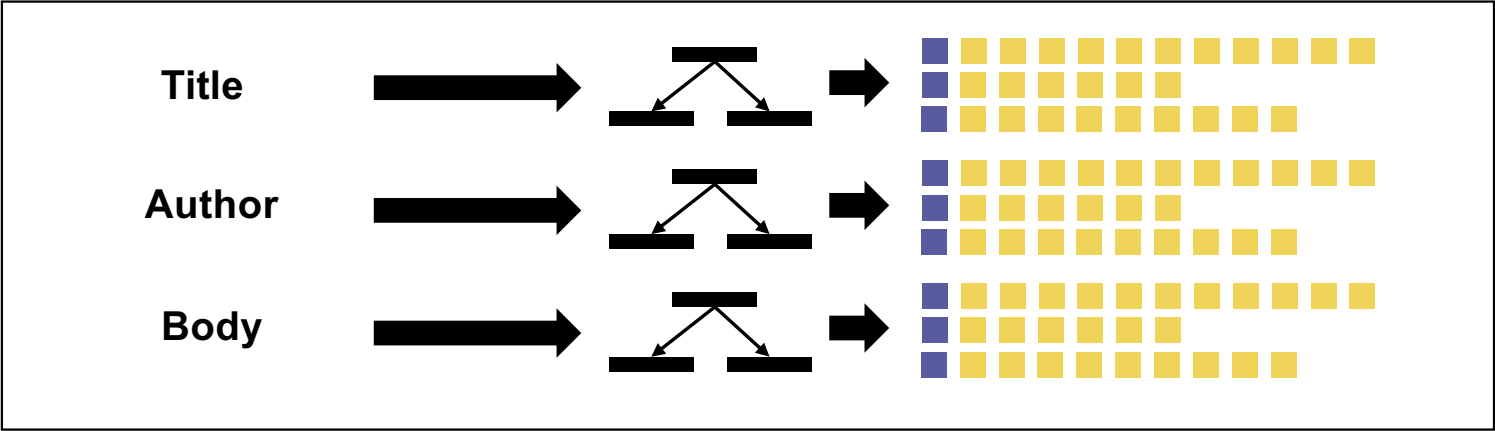
Indices (old)



Indices (new)

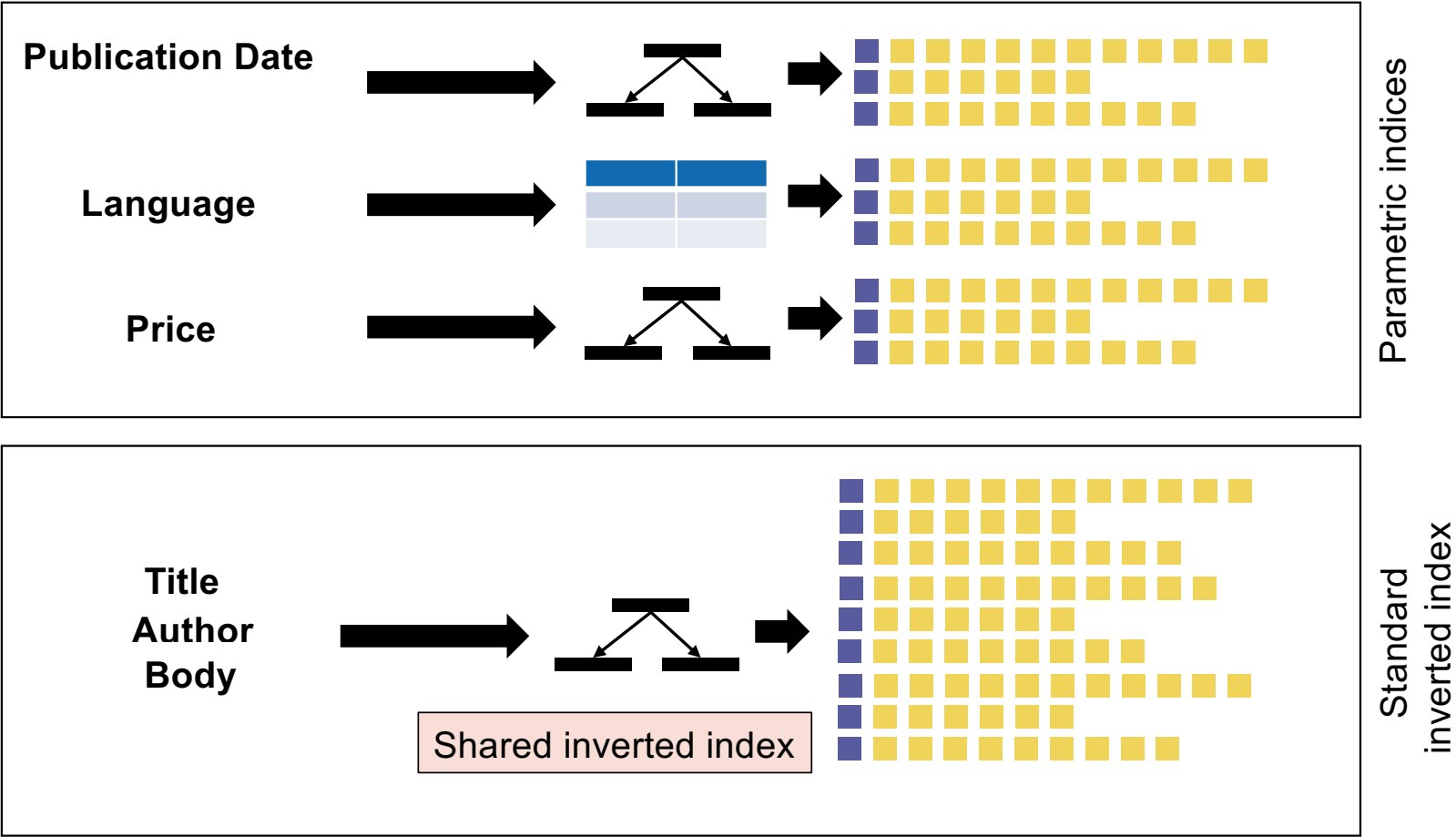


Parametric indices



Standard
inverted indices

Indices (new)



Shared inverted index

ETH	➡	6	1	2	3	5	6
Zürich	➡	5	3	4	7	8	
computer	➡	5	1	2	4	5	7
data	➡	5	1	3	5	8	
CPU	➡	4	2	3	4		
information	➡	6	1	2	4	5	8
retrieval	➡	4	3	5	7	8	

Shared inverted index

ETH	➡	6	1	2	3	5	6
Zürich	➡	5	3	4	7	8	
computer	➡	5	1	2	4	5	7
data	➡	5	1	3	5	8	
CPU	➡	4	2	3	4		
information	➡	6	1	2	4	5	8
retrieval	➡	4	3	5	7	8	

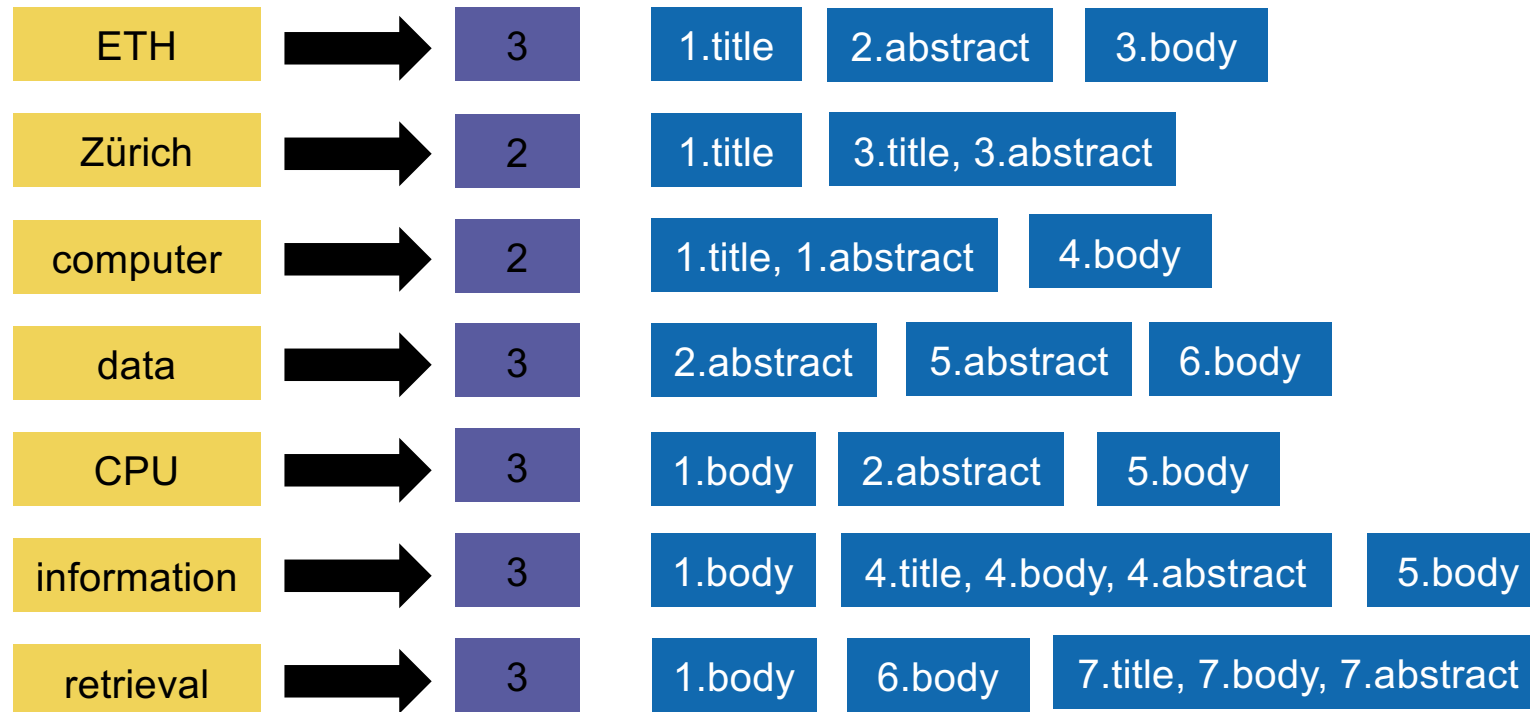
Shared inverted index

ETH.body	➡	6	1	2	3	5	6
Zürich.body	➡	5	3	4	7	8	
computer.body	➡	5	1	2	4	5	7
data.body	➡	5	1	3	5	8	
CPU.body	➡	4	2	3	4		
information.body	➡	6	1	2	4	5	8
retrieval.body	➡	4	3	5	7	8	

Shared inverted index (zones in terms)

ETH.body	→	6	1	2	3	5	6
ETH.title	→	5	3	4	7	8	
computer.body	→	5	1	2	4	5	7
computer.title	→	5	1	3	5	8	
albert.title	→	4	2	3	4		
albert.author	→	6	1	2	4	5	8
albert.body	→	4	3	5	7	8	

Shared inverted index (zones in postings)





Single-term query



Score of a document:

$$\sum_{i=1}^{i=l} g_i s_i$$

Single-term query



Score of a document:

$$\sum_{i=1}^{i=l} \underbrace{g_i s_i}_{\text{Zone weights}}$$

Single-term query



Score of a document:

$$\sum_{i=1}^{i=l} g_i s_i$$

Boolean results

1 = zone i contains term

0 = zone i does not contain term

Single-term query



Score of a document:

$$\vec{g} \cdot \vec{s}$$

Single-term query



Zone	weight
Title	0.3
Abstract	0.2
Body	0.5

Single-term query



Zone	weight
Title	0.3
Abstract	0.2
Body	0.5

Single-term query



Zone	weight
Title	0.3
Abstract	0.2
Body	0.5



weight
0.5

Single-term query



Zone	weight
Title	0.3
Abstract	0.2
Body	0.5

weight
0.5

Total score	0.5
-------------	-----

Single-term query



Zone	weight
Title	0.3
Abstract	0.2
Body	0.5

Single-term query



Zone	weight		weight
Title	0.3	→	0.3
Abstract	0.2		
Body	0.5	→	0.5

Single-term query



Zone	weight		weight
Title	0.3	→	0.3
Abstract	0.2		
Body	0.5	→	0.5

Total score	0.8
-------------	-----

Single-term query



Zone	weight
Title	0.3
Abstract	0.2
Body	0.5

Single-term query

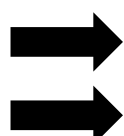


Zone	weight		weight
Title	0.3		
Abstract	0.2	→	0.2
Body	0.5	→	0.5

Single-term query



Zone	weight
Title	0.3
Abstract	0.2
Body	0.5



weight
0.2
0.5

Total score	0.7
-------------	-----

Single-term query



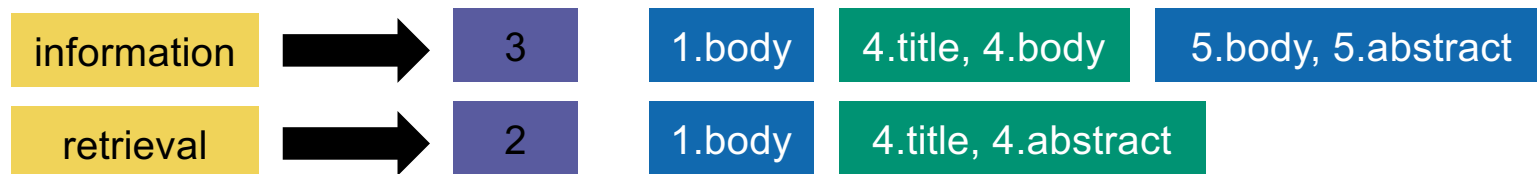
Document	Score
1	0.5
4	0.8
5	0.7

Single-term query



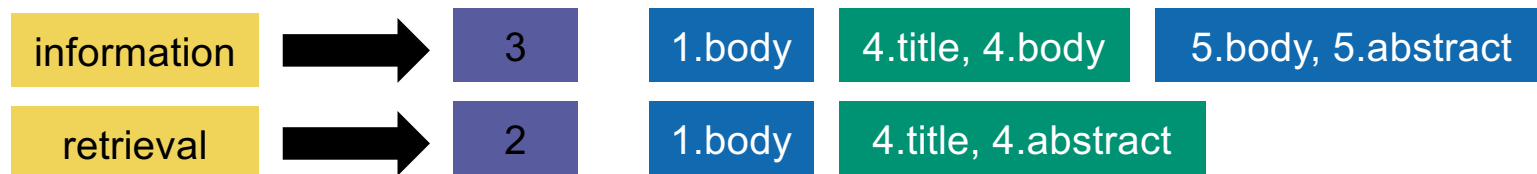
Document	Score
4	0.8
5	0.7
1	0.5

Two-term query



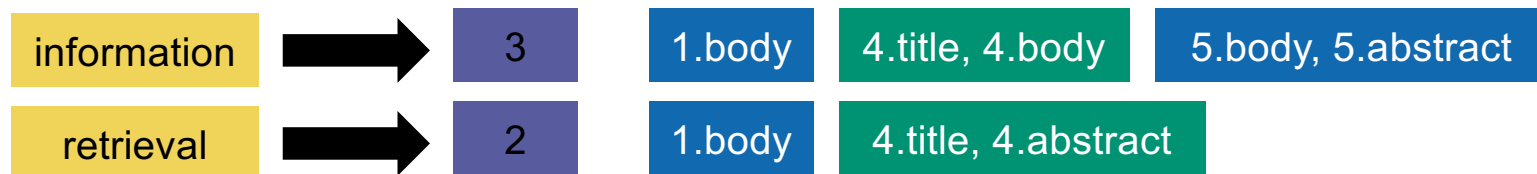
Zone	weight		information
Title	0.3	→	0.3
Abstract	0.2	→	
Body	0.5	→	0.5

Two-term query



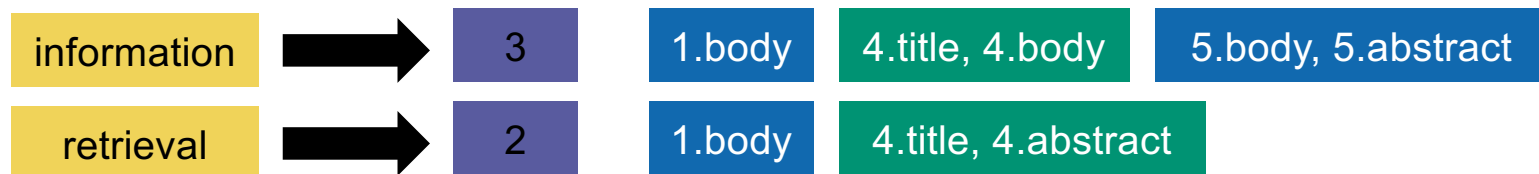
Zone	weight		information	retrieval
Title	0.3	→	0.3	
Abstract	0.2	→		
Body	0.5	→	0.5	

Two-term query



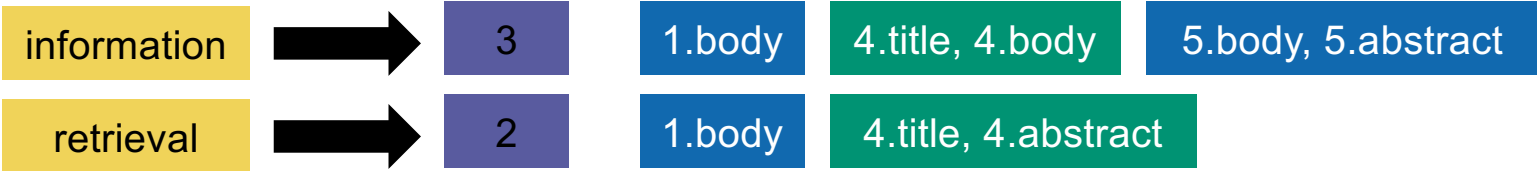
Zone	weight		information	retrieval
Title	0.3	→	0.3	0.3
Abstract	0.2	→		0.2
Body	0.5	→	0.5	

Two-term query



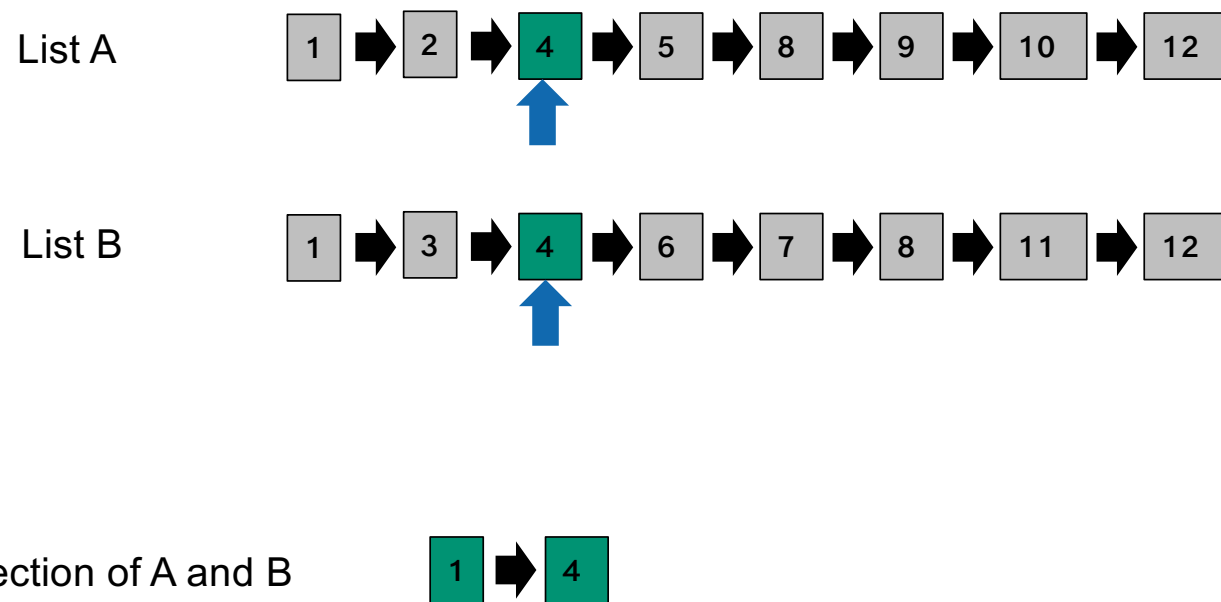
Zone	weight		information	retrieval	Total
Title	0.3	→	0.3	0.3	0.6
Abstract	0.2	→		0.2	0.2
Body	0.5	→	0.5		0.5

Two-term query

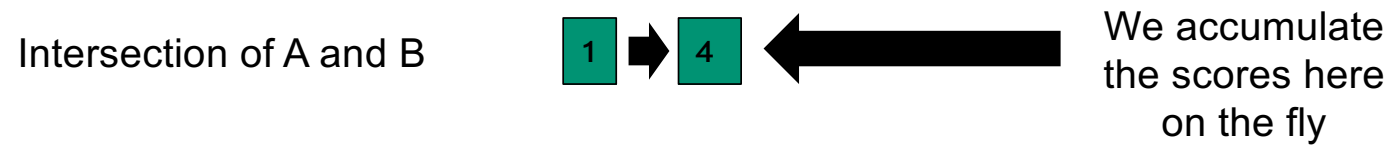
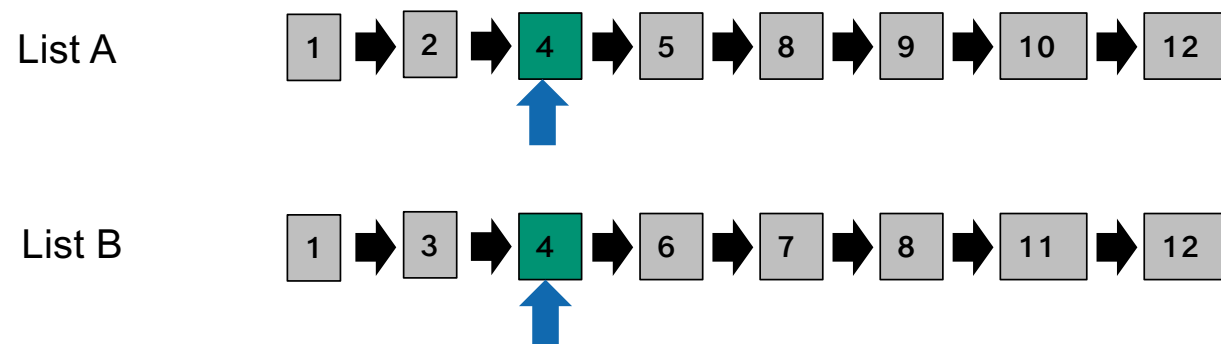


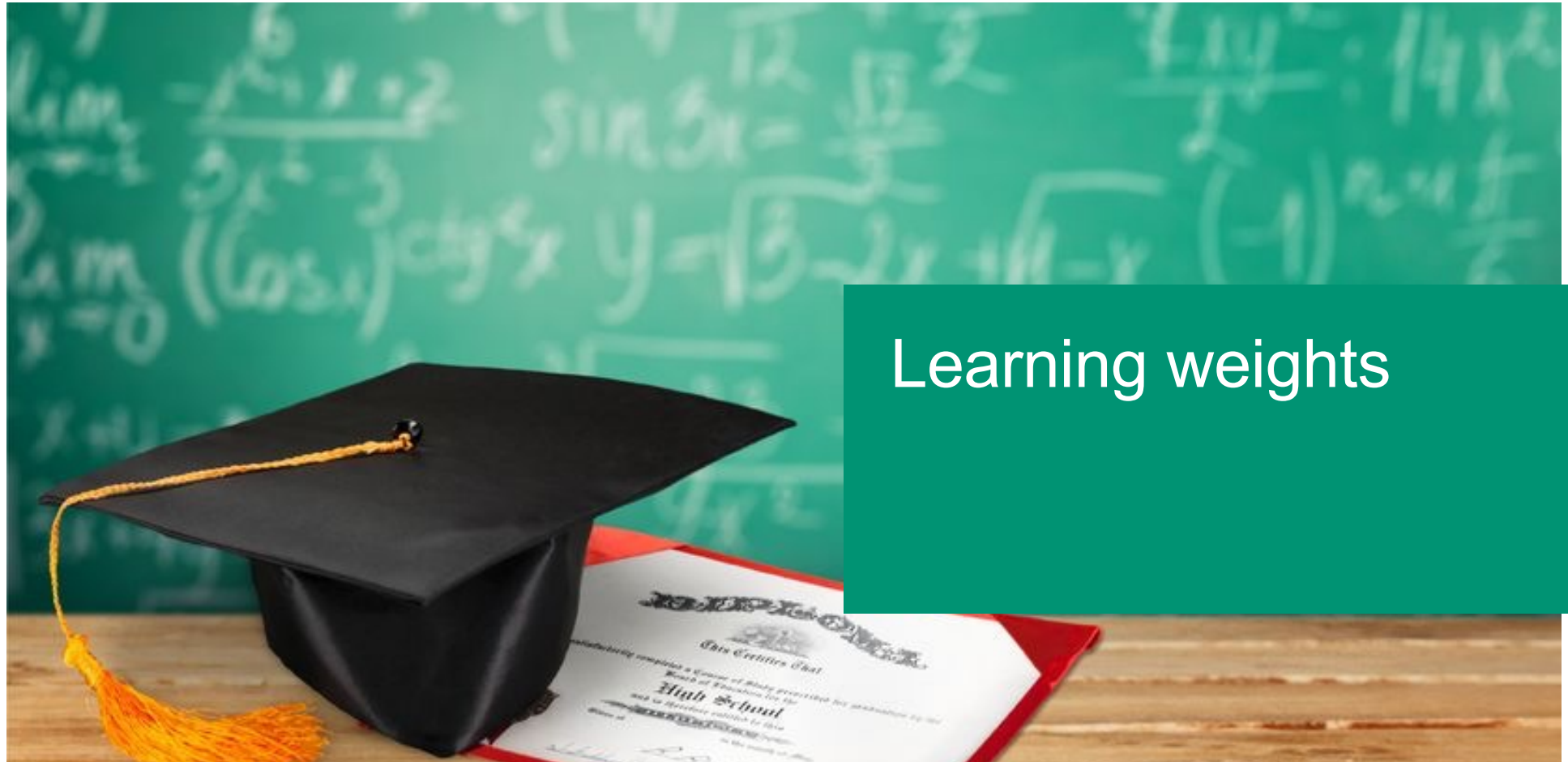
Zone	weight		information	retrieval	Total
Title	0.3	→	0.3	0.3	0.6
Abstract	0.2	→		0.2	0.2
Body	0.5	→	0.5		0.5
Total					1.3

Intersection algorithm



Intersection algorithm





Learning weights

Samples

$$\vec{s}_1, r_1$$

$$\vec{s}_2, r_2$$

$$\vec{s}_3, r_3$$

...

Samples

One query+result

Example:

Query "ETH" on document A

$$\vec{s}_1, r_1$$

$$\vec{s}_2, r_2$$

$$\vec{s}_3, r_3$$

...

Samples

	$\vec{s}_1, r_1$
	$\vec{s}_2, r_2$
0 (title)	$\vec{s}_3, r_3$
1 (abstract)	
0 (author)	
1 (body)	...

Boolean vector (zones that contain term)

Samples

$$\vec{s}_1, r_1$$

$$\vec{s}_2, \boxed{r_2}$$

1 Relevance (1 or 0, human input)

$$\vec{s}_3, r_3$$

...

Score formula

$$\vec{g}_j \cdot \vec{s}_j = \sum_{i=1}^{i=l} g_{ij} s_{ij}$$

Predicting relevance

$$\vec{g}_j \cdot \vec{s}_j = \sum_{i=1}^{i=l} g_{ij} s_{ij} \longleftrightarrow r_j$$

In an ideal world, equal

Predicting relevance

$$\vec{g}_j \cdot \vec{s}_j = \sum_{i=1}^{i=l} g_{ij} s_{ij} \longleftrightarrow r_j$$

In an ideal world, equal

$$\text{error}_j(g) = (r_j - \vec{g}_j \cdot \vec{s}_j)^2$$

Finding the minimal error

$$\operatorname{argmin}_g \sum_j \operatorname{error}_j(g) = \operatorname{argmin}_g \sum_j (r_j - \vec{g}_j \cdot \vec{s}_j)^2$$

This is
Machine Learning
(see other lectures by Prof. Krause, Prof. Buhmann)

Sometimes, we don't actually need machine learning...

$$\frac{n_{10r} + n_{01n}}{n_{10r} + n_{10n} + n_{01r} + n_{01n}}$$

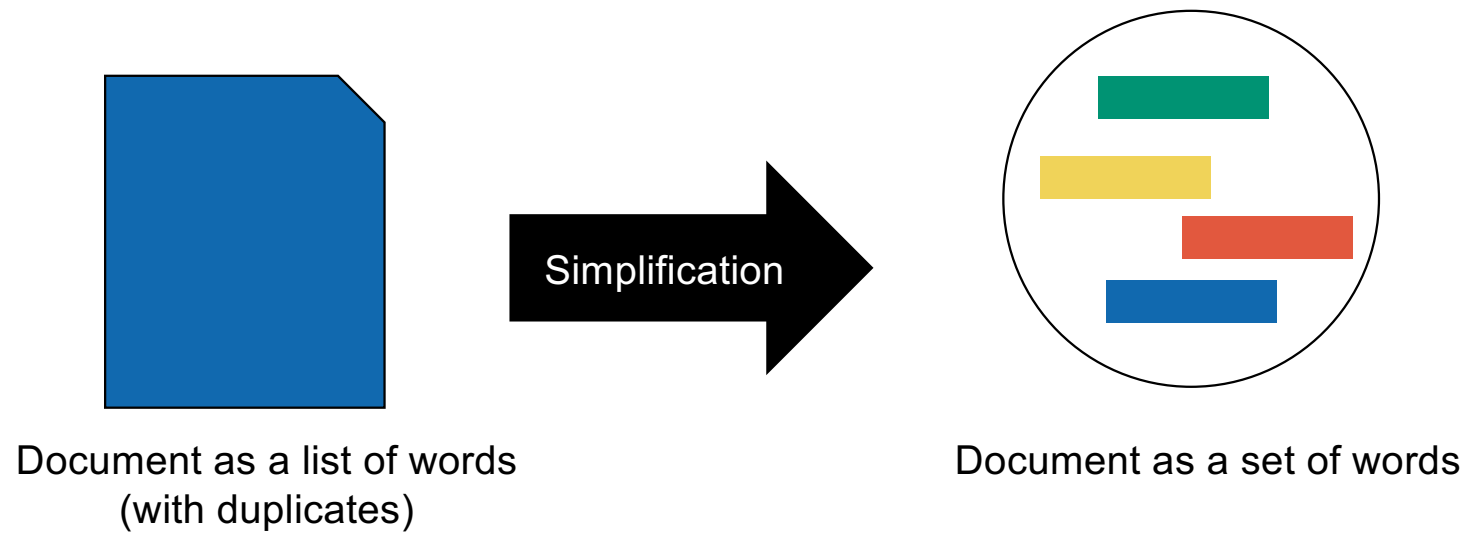
Analysis on polynomials is sometimes enough

(Here: two zones, using the document counts per zone and relevance)

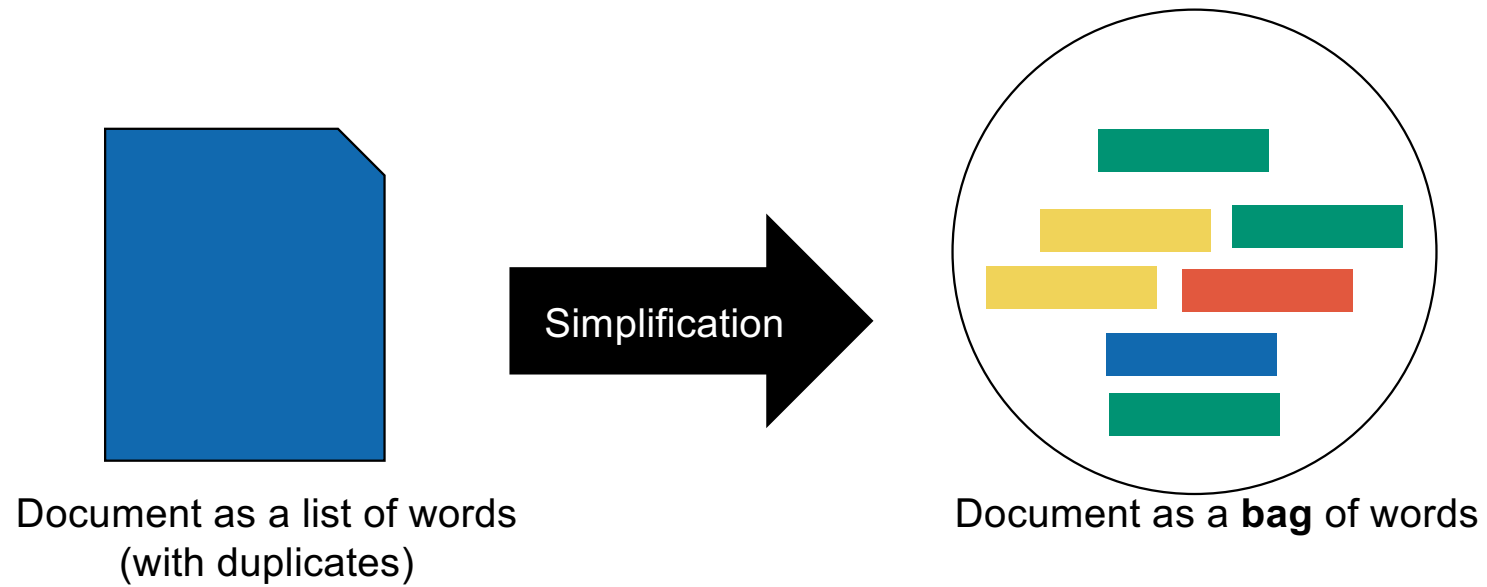


Weights

Model and abstraction



Model and abstraction



Term frequency

foo
bar
foobar
foo
foo
foobar
bar

d

Term frequency

foo
bar
foobar
foo
foo
foobar
bar

d



t	$tf_{t,d}$
foo	3
bar	2
foobar	2

Score (first tentative)

tf	A	B
foo	5	1
bar	0	4
foobar	2	1

Score (first tentative)

tf	A	B
foo	5	1
bar	0	4
foobar	2	1

score	A	B
	7	6

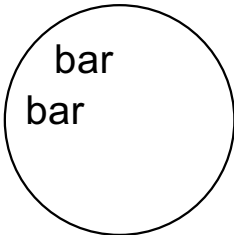
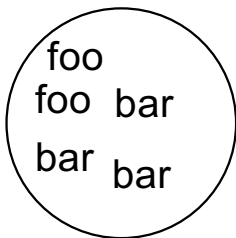
Score (first tentative)

tf	A	B
foo	5	1
bar	0	4
foobar	2	1

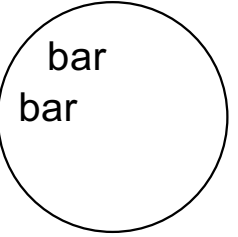
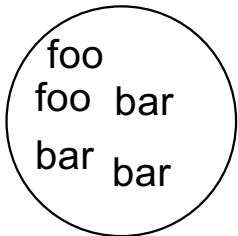
score	A	B
	7	6

But is B not more relevant than A as **bar** is rare?

Collection frequency

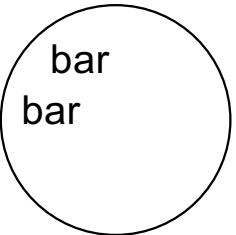
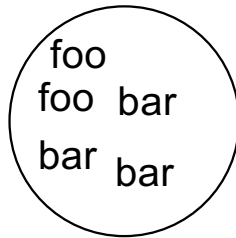


Collection frequency



t	cf _t
foo	4
bar	5
foobar	5

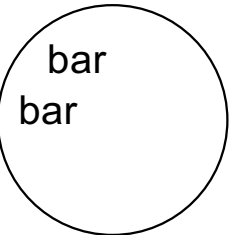
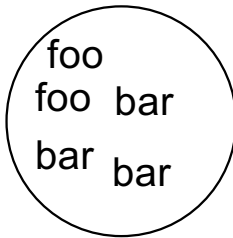
Collection frequency



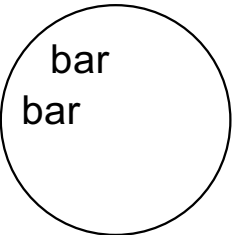
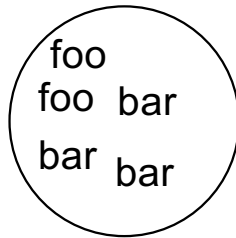
t	cf _t
foo	4
bar	5
foobar	5

But foobar is rare, in that it appears only in one document!

Document frequency

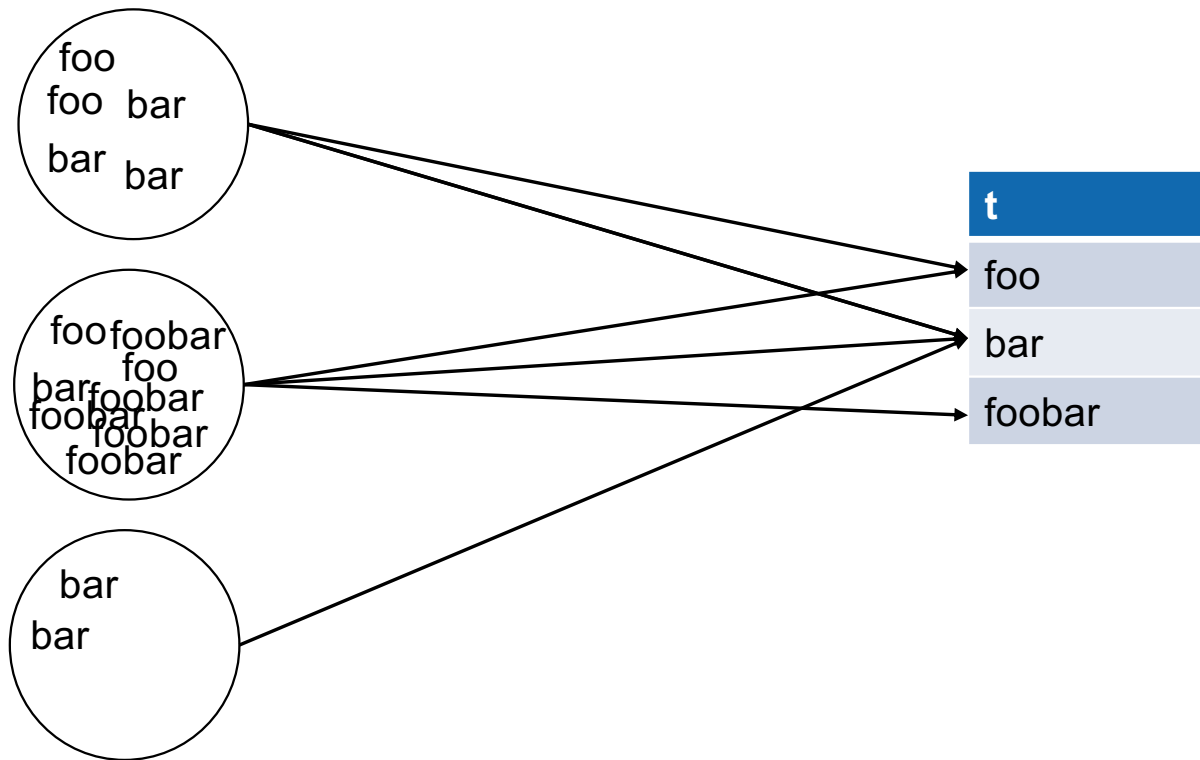


Document frequency

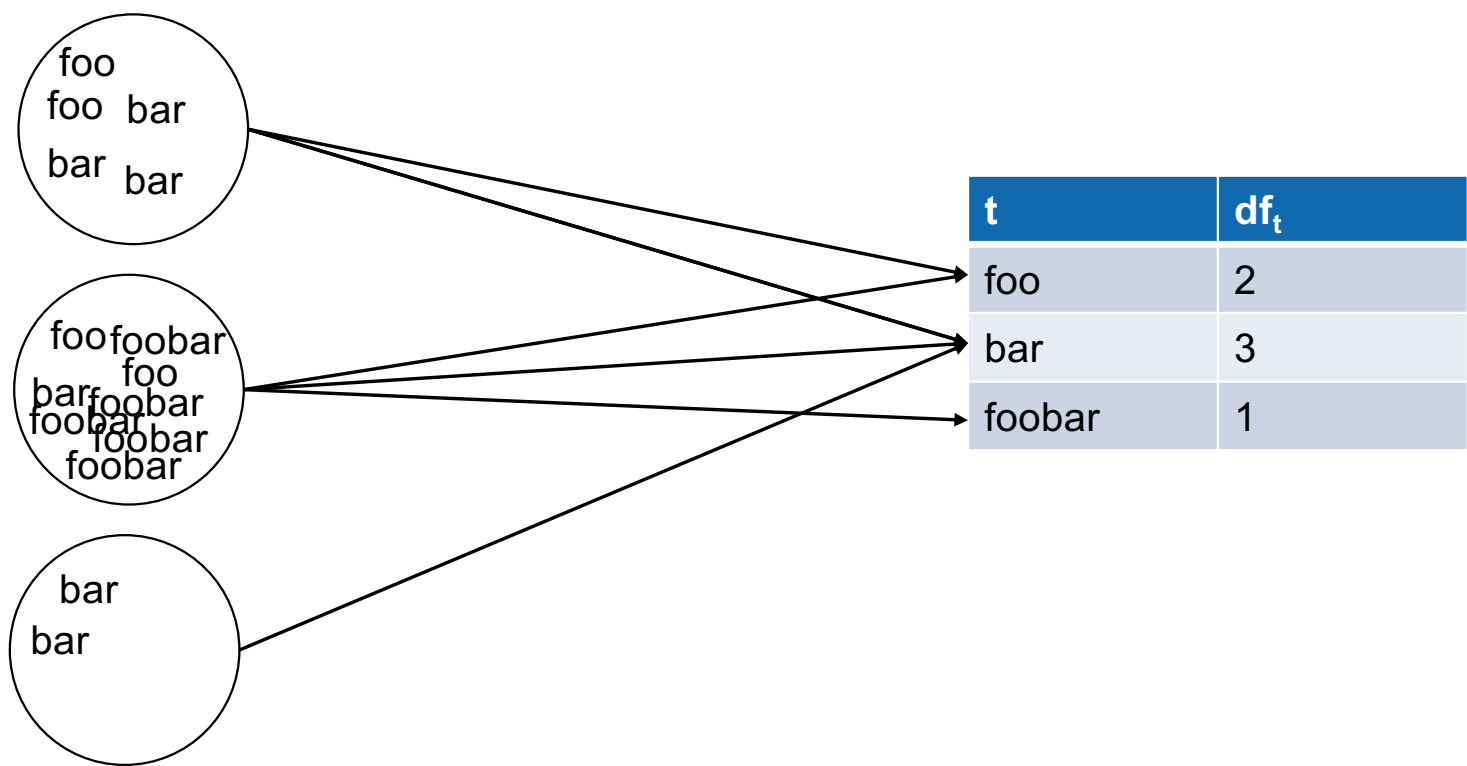


t
foo
bar
foobar

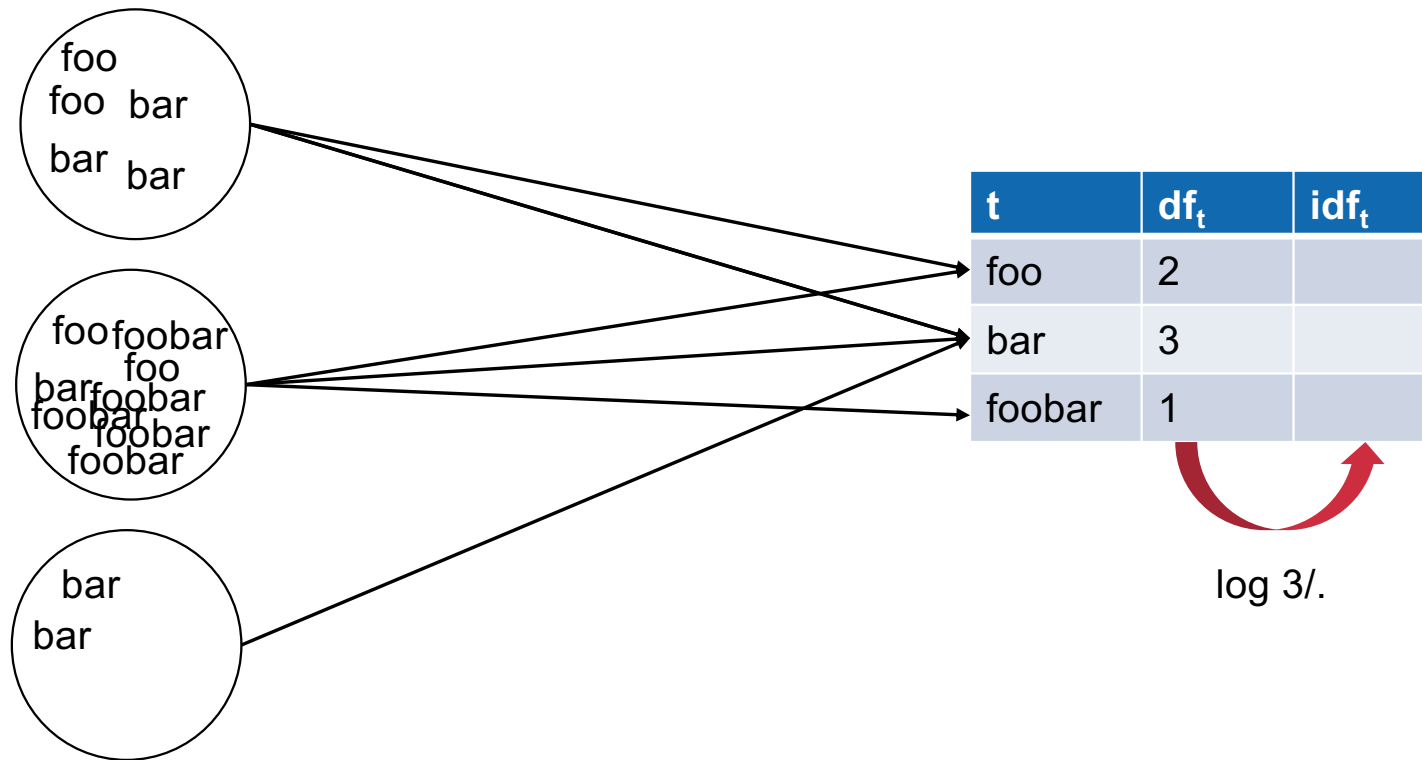
Document frequency



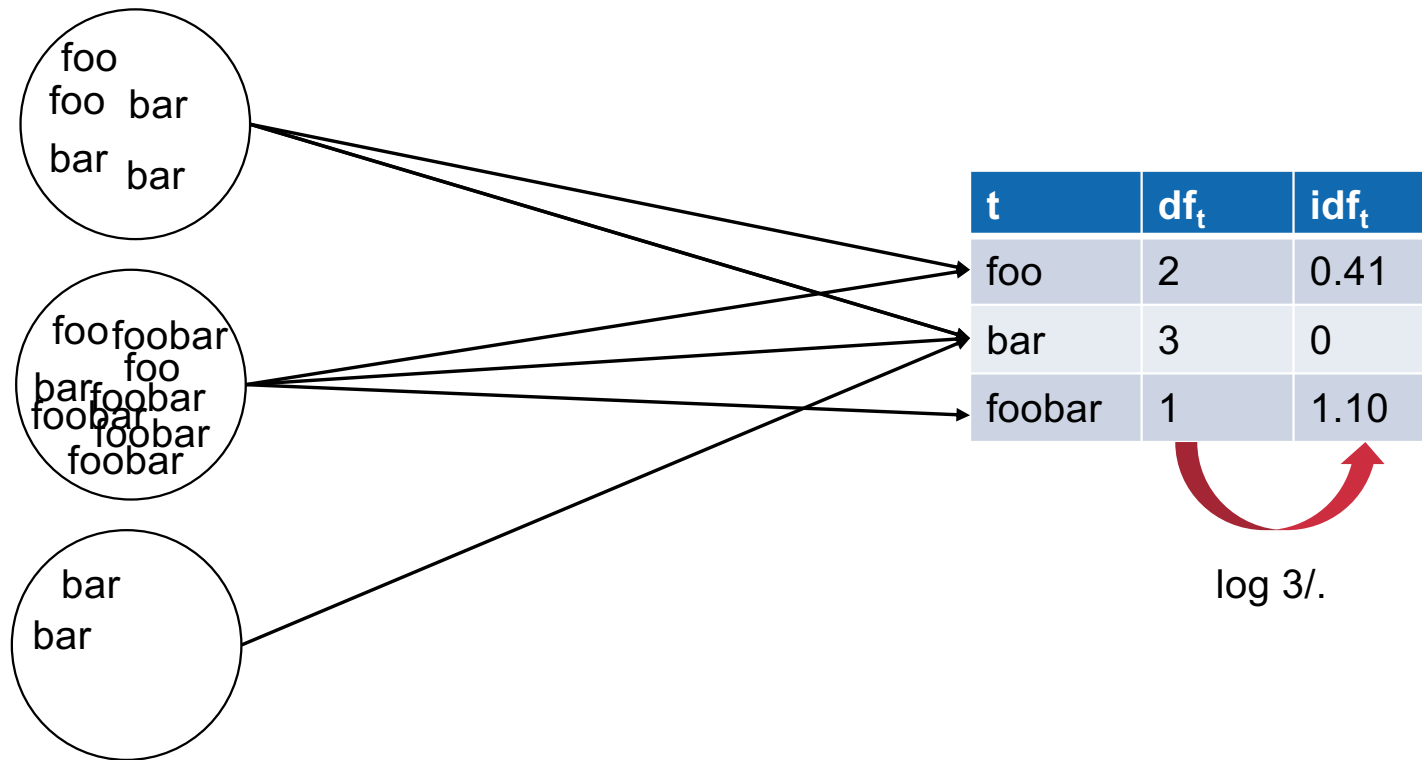
Document frequency



Inverse document frequency



Inverse document frequency



Putting it all together (tf-idf)

tf	A	B
foo	5	1
bar	0	4
foobar	2	1

Putting it all together (tf-idf)

tf	A	B
foo	5	1
bar	0	4
foobar	2	1

idf	
foo	5
bar	10
foobar	3

Putting it all together (tf-idf)

tf	A	B
foo	5	1
bar	0	4
foobar	2	1



tf-idf	A	B
foo	25	5
bar	0	40
foobar	6	3



idf	
foo	5
bar	10
foobar	3

Score

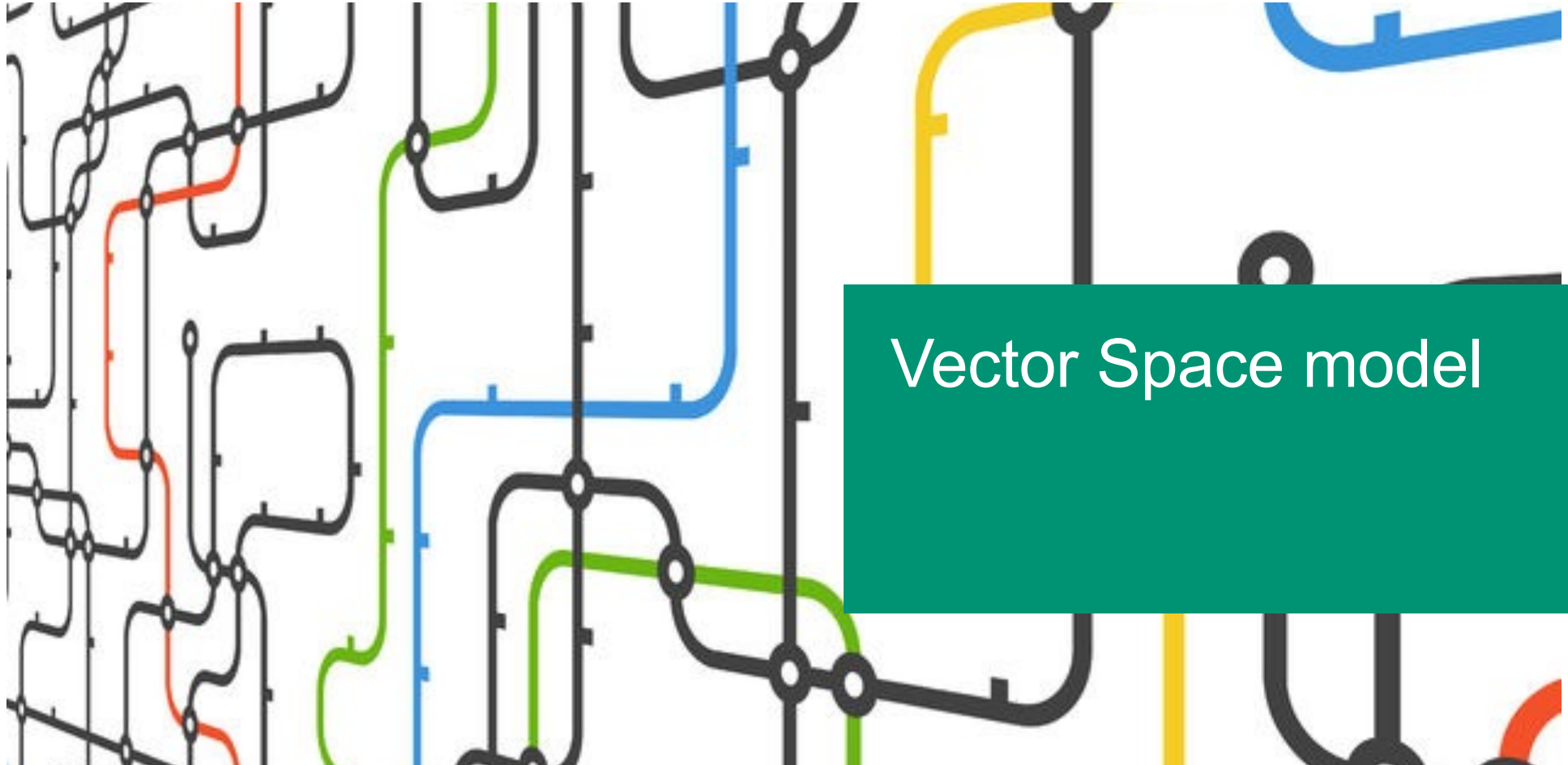
tf-idf	A	B
foo	25	5
bar	0	40
foobar	6	3

score	A	B
	31	48

Score

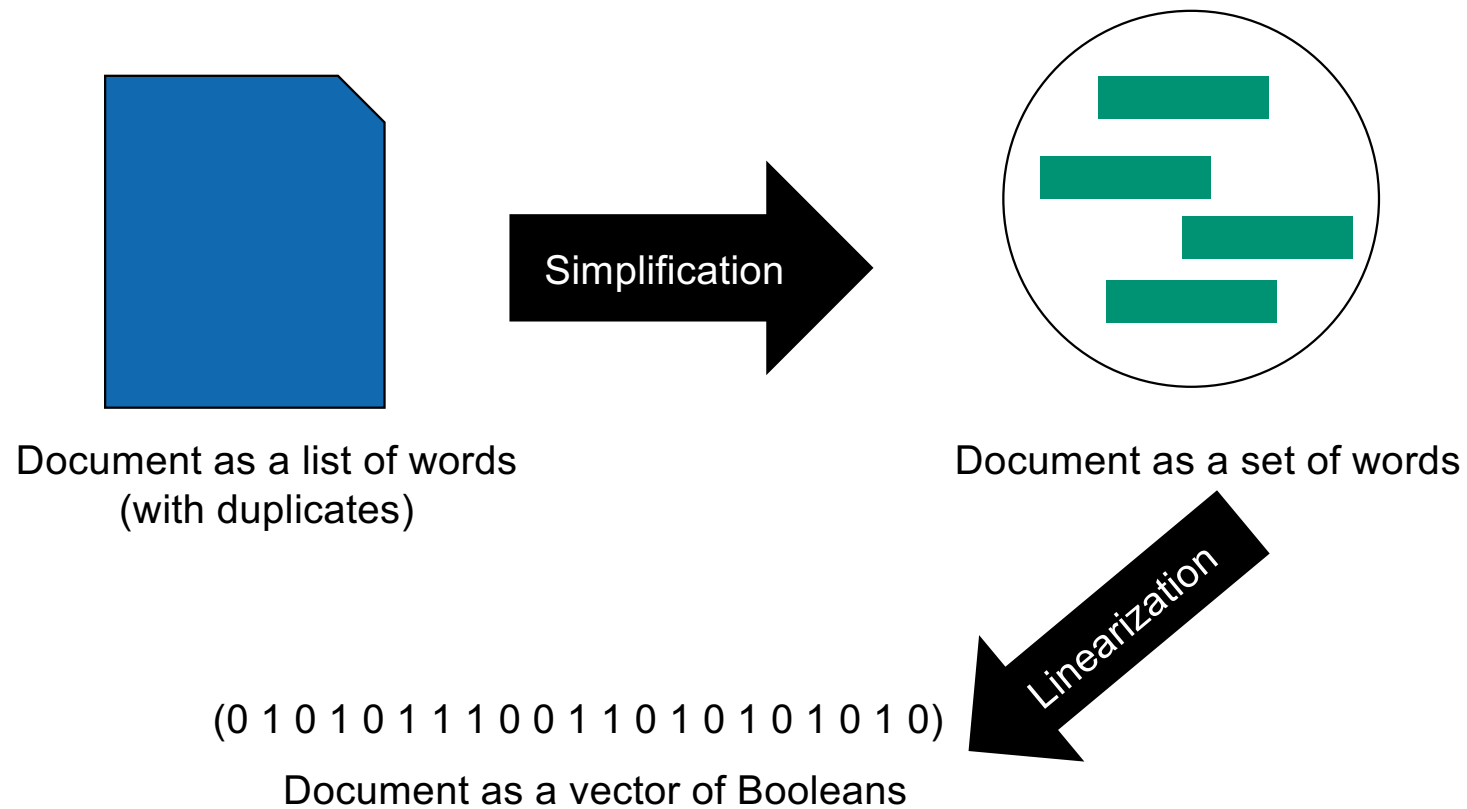
tf-idf	A	B
foo	25	5
bar	0	40
foobar	6	3

score	A	B
	31	48

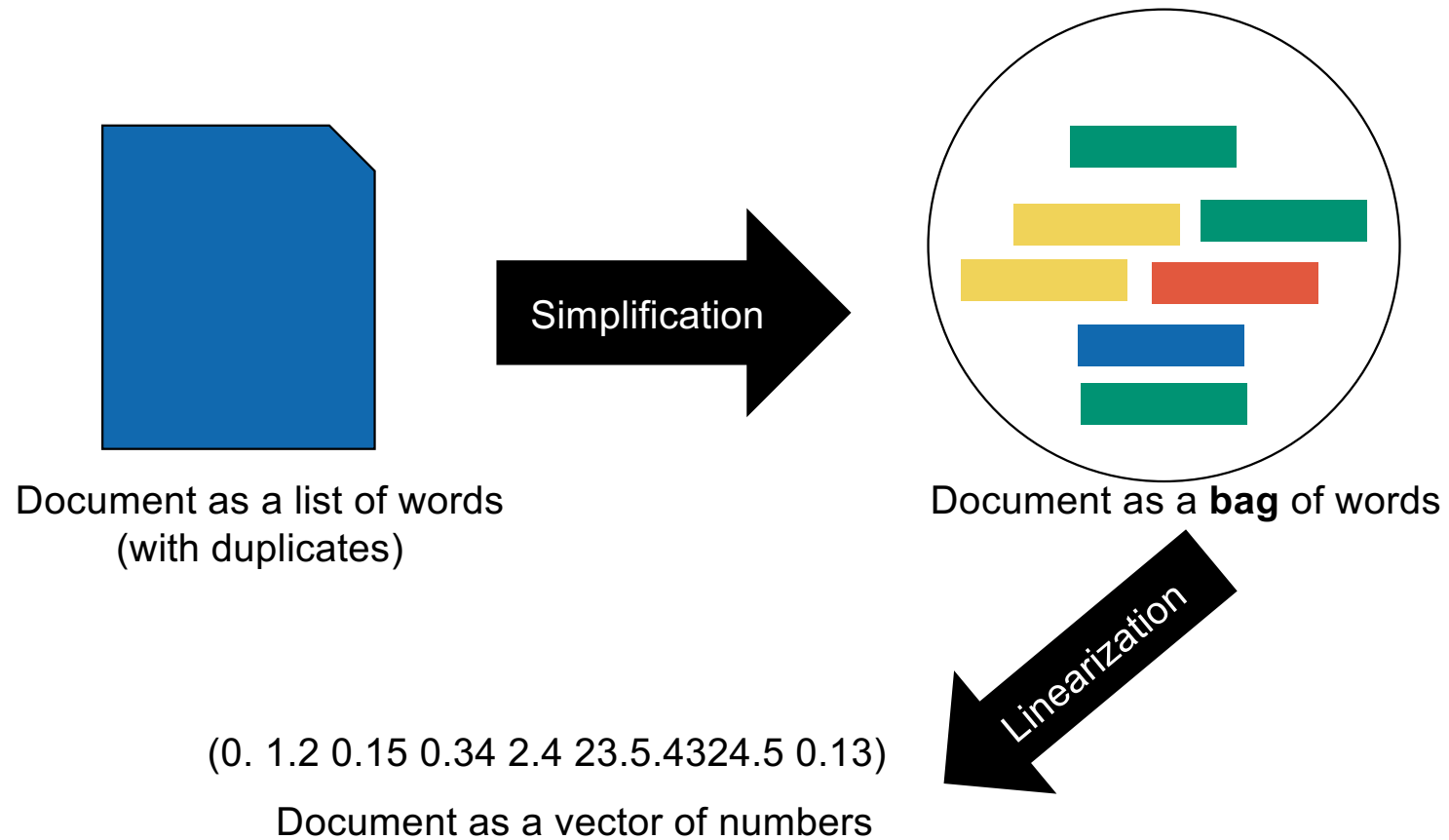


Vector Space model

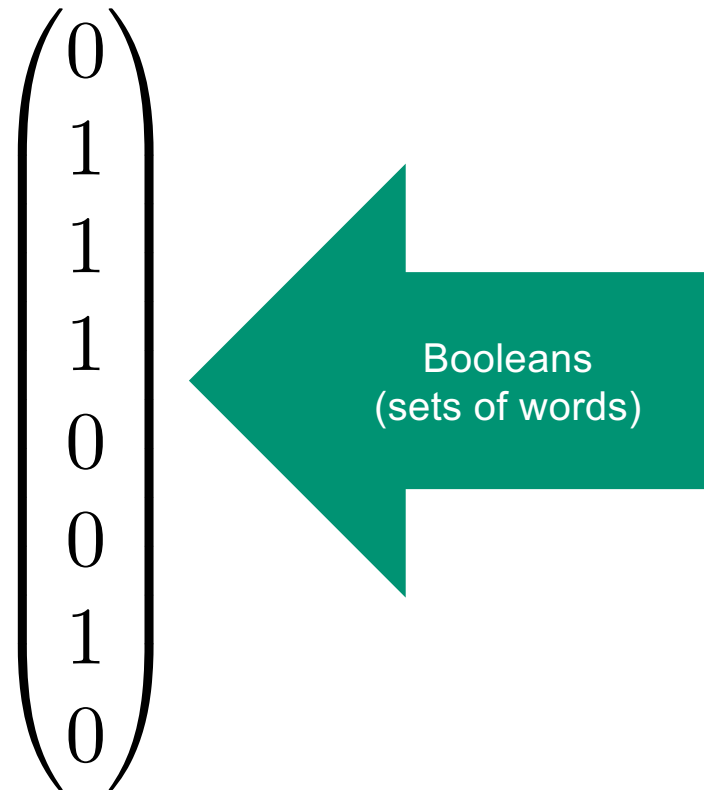
Model and abstraction



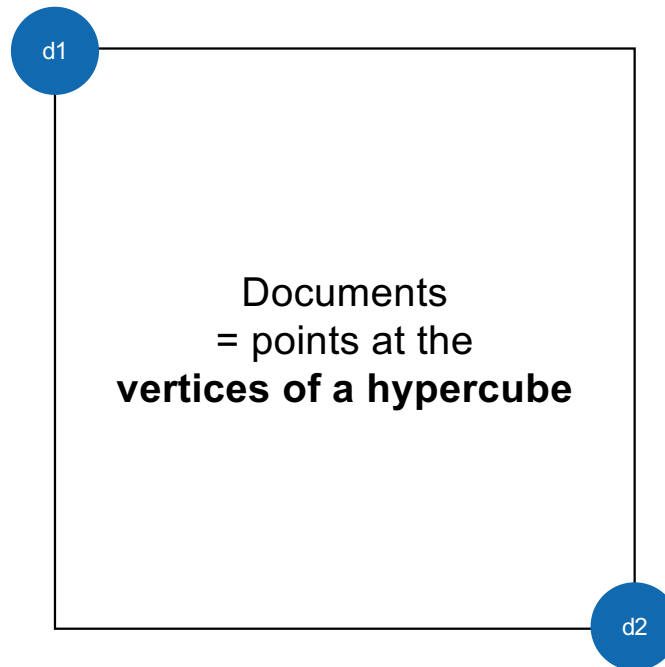
Model and abstraction



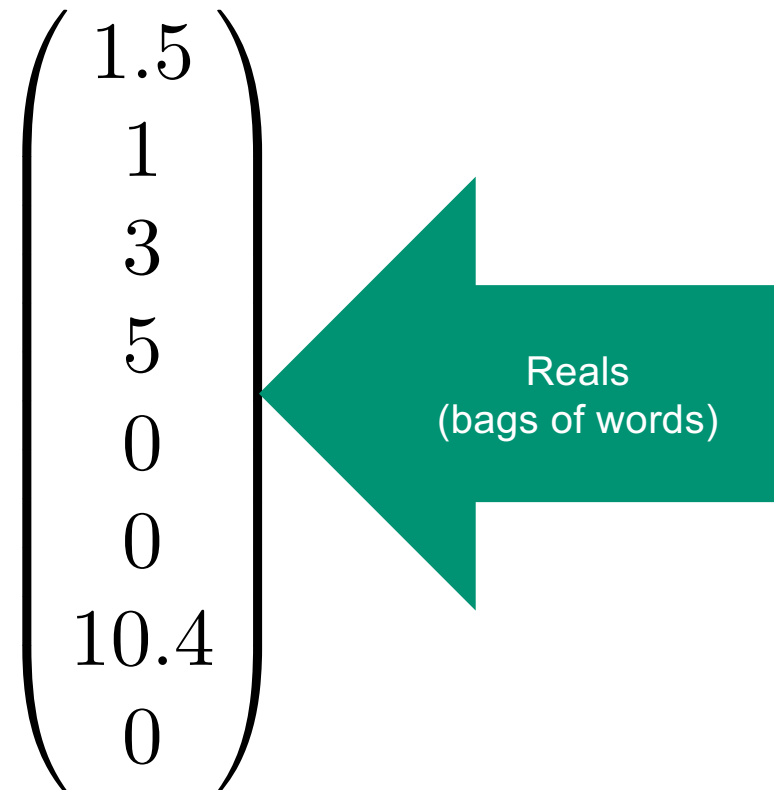
Documents as vectors



Documents as vectors



Documents as vectors



Documents as vectors

Could be:

- term frequency
- **td-idf [preferred]**
- other weighting schemes

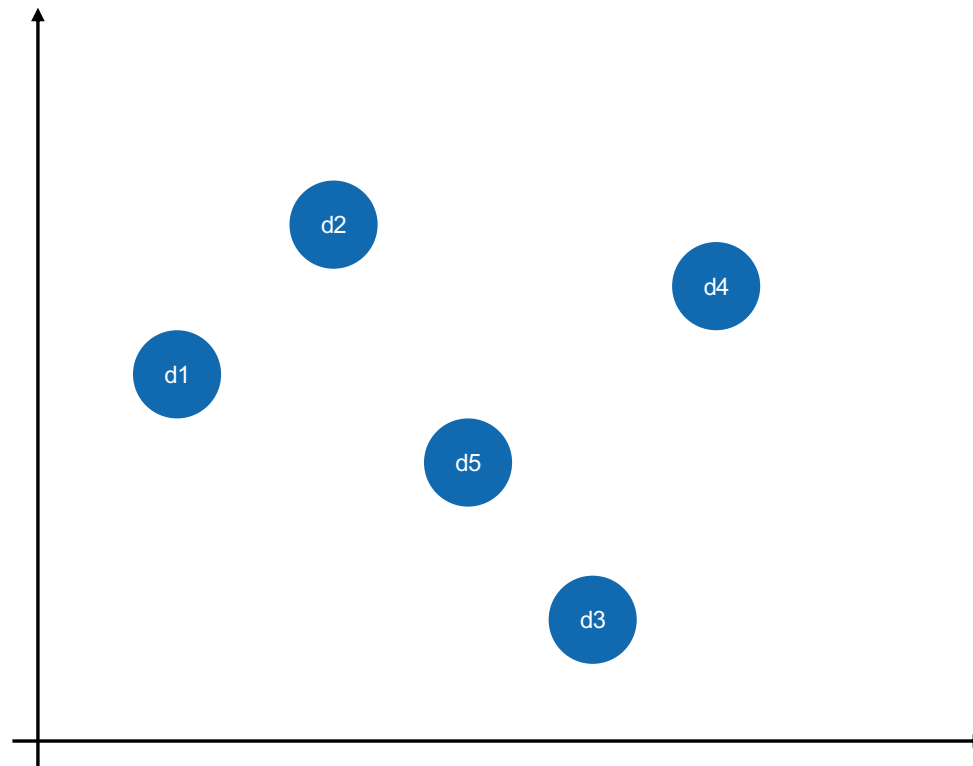
$$\begin{pmatrix} 1.5 \\ 1 \\ 3 \\ 5 \\ 0 \\ 0 \\ 10.4 \\ 0 \end{pmatrix}$$

Documents as vectors

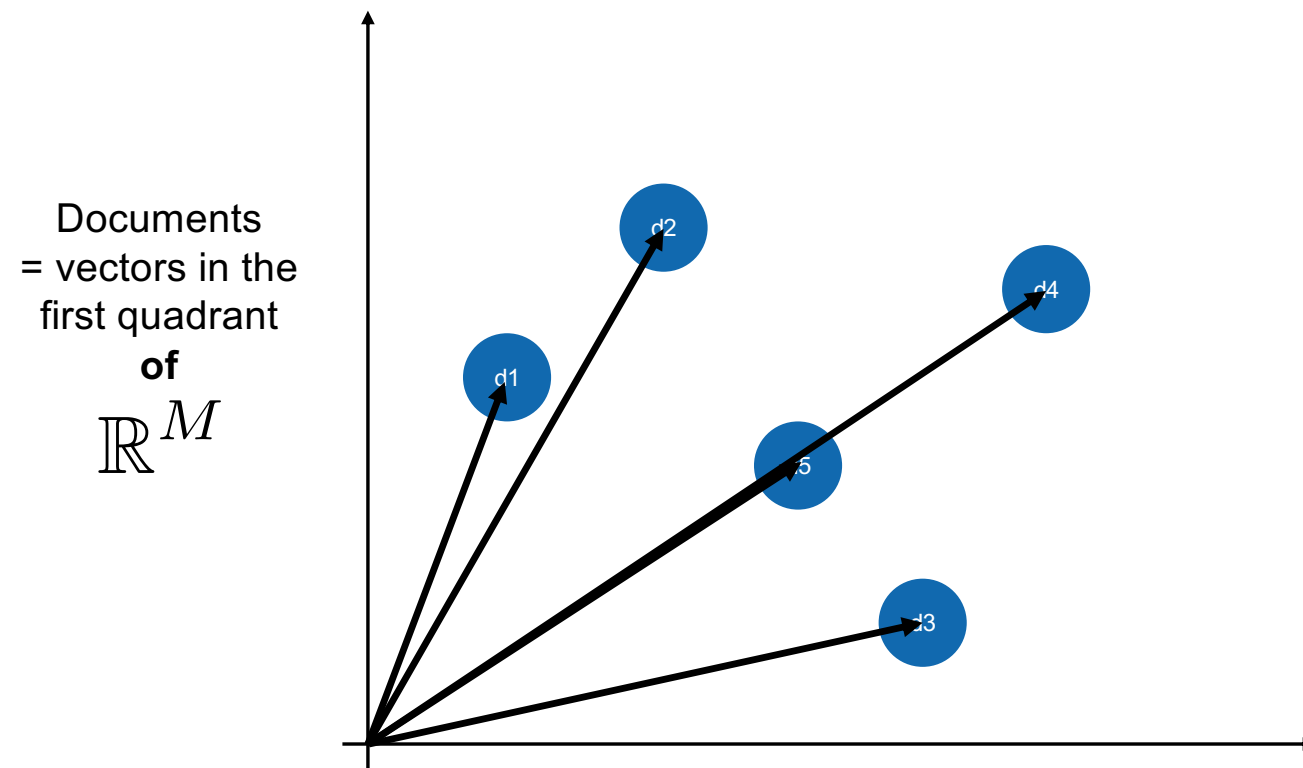


Documents as vectors

Documents
= points in the
first quadrant
of
 $\mathbb{R}^M$



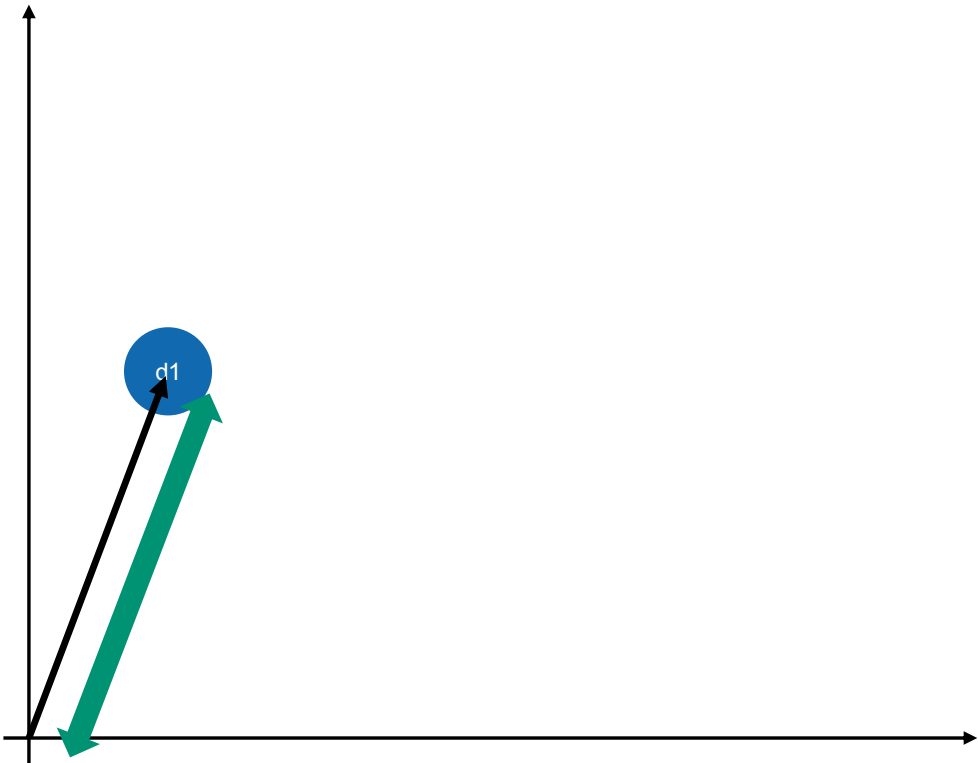
Documents as vectors



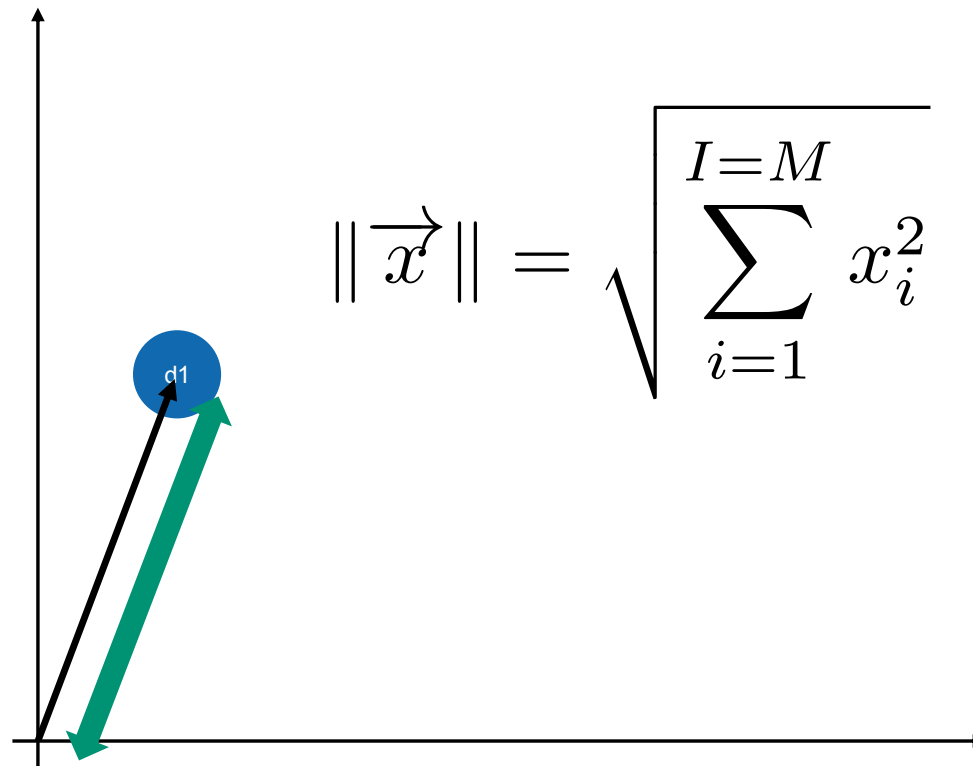
Vector notation

$$\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_M \end{pmatrix}$$

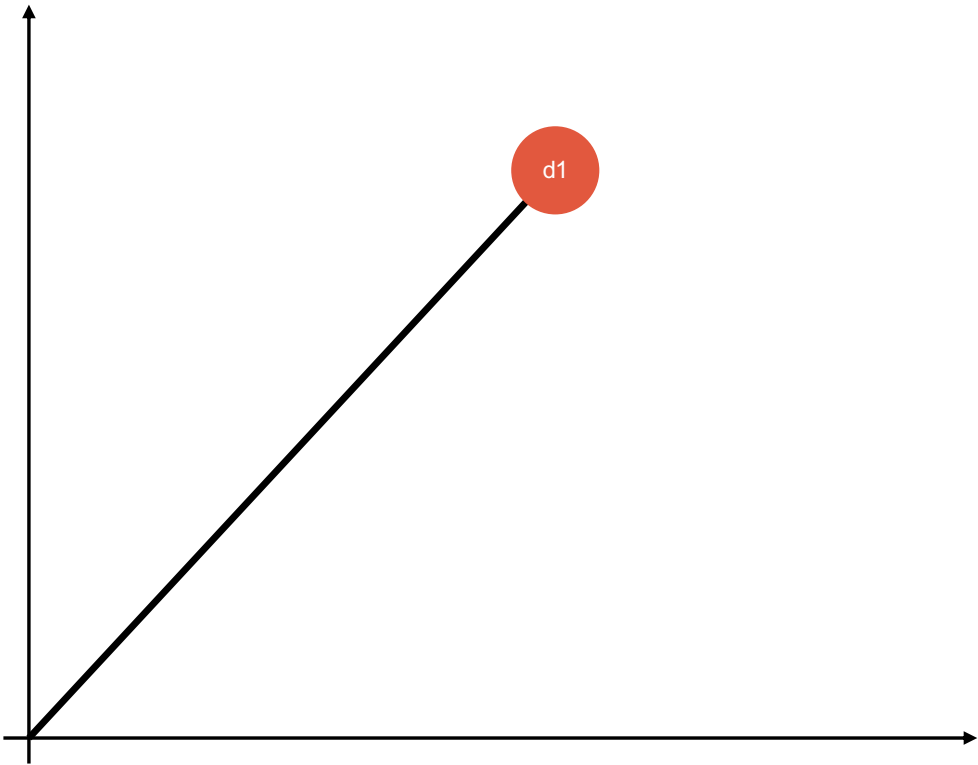
Euclidian norm



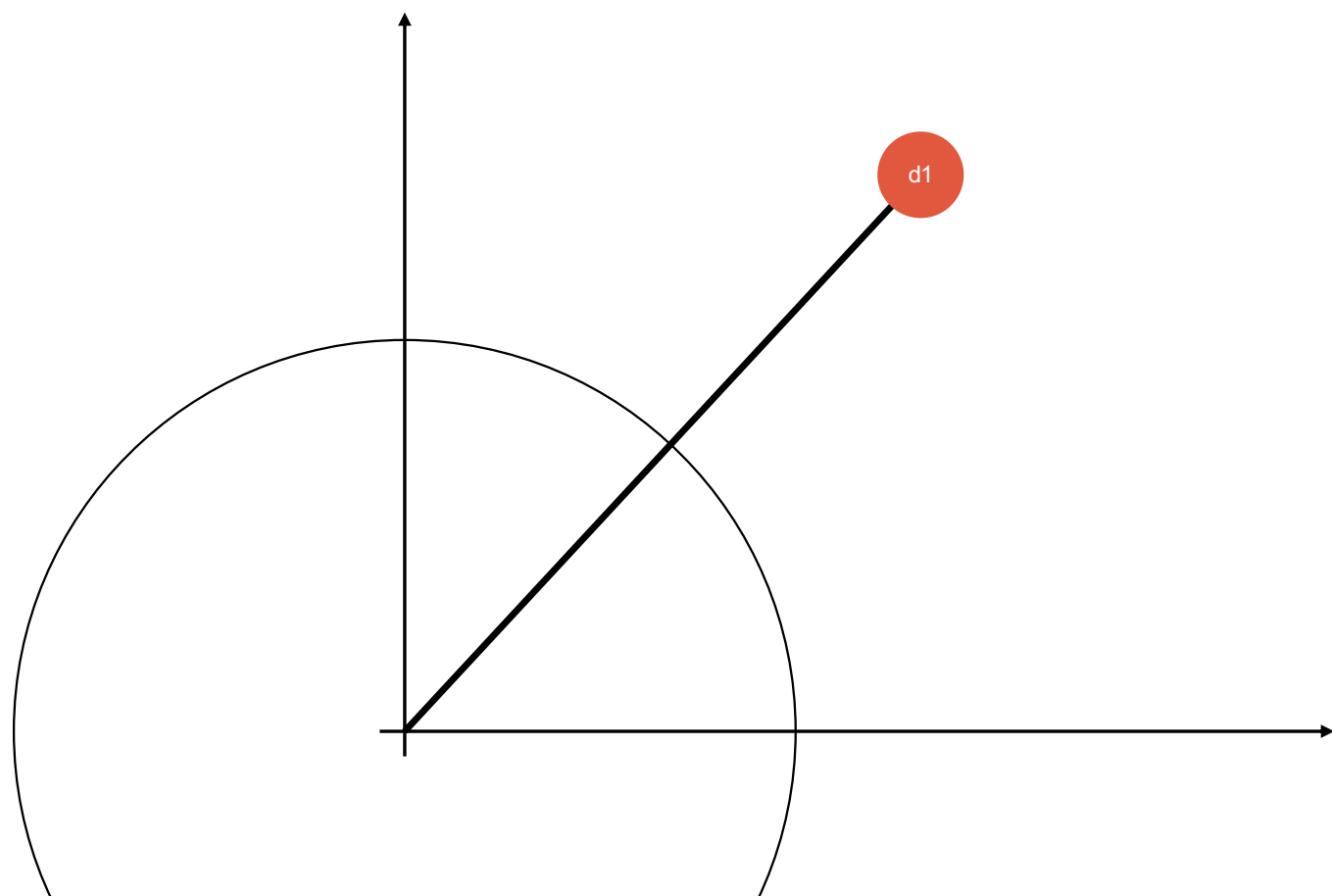
Euclidian norm



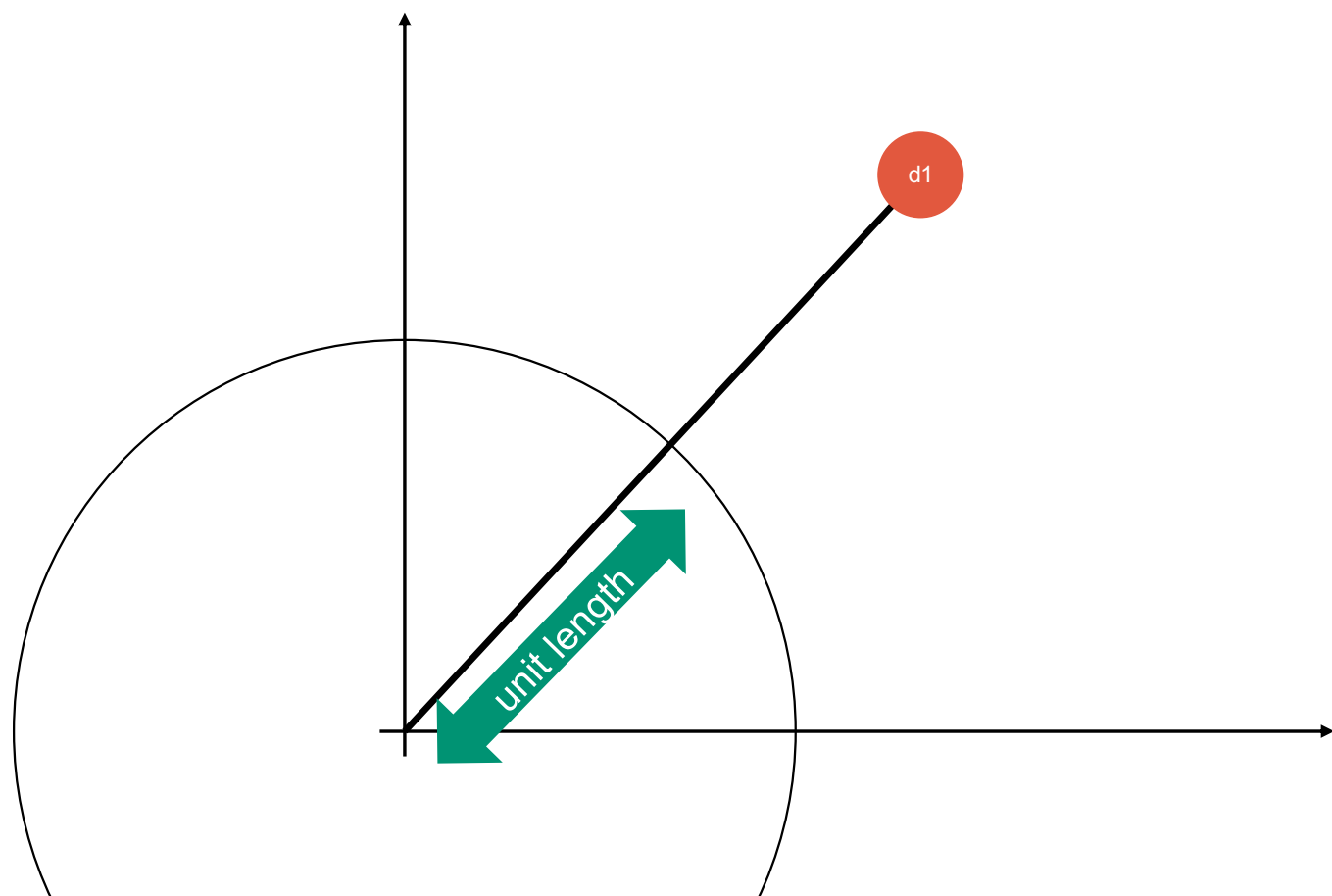
Renormalization



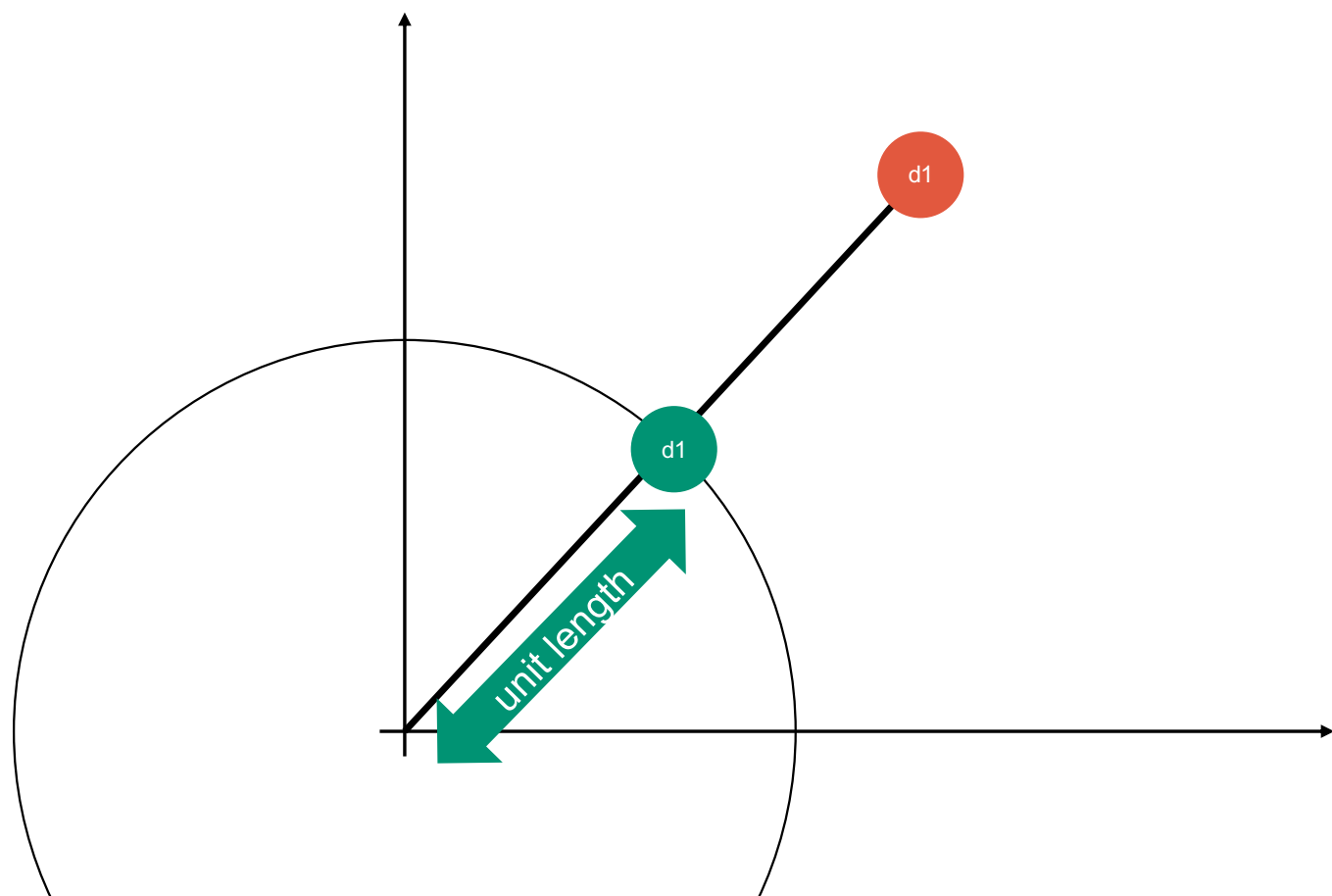
Renormalization



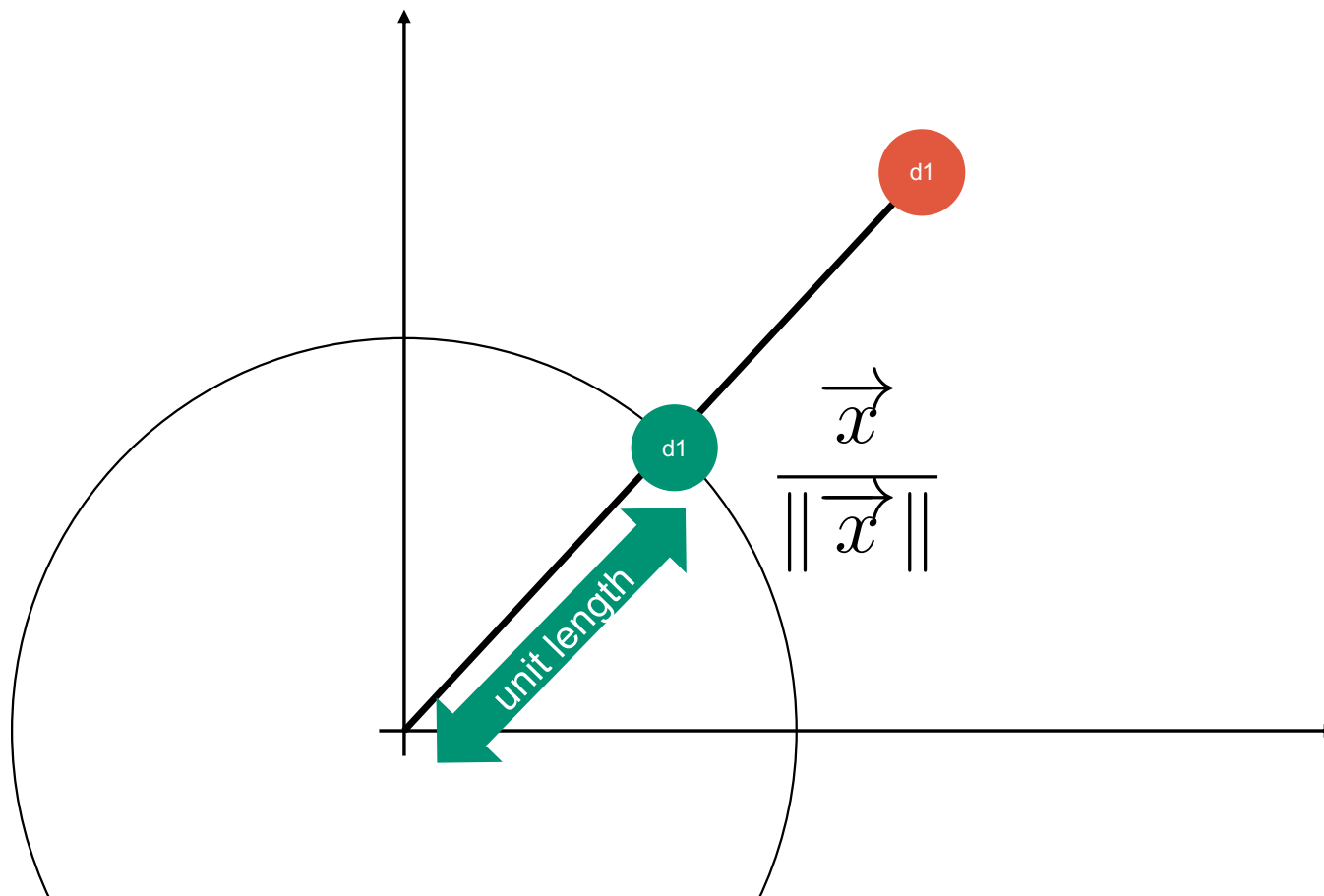
Renormalization



Renormalization



Renormalization

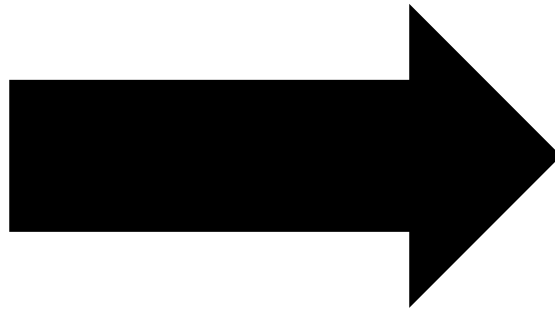


Normalization

$$\begin{pmatrix} 1.5 \\ 1 \\ 3 \\ 5 \\ 0 \\ 0 \\ 10.4 \\ 0 \end{pmatrix}$$

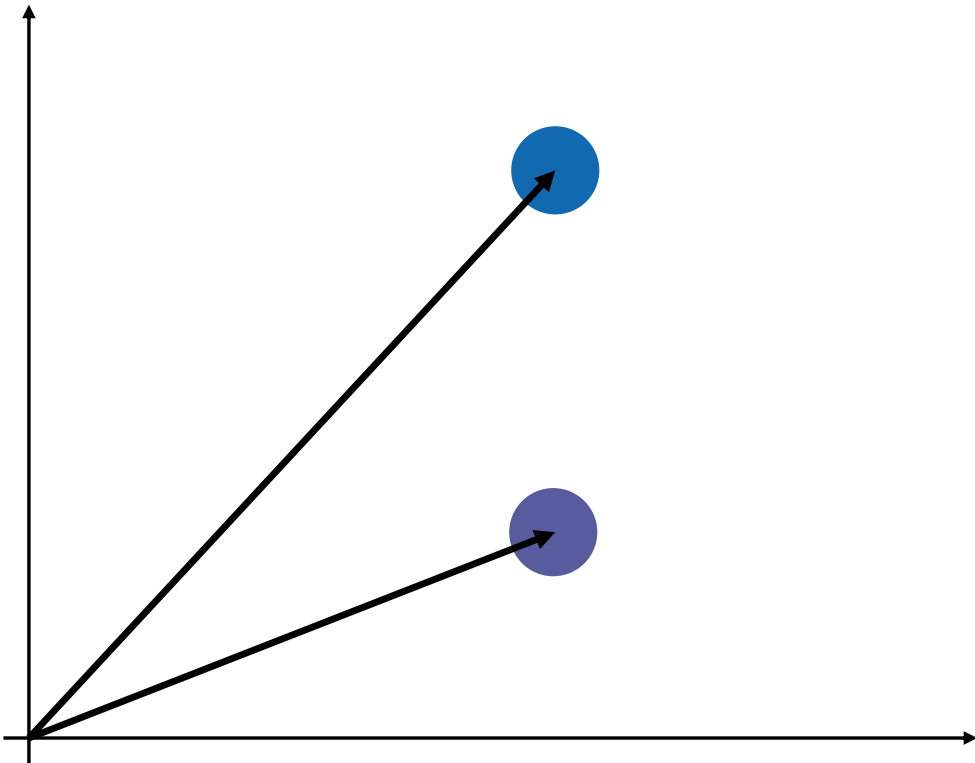
Normalization

$$\begin{pmatrix} 1.5 \\ 1 \\ 3 \\ 5 \\ 0 \\ 0 \\ 10.4 \\ 0 \end{pmatrix}$$

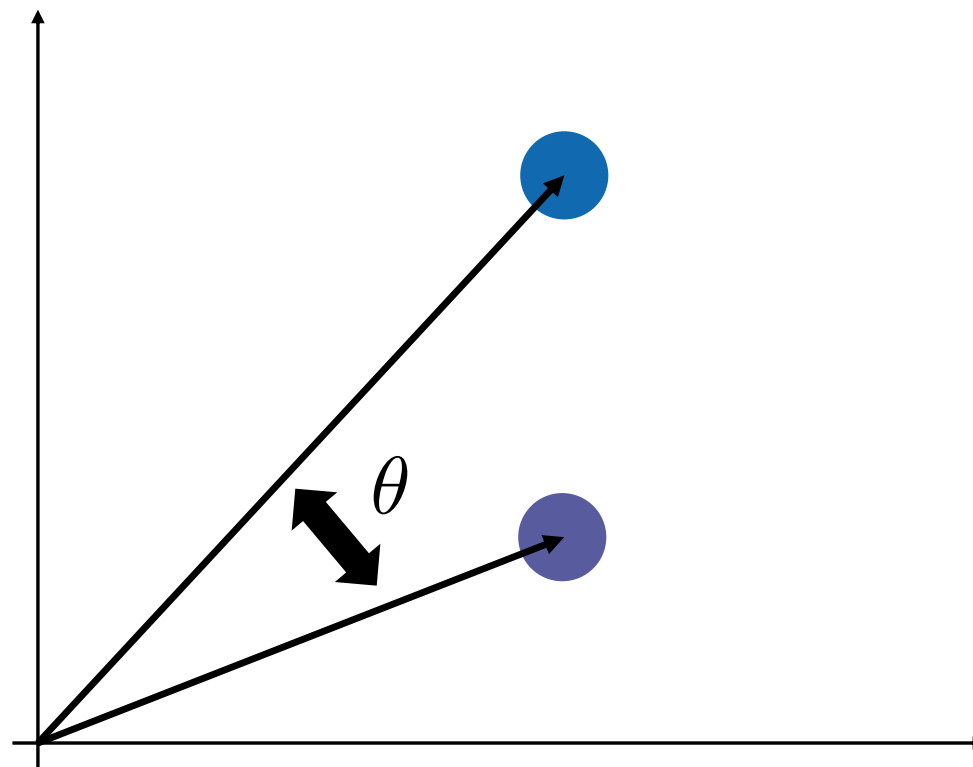


$$\begin{pmatrix} 0.12 \\ 0.08 \\ 0.25 \\ 0.41 \\ 0 \\ 0 \\ 0.86 \\ 0 \end{pmatrix}$$

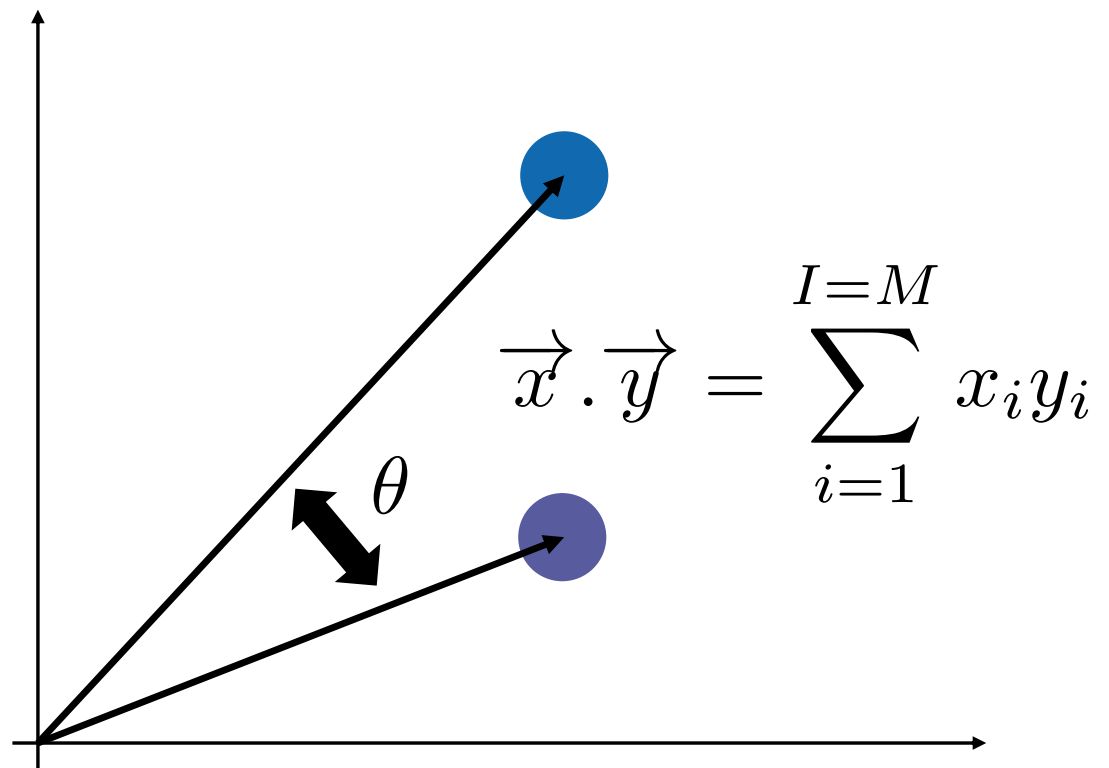
Inner product



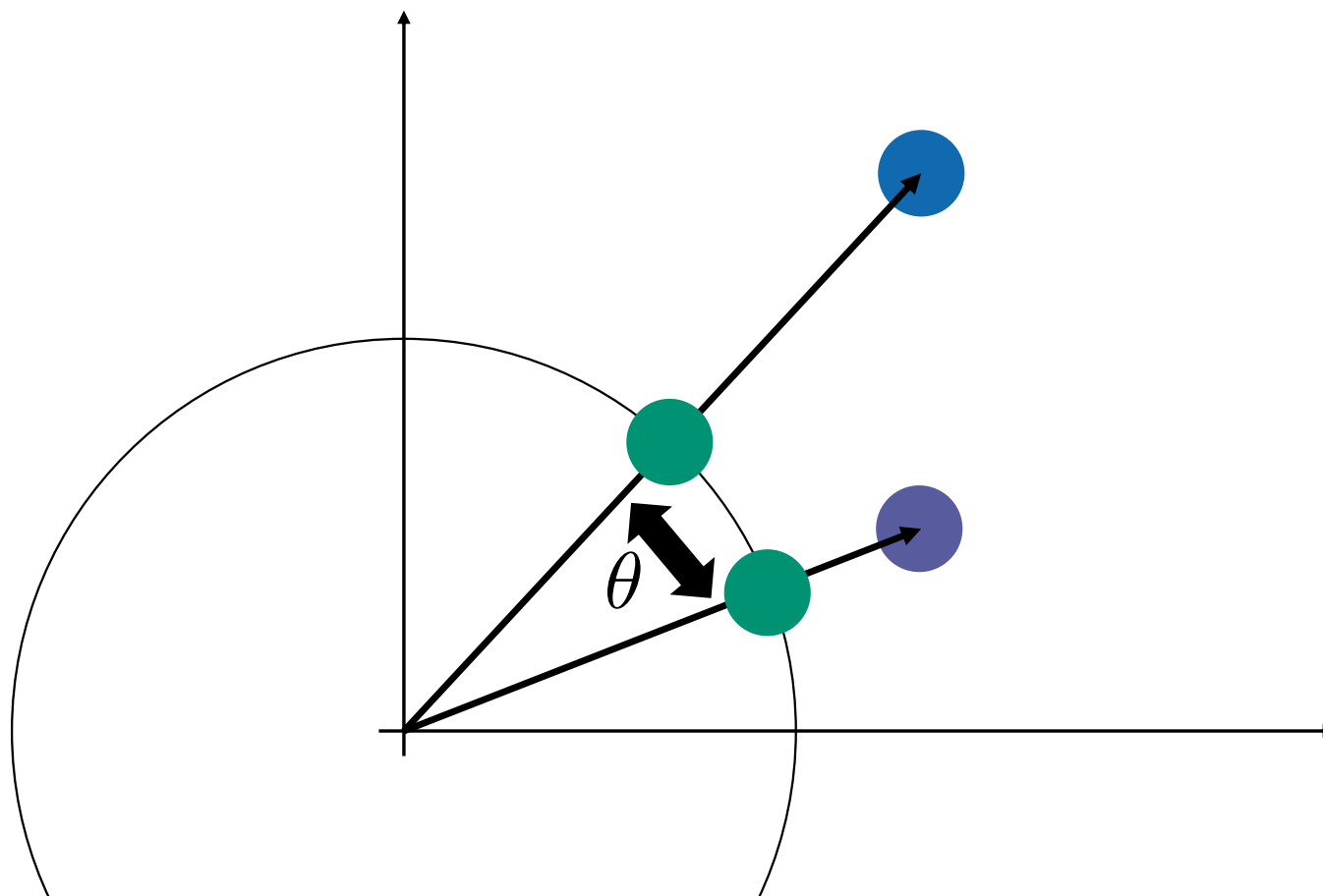
Inner product



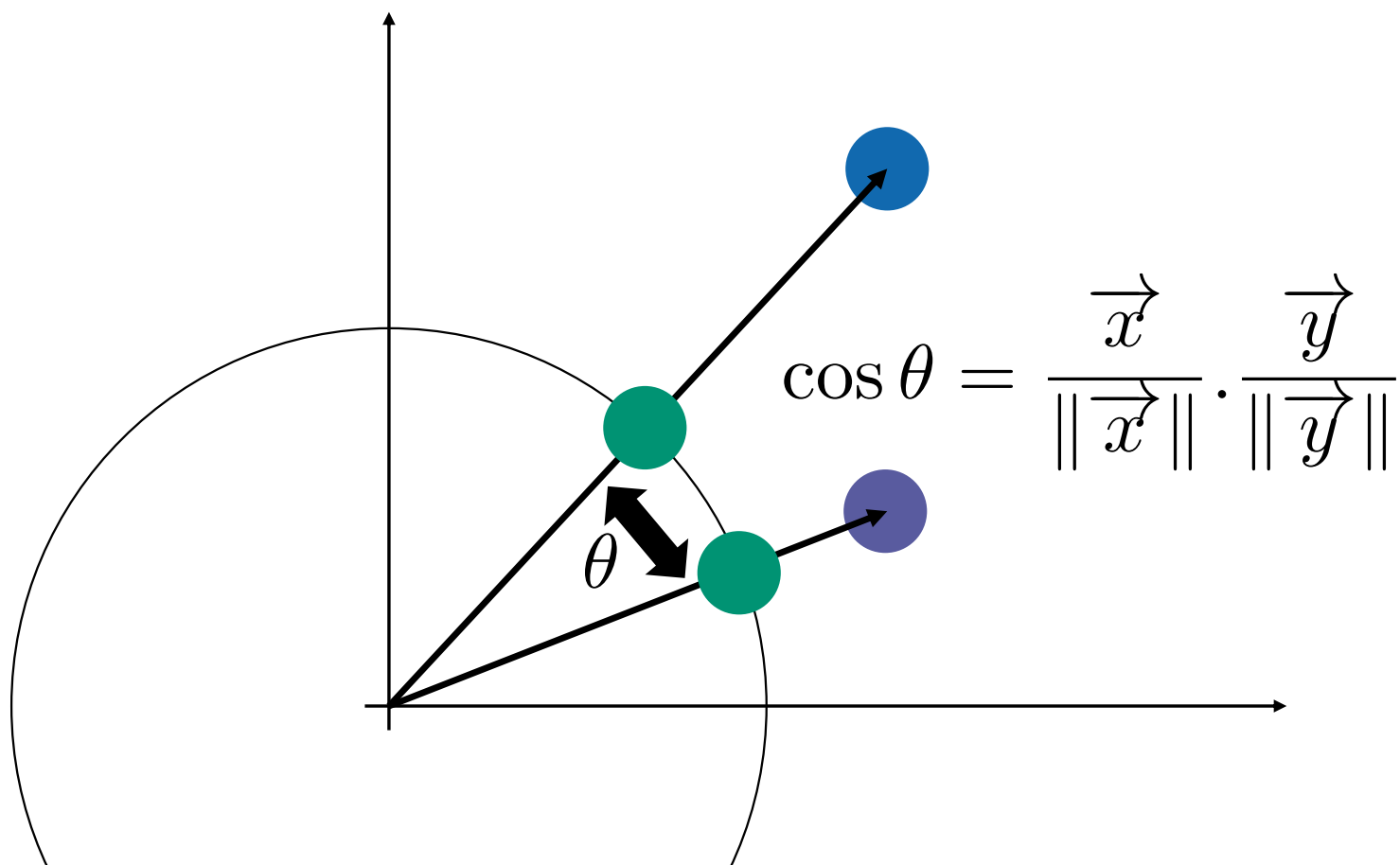
Inner product



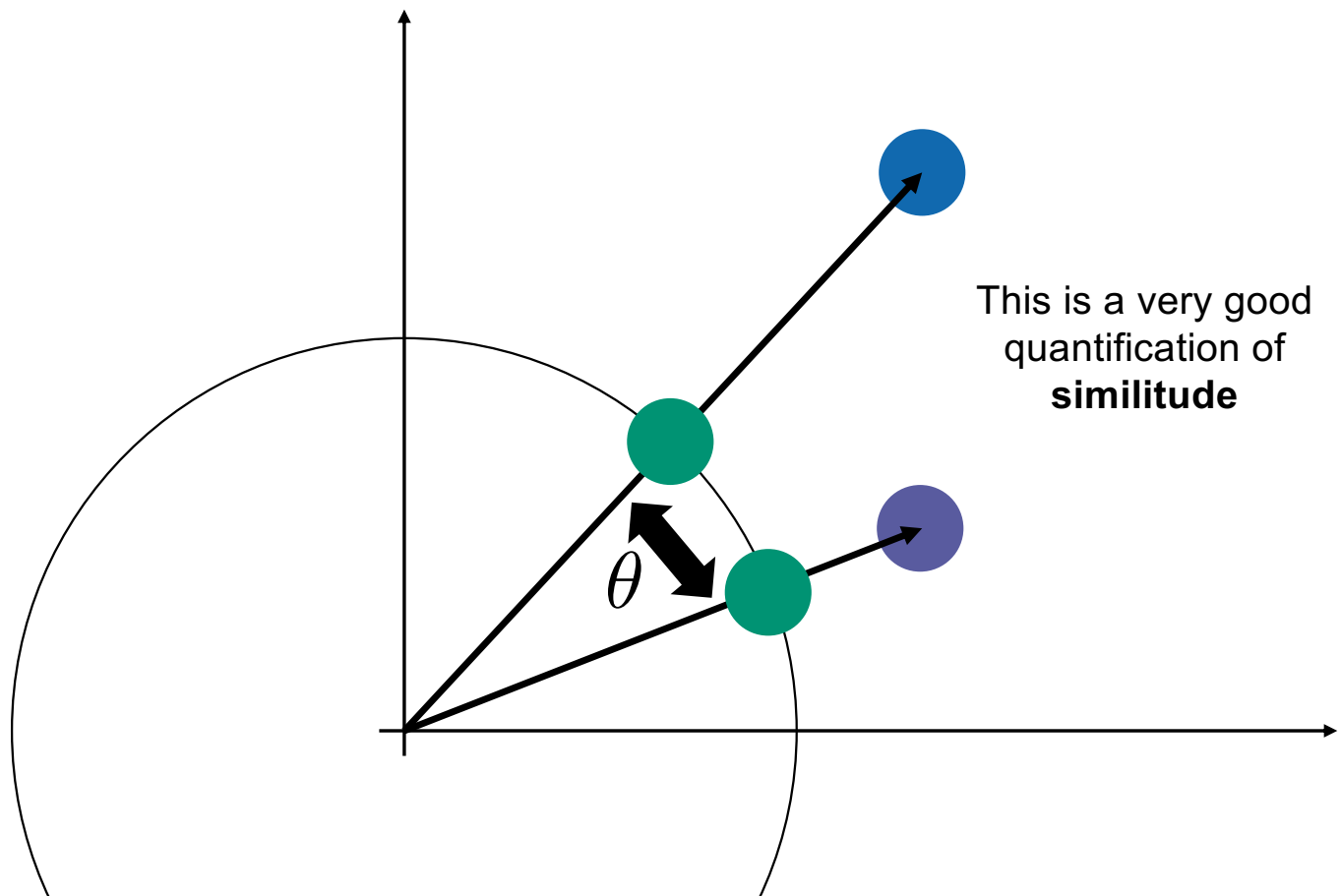
Inner product (renormalized)



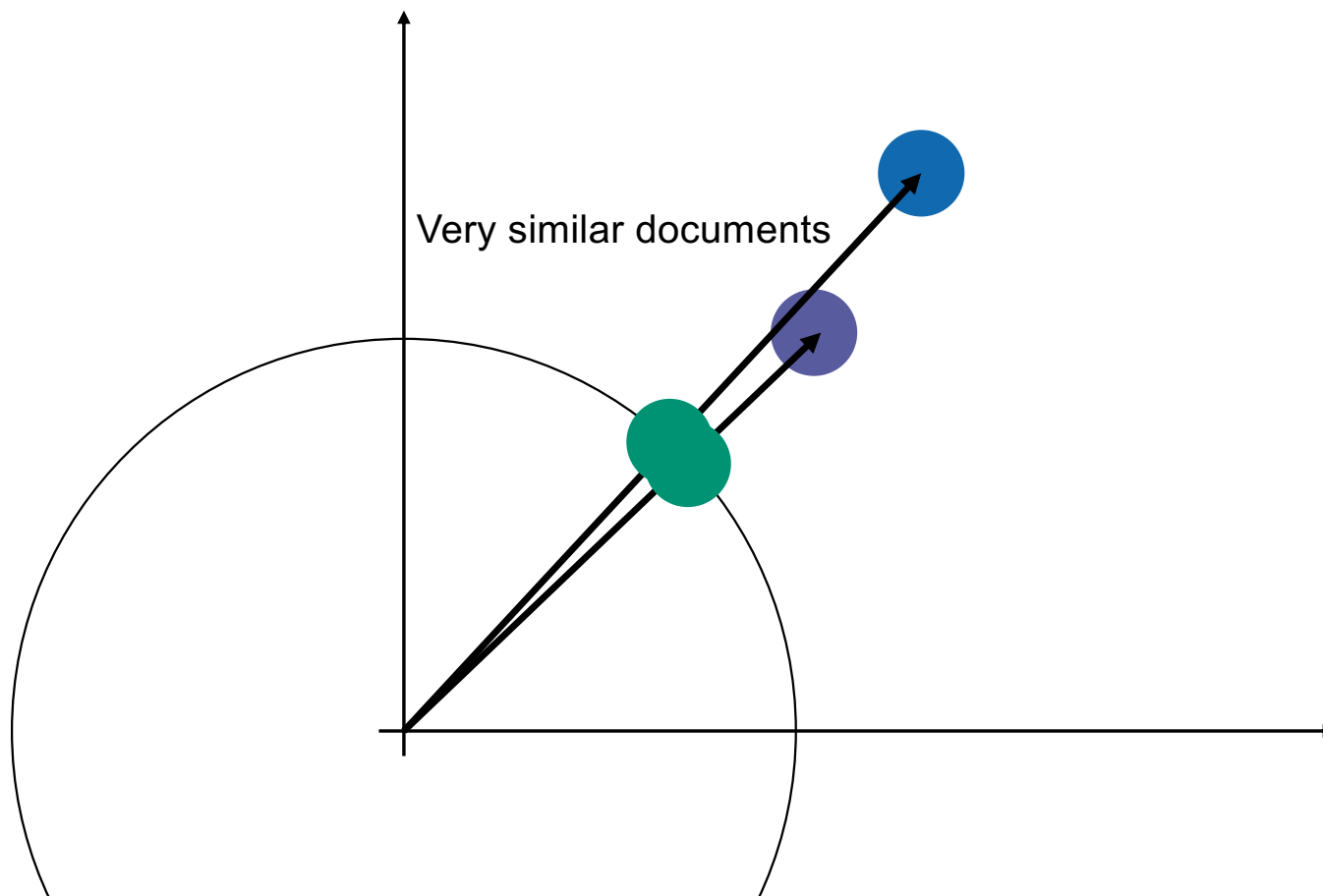
Inner product (renormalized)



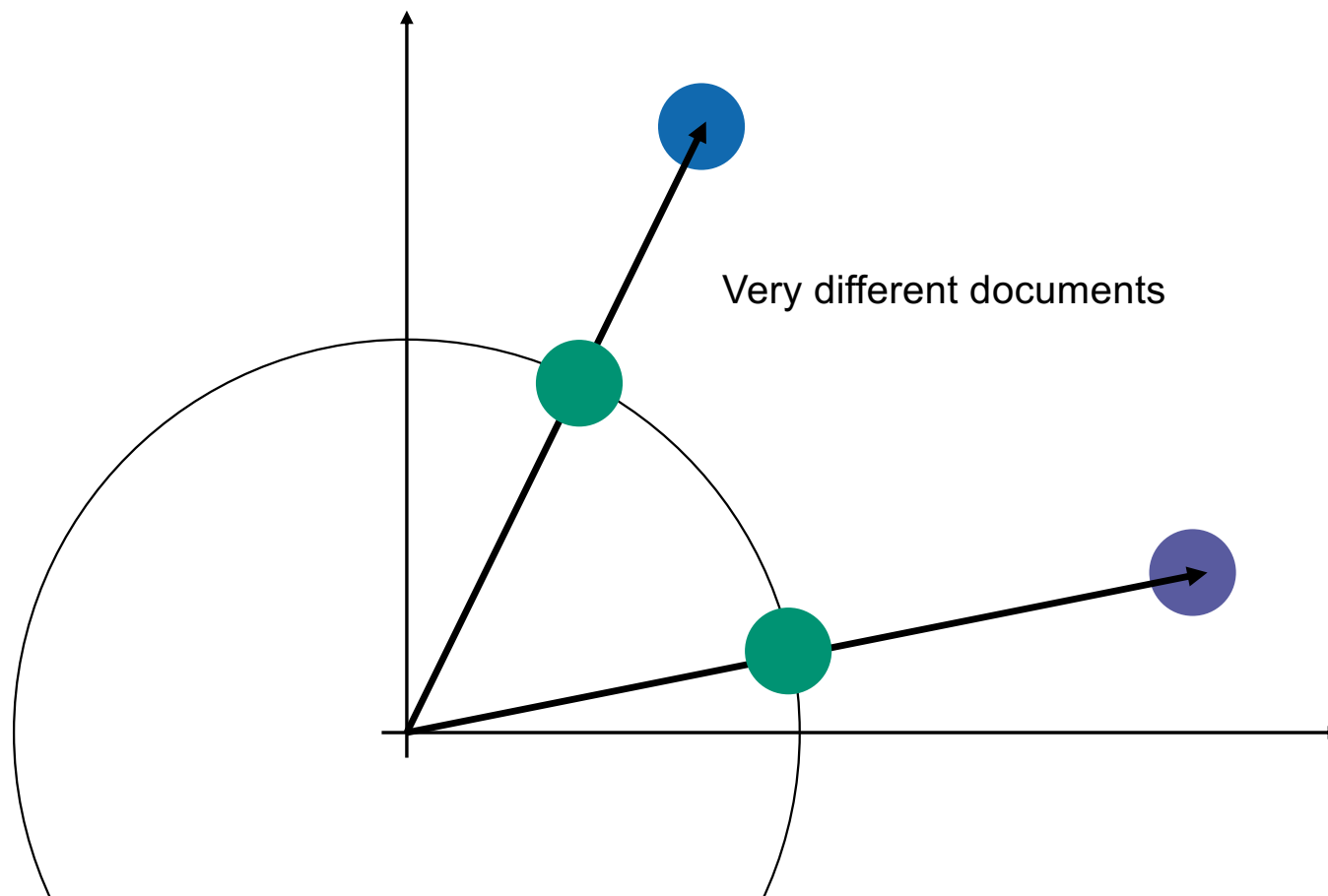
Inner product (renormalized)



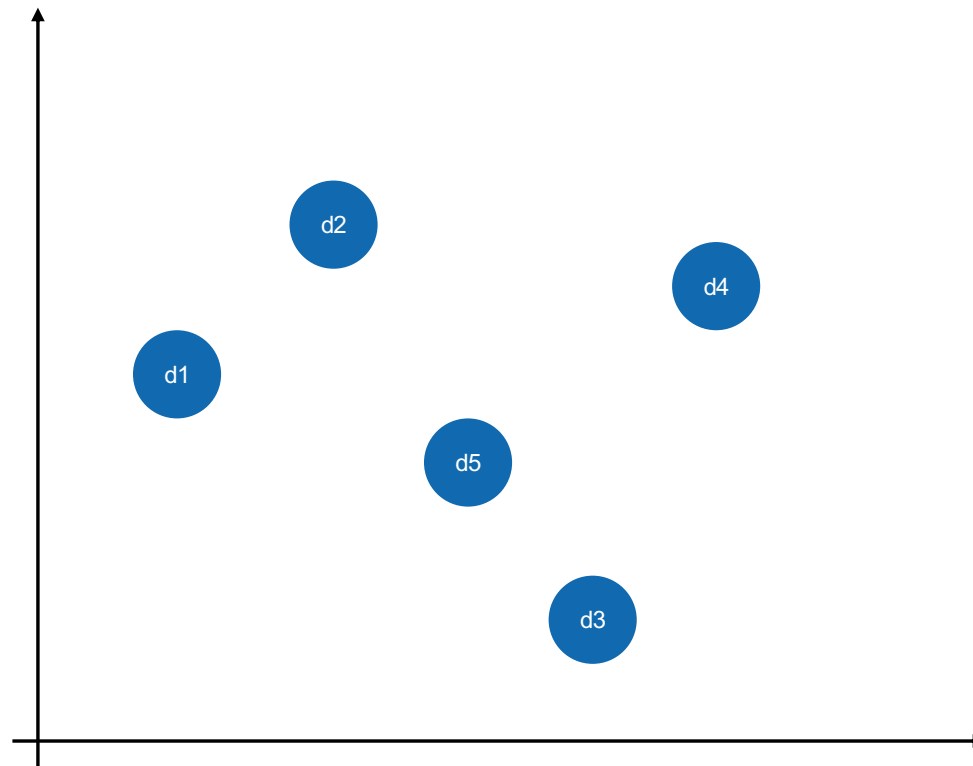
Inner product (renormalized)



Inner product (renormalized)

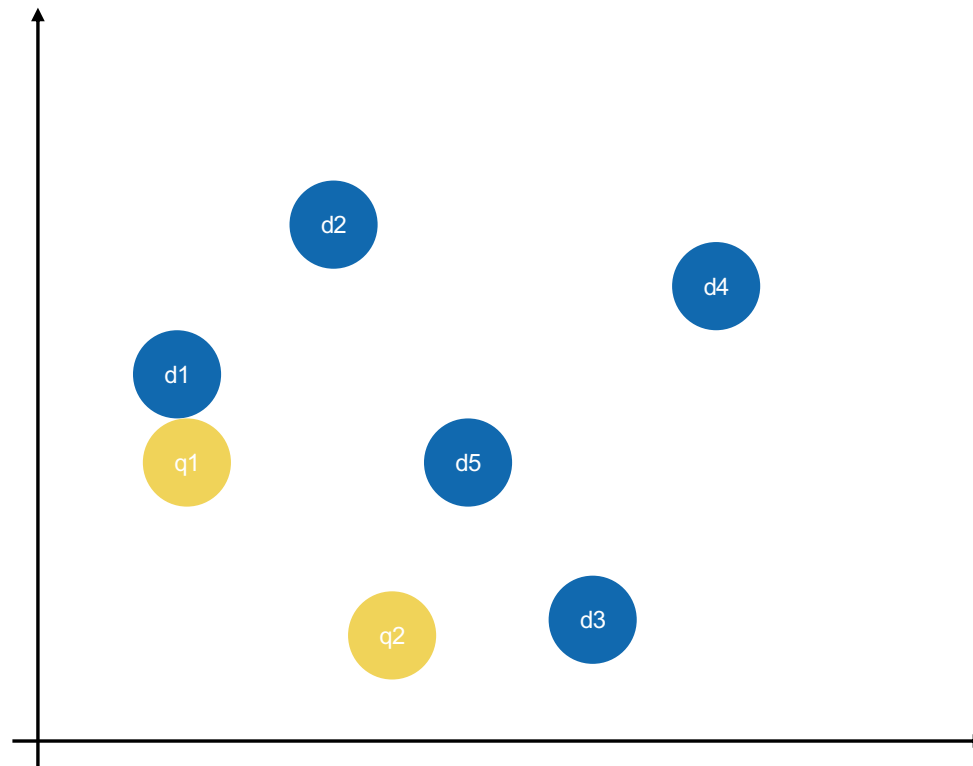


Queries as vectors



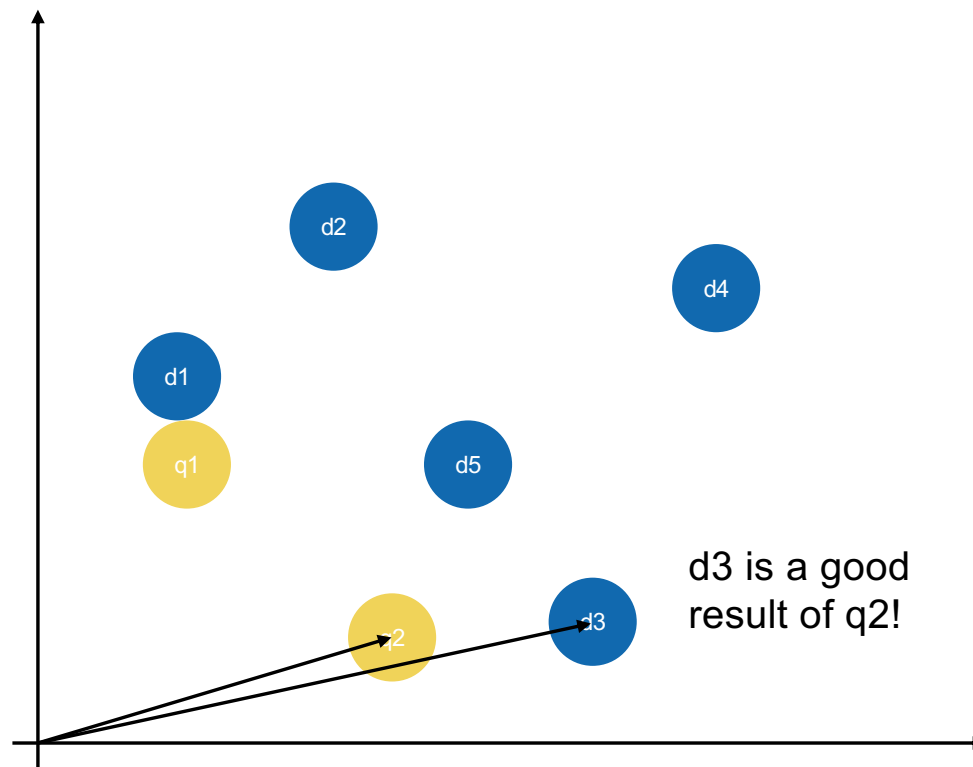
Queries as vectors

Queries
= points in the
first quadrant
of
 $\mathbb{R}^M$



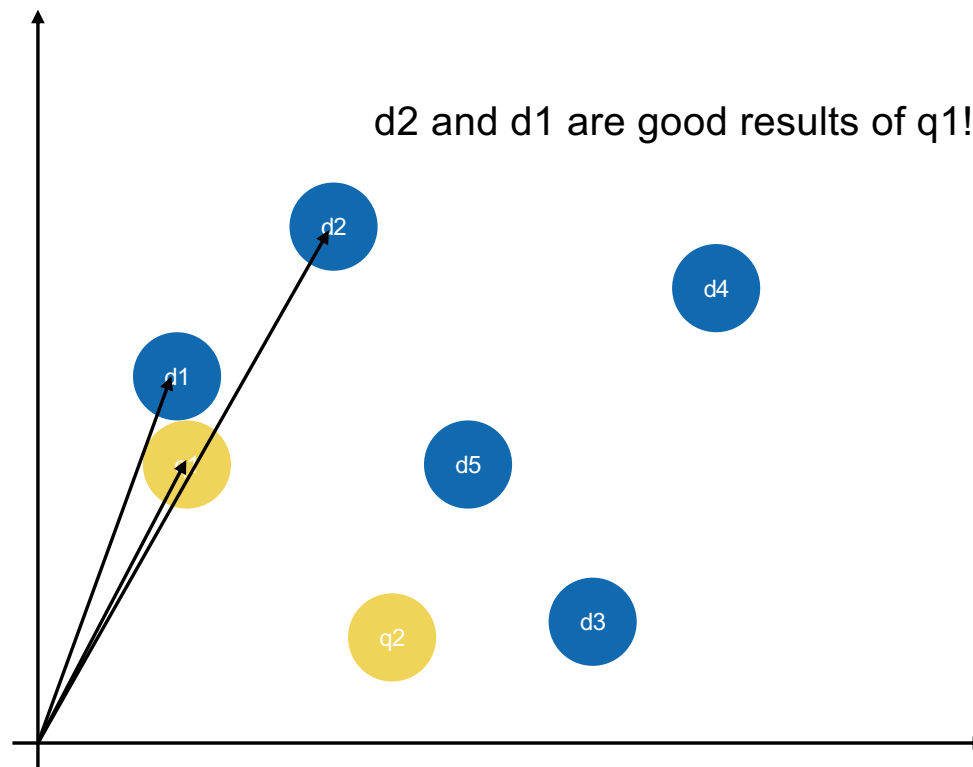
Queries as vectors

Queries
= points in the
first quadrant
of
 $\mathbb{R}^M$

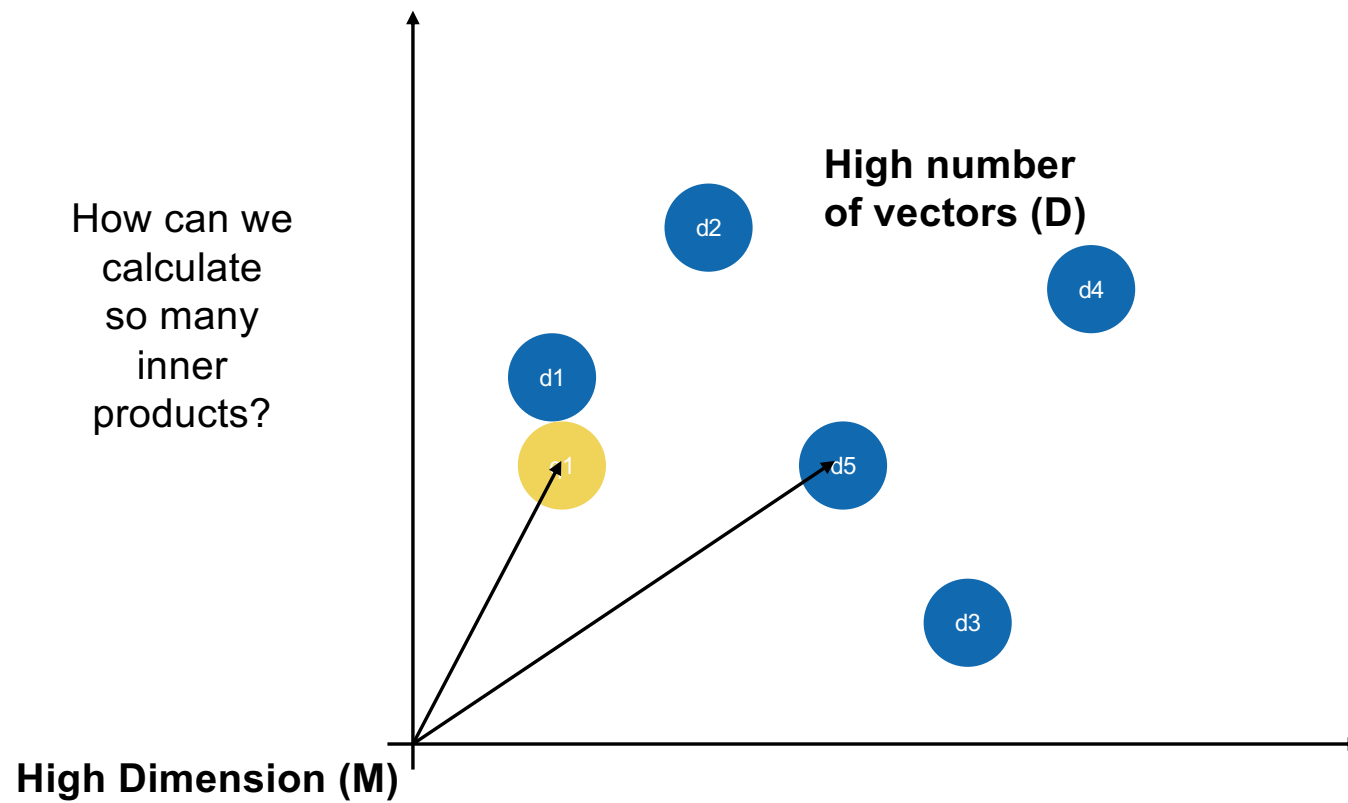


Queries as vectors

Queries
= points in the
first quadrant
of
 $\mathbb{R}^M$



Evaluating a query



Evaluating a query

Evaluating a query

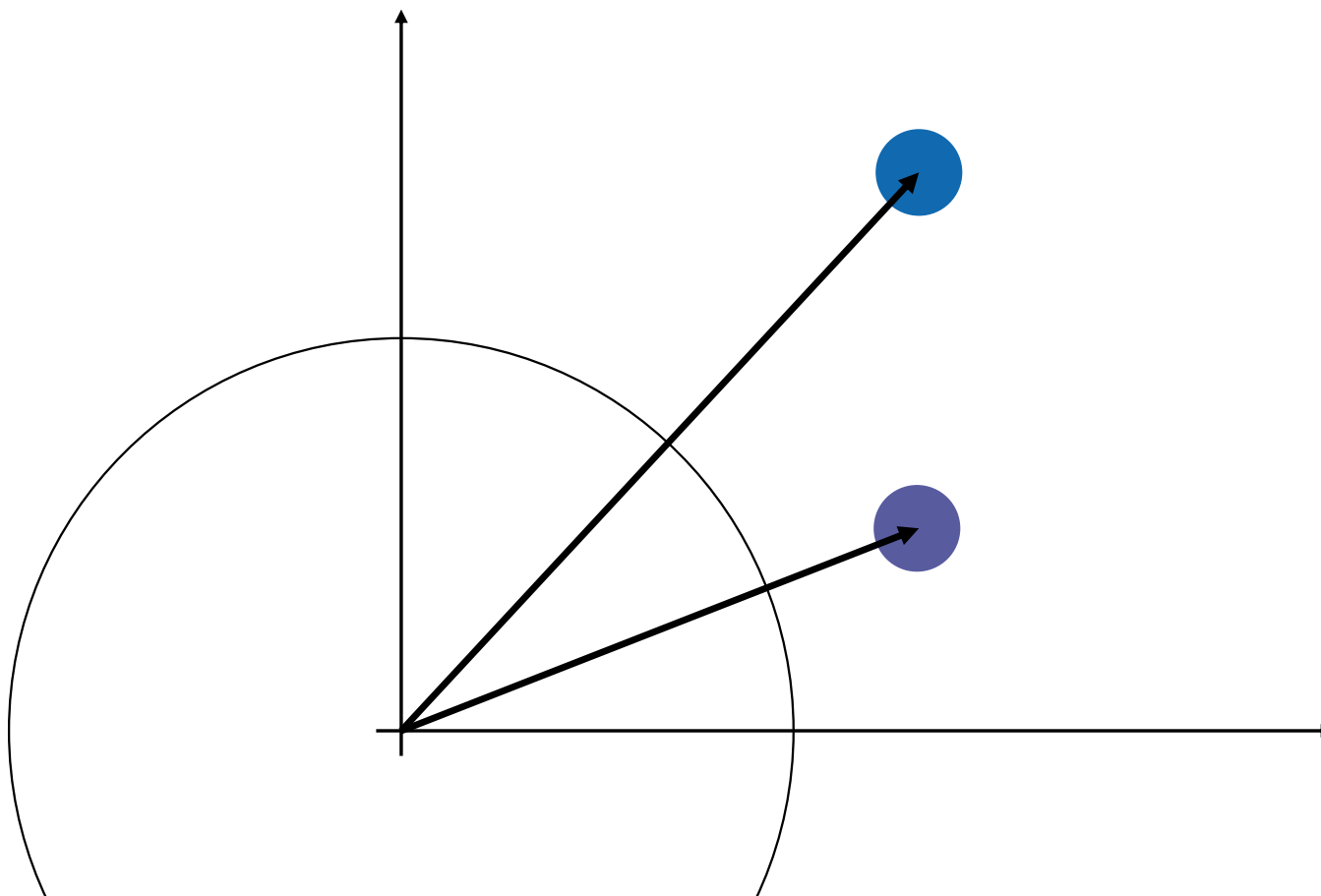
$$\begin{pmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \\ q_7 \\ q_8 \\ \dots \\ q_M \end{pmatrix} \cdot \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \\ d_6 \\ d_7 \\ d_8 \\ \dots \\ d_M \end{pmatrix}$$

Evaluating a query

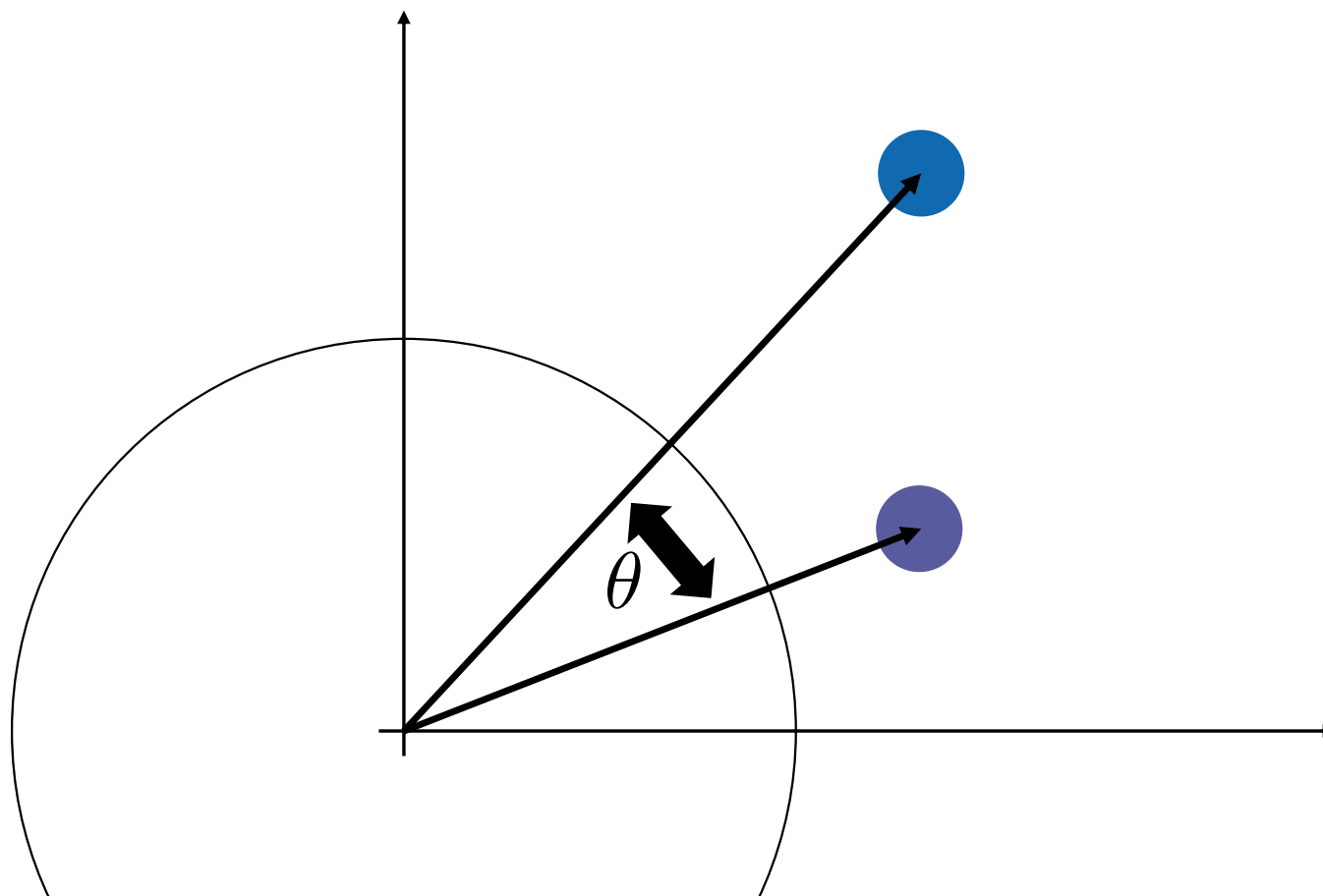
$$\begin{pmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ 0 \\ 0 \\ 0 \\ 0 \\ \dots \\ 0 \end{pmatrix} \cdot \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \\ d_6 \\ d_7 \\ d_8 \\ \dots \\ d_M \end{pmatrix}$$

The dimensionality is not that big!
(And constant for a given query)

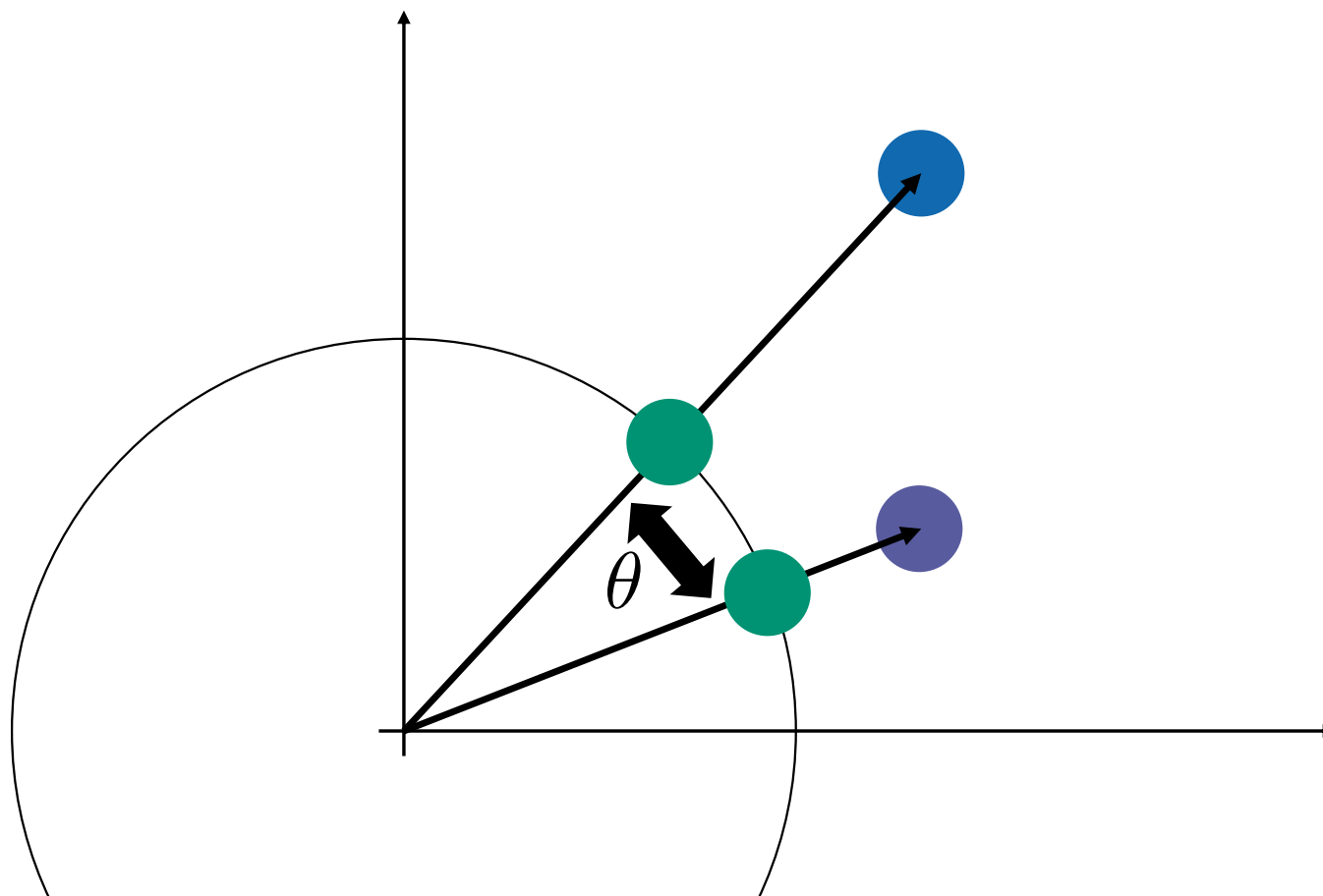
Inner product (renormalized)



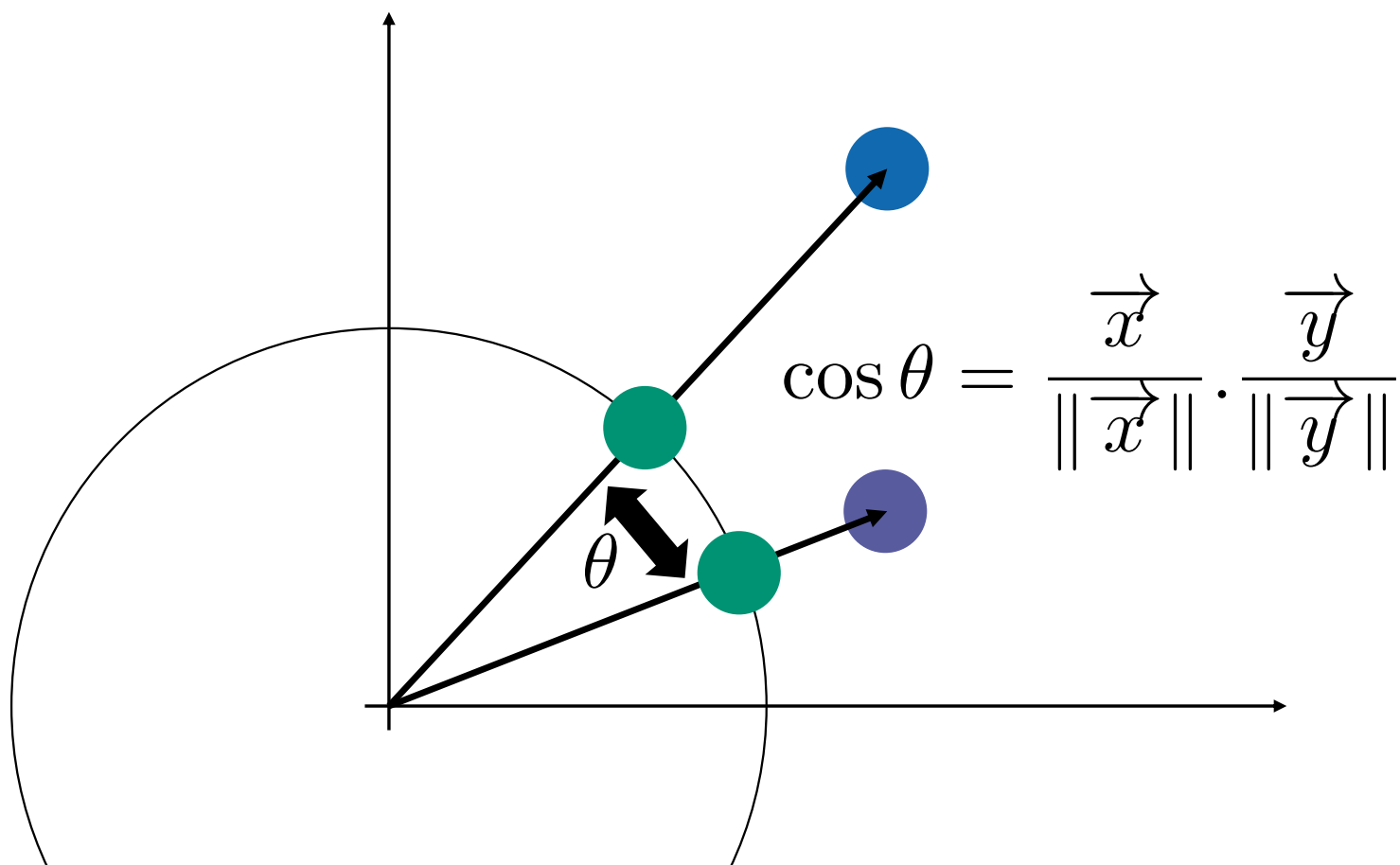
Inner product (renormalized)



Inner product (renormalized)

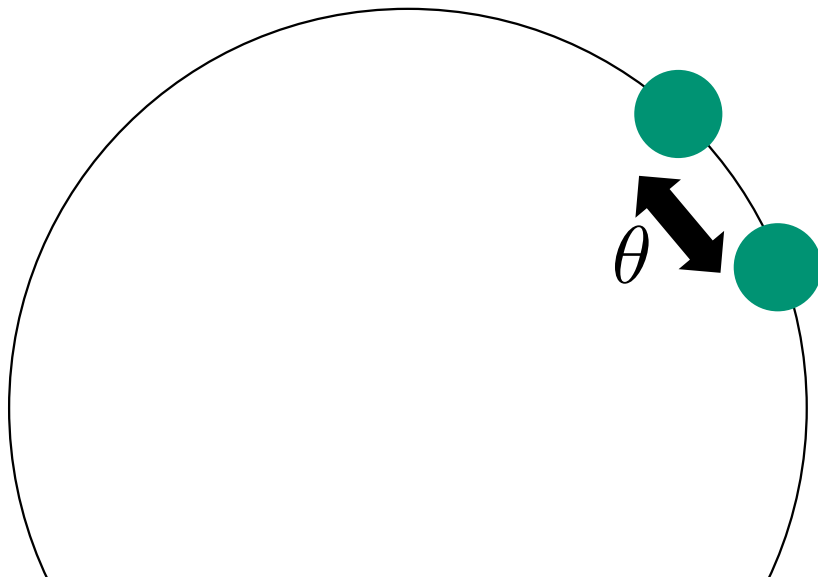


Inner product (renormalized)



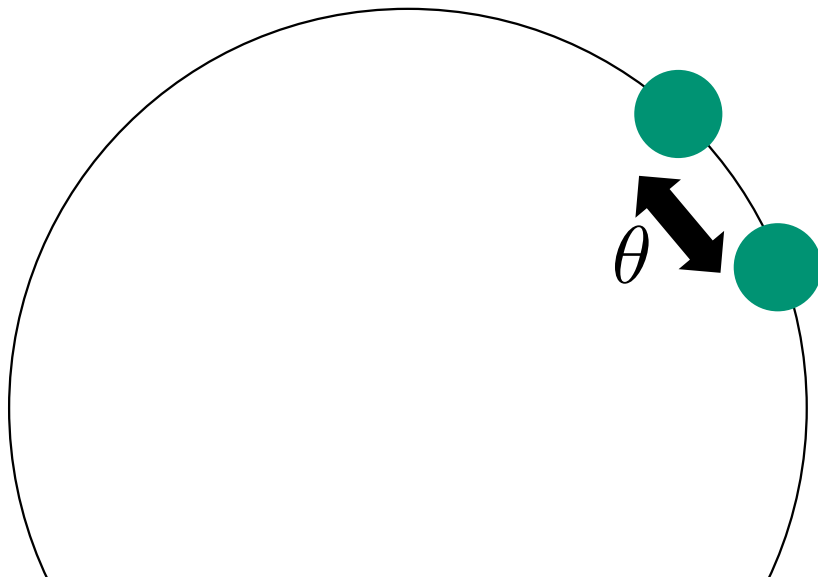
Inner product (renormalized)

$$\text{sim}(\vec{x}, \vec{y}) = \sum_i \frac{x_i \times y_i}{\|\vec{x}\| \times \|\vec{y}\|}$$



Inner product (renormalized)

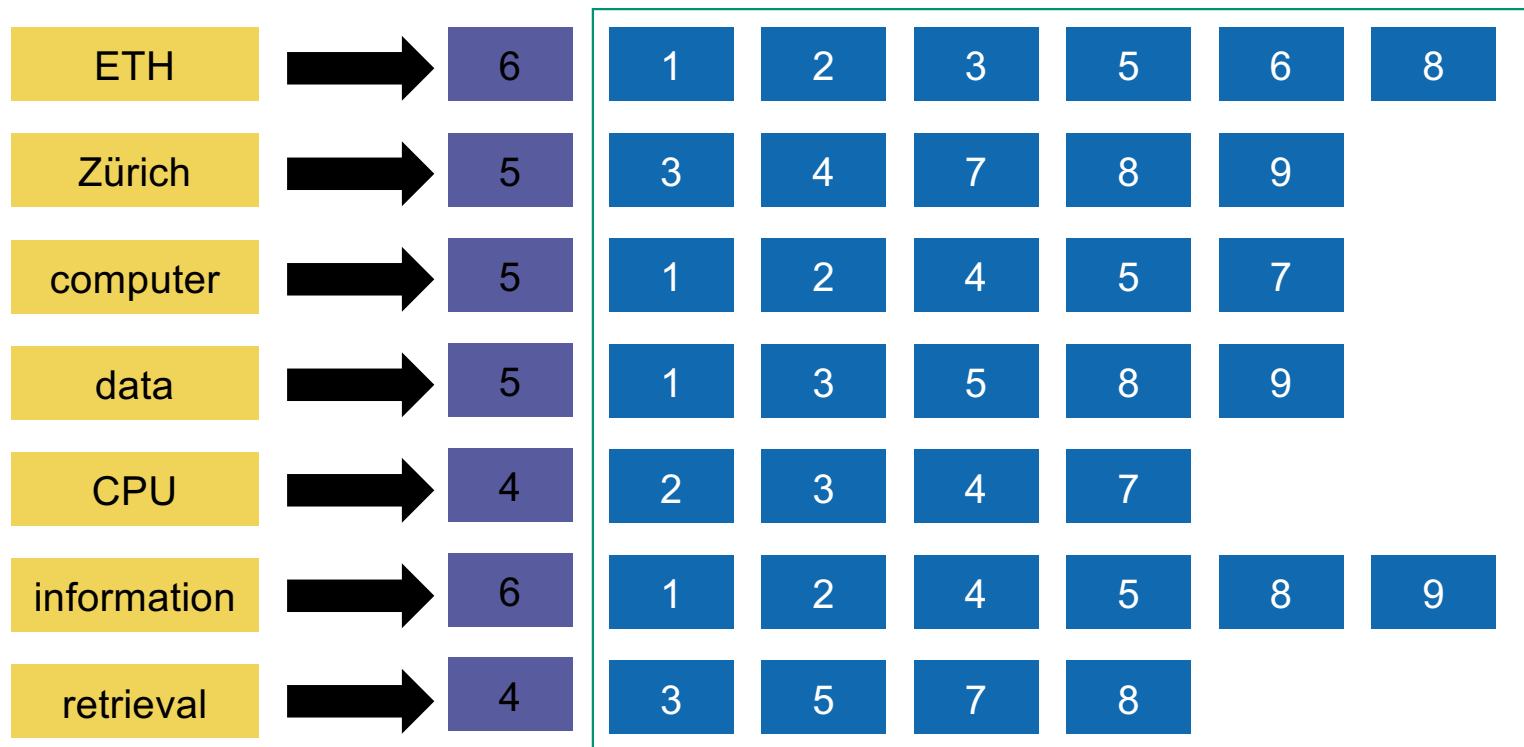
$$\text{sim}(\vec{d}, \vec{q}) = \sum_t \frac{\text{tf-idf}_{td} \times \text{tf-idf}_{tq}}{\|\vec{q}\| \times \|\vec{d}\|}$$



Inverted index adapted to ranked retrieval

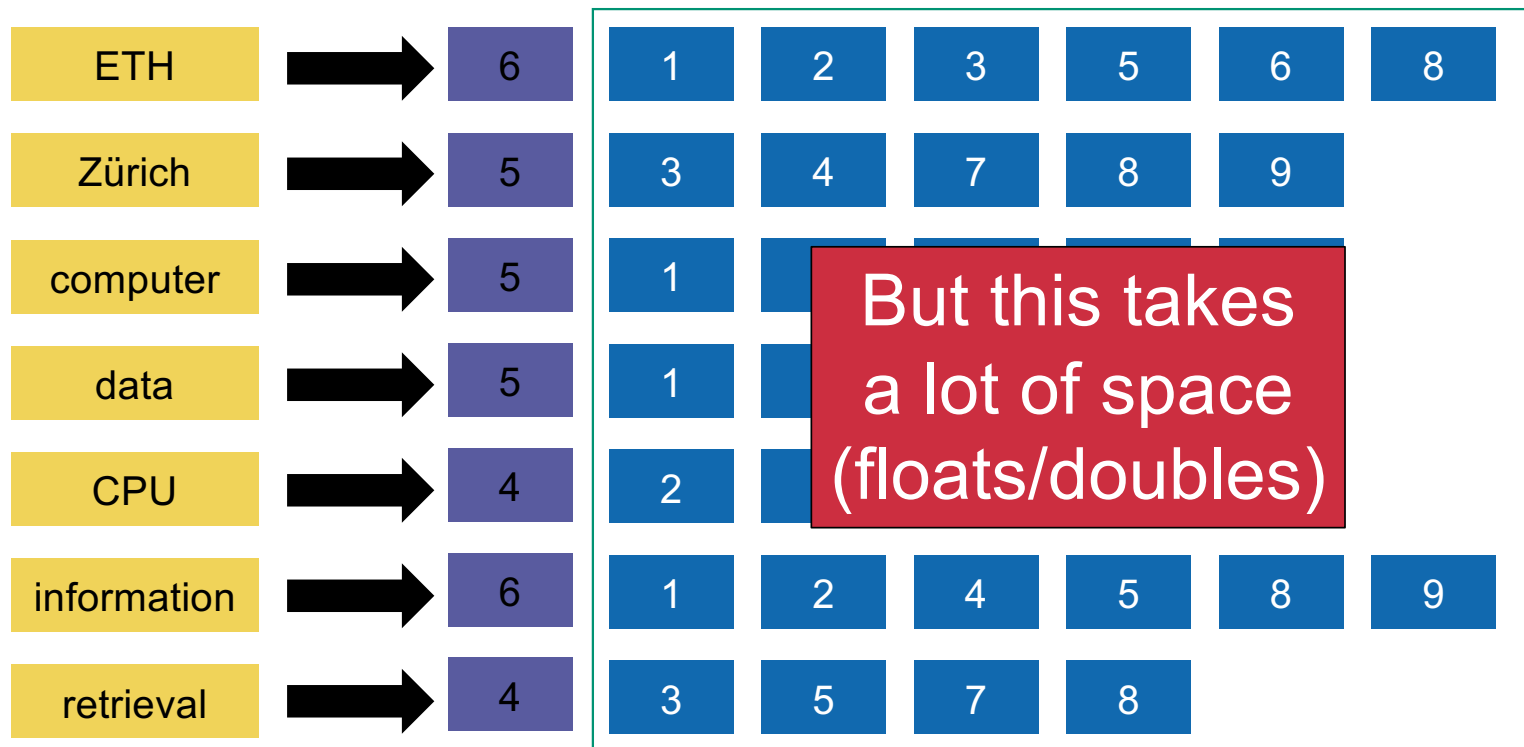
ETH	➡	6	1	2	3	5	6	8
Zürich	➡	5	3	4	7	8	9	
computer	➡	5	1	2	4	5	7	
data	➡	5	1	3	5	8	9	
CPU	➡	4	2	3	4	7		
information	➡	6	1	2	4	5	8	9
retrieval	➡	4	3	5	7	8		

Inverted index adapted to ranked retrieval



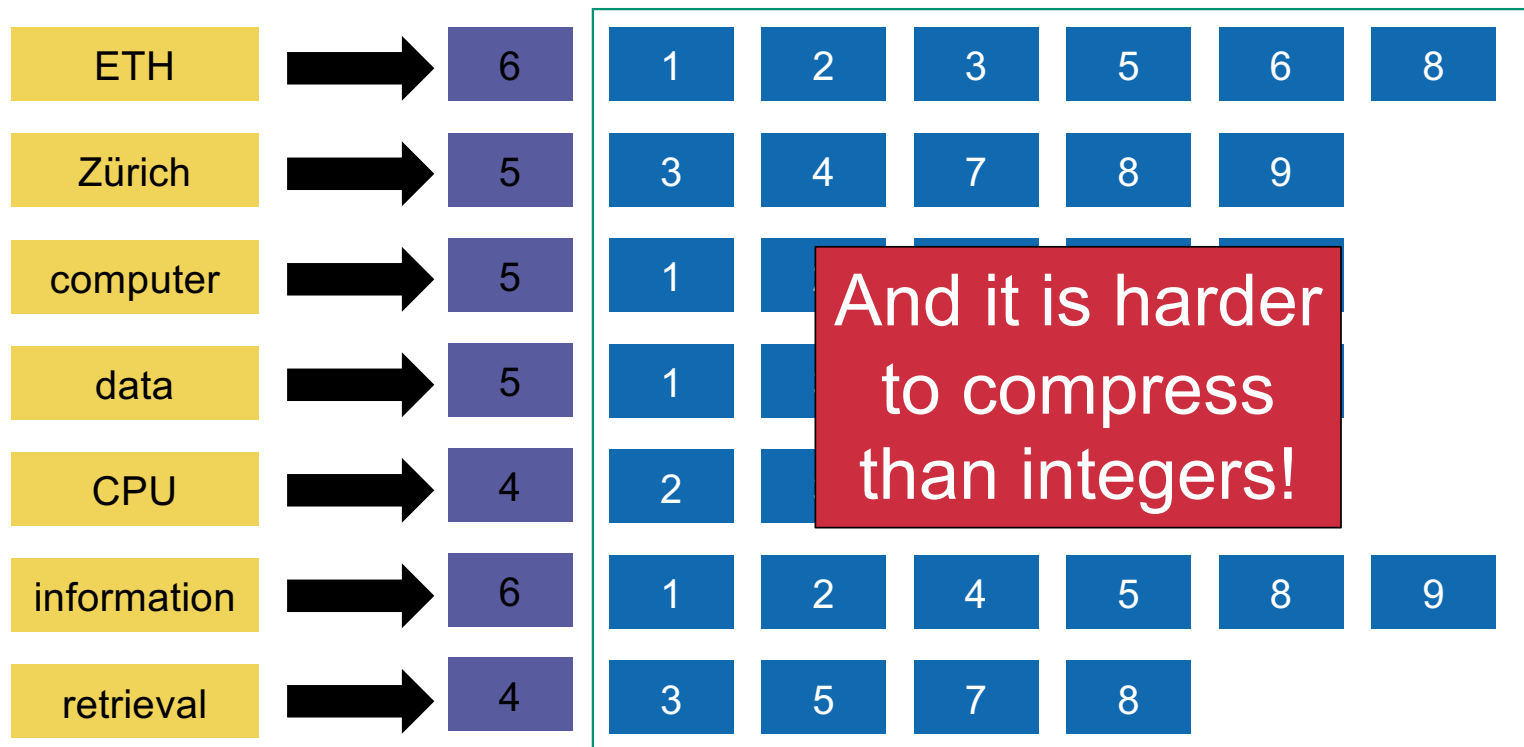
We could store the precomputed $\text{tf-idf}_{\text{td}}$ here

Inverted index adapted to ranked retrieval



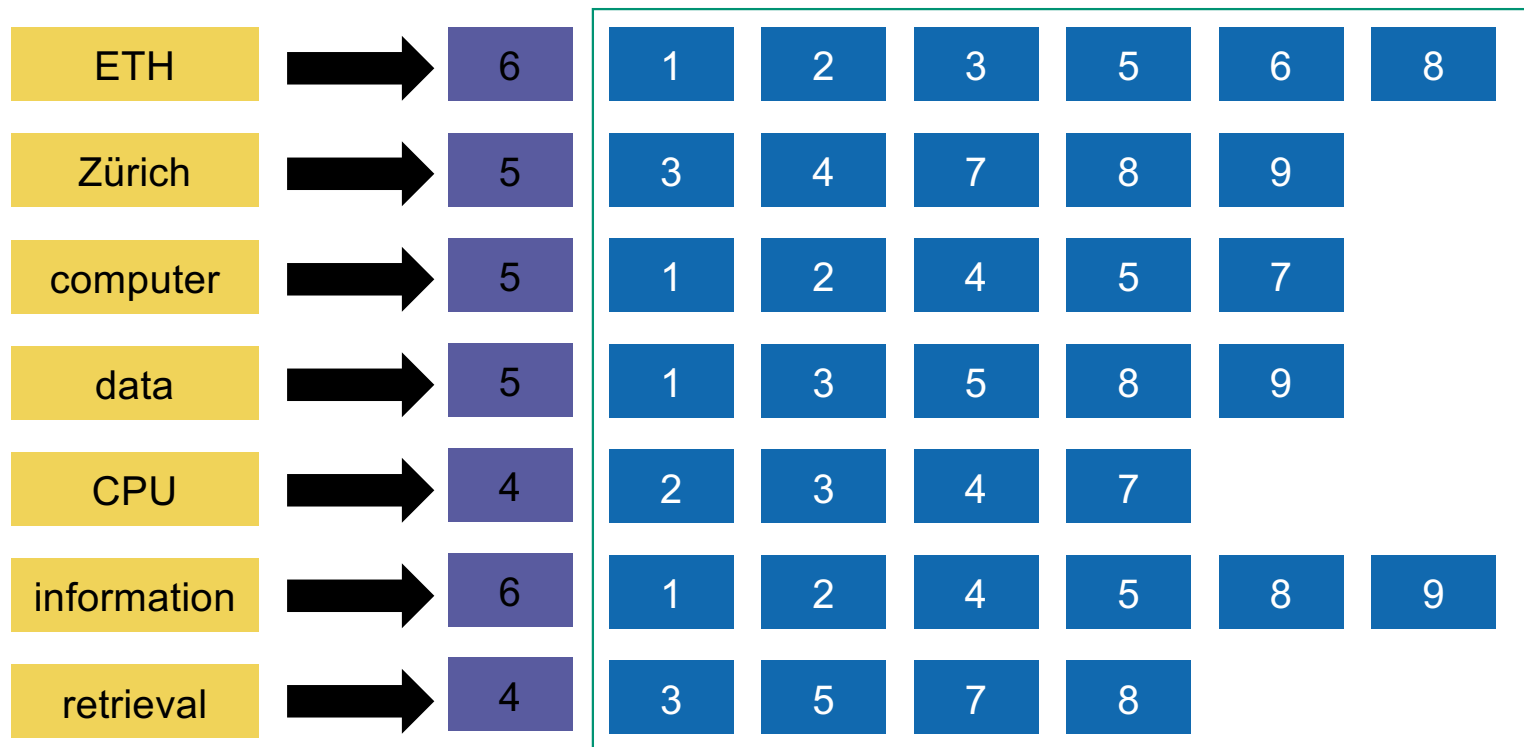
We could store the precomputed $\text{tf-idf}_{\text{td}}$ here

Inverted index adapted to ranked retrieval



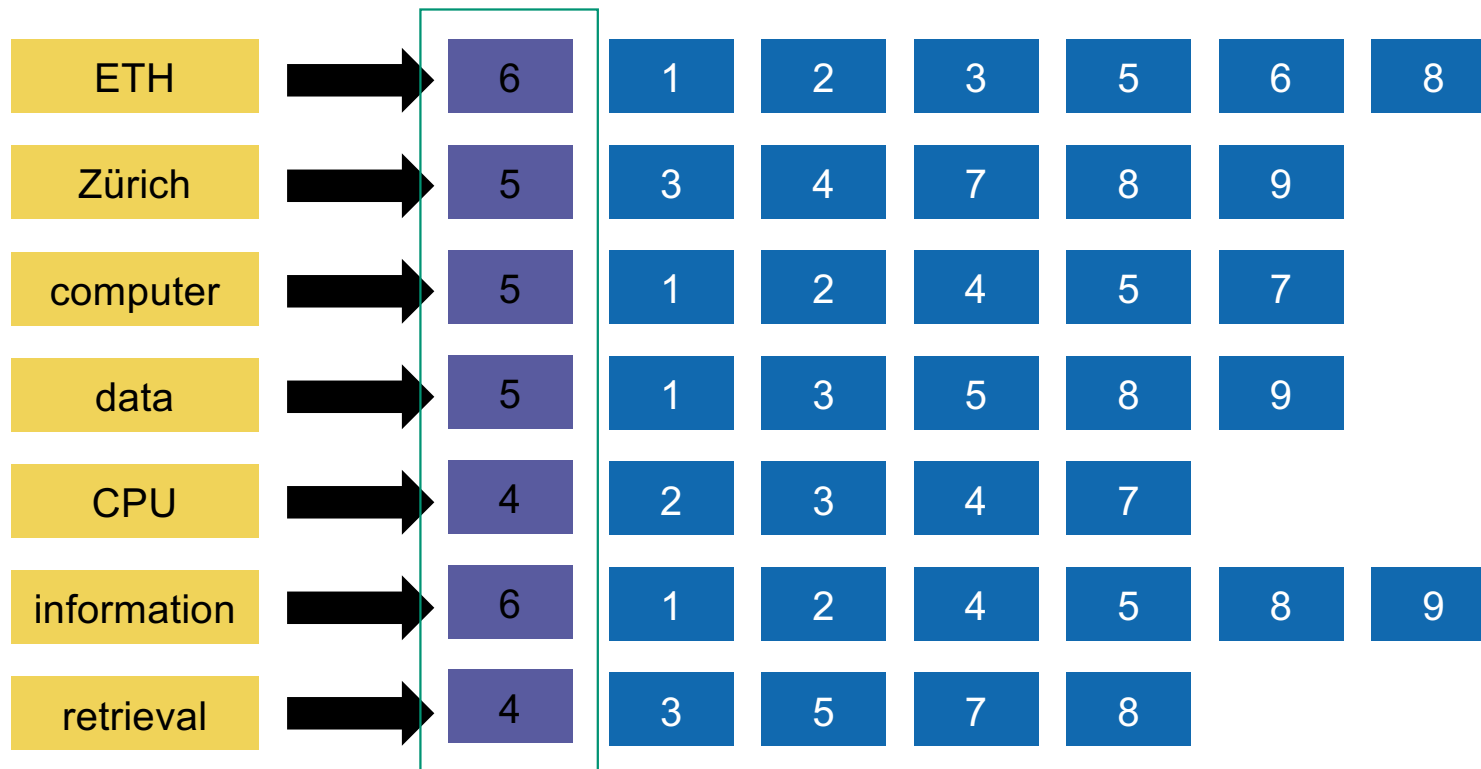
We could store the precomputed $\text{tf-idf}_{\text{td}}$ here

Inverted index adapted to ranked retrieval



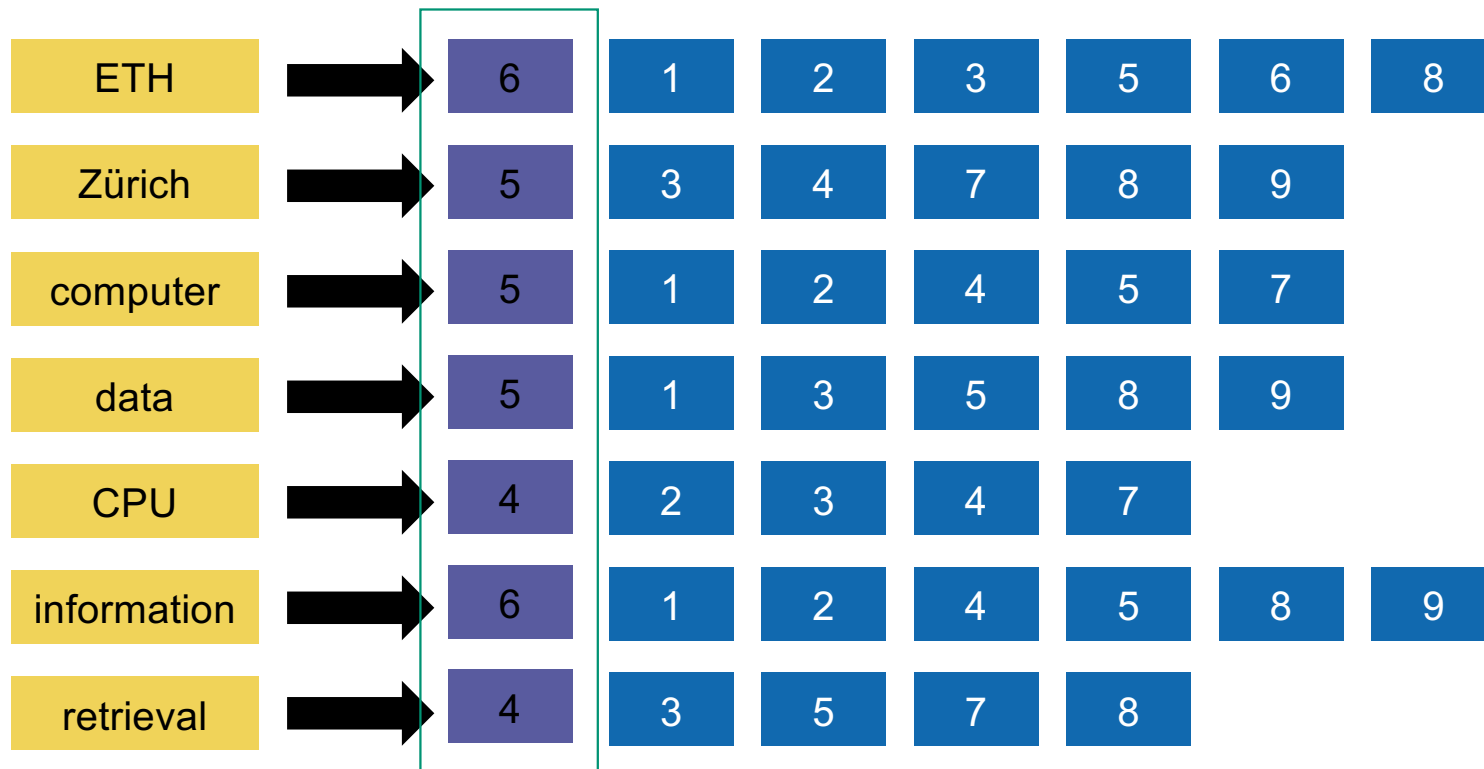
We can store the precomputed tf_{td} here

Inverted index adapted to ranked retrieval



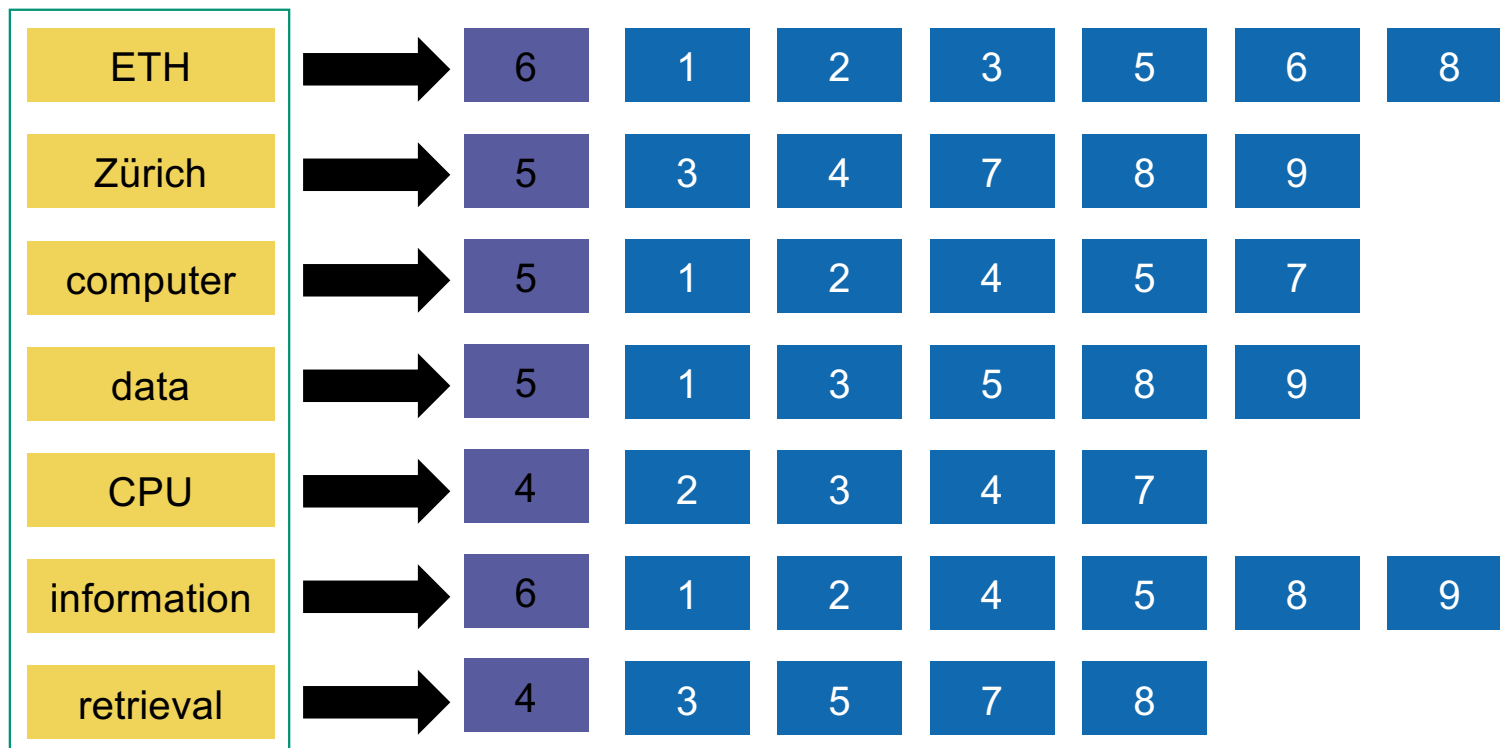
We can store the pre-computed idf_t here as well as document vector norm

Inverted index adapted to ranked retrieval



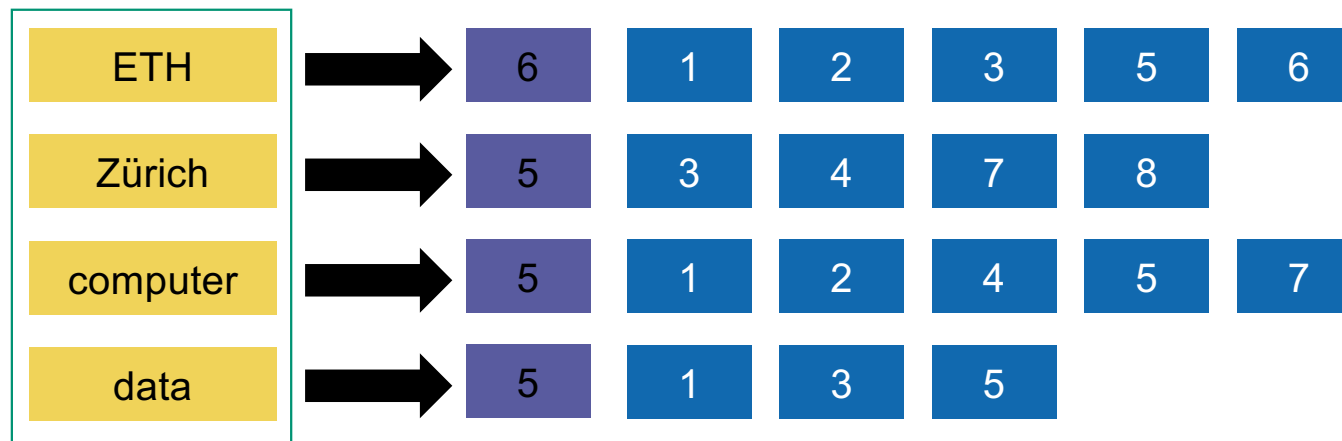
We can store the pre-computed N/df_t here as well as document vector norm

Inverted index adapted to ranked retrieval

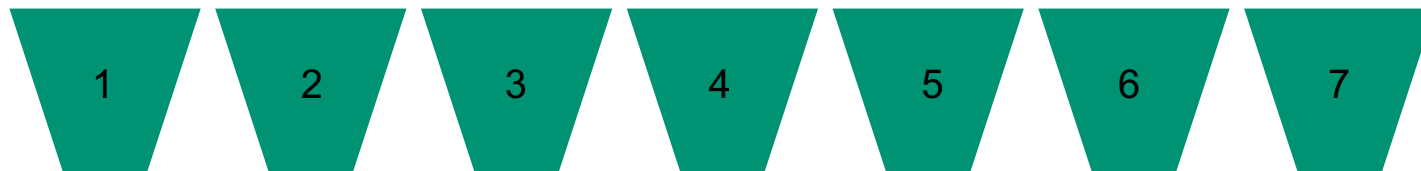


We can compute the tf_{tq} for the query at hand

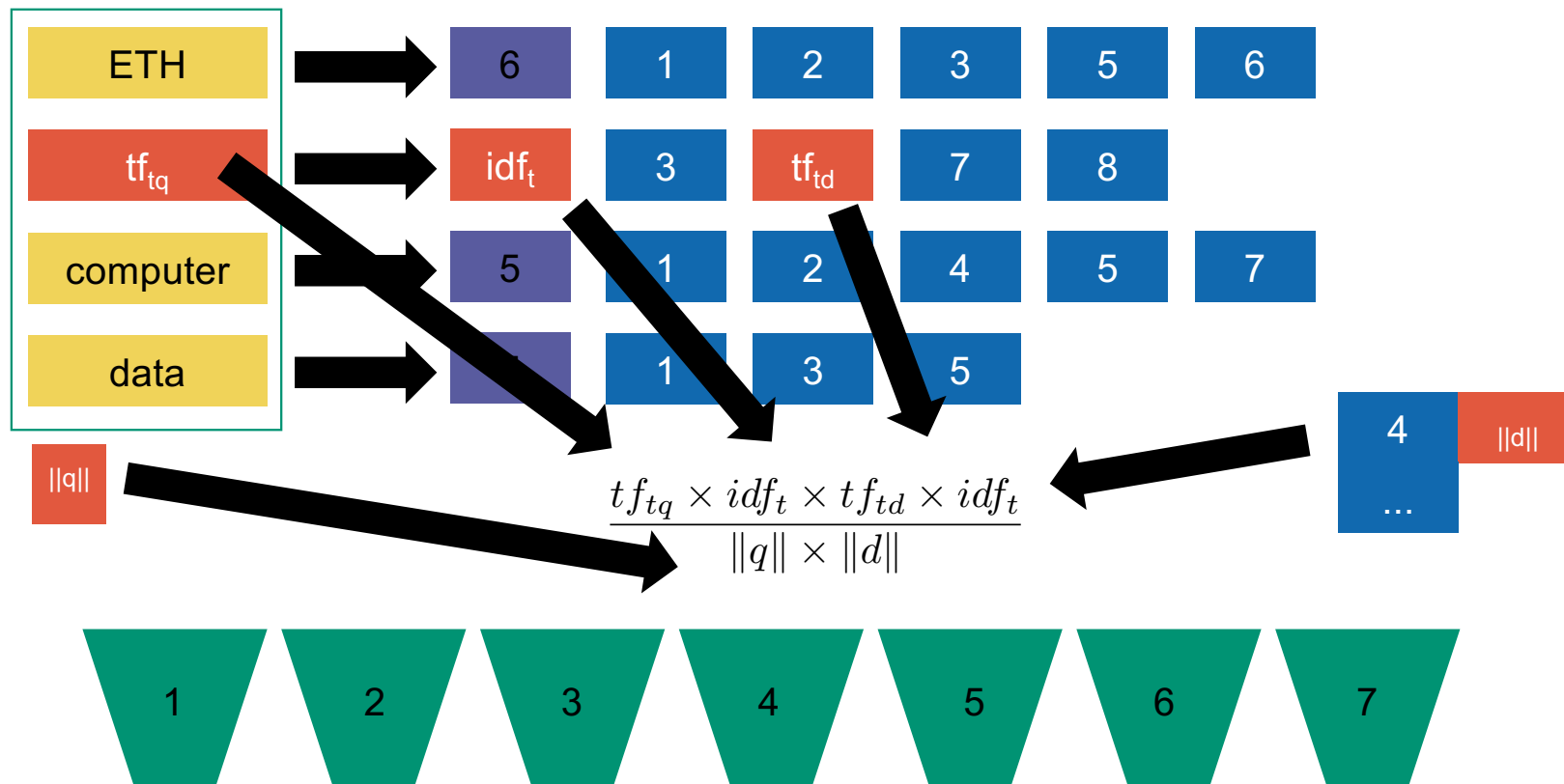
Inverted index adapted to ranked retrieval



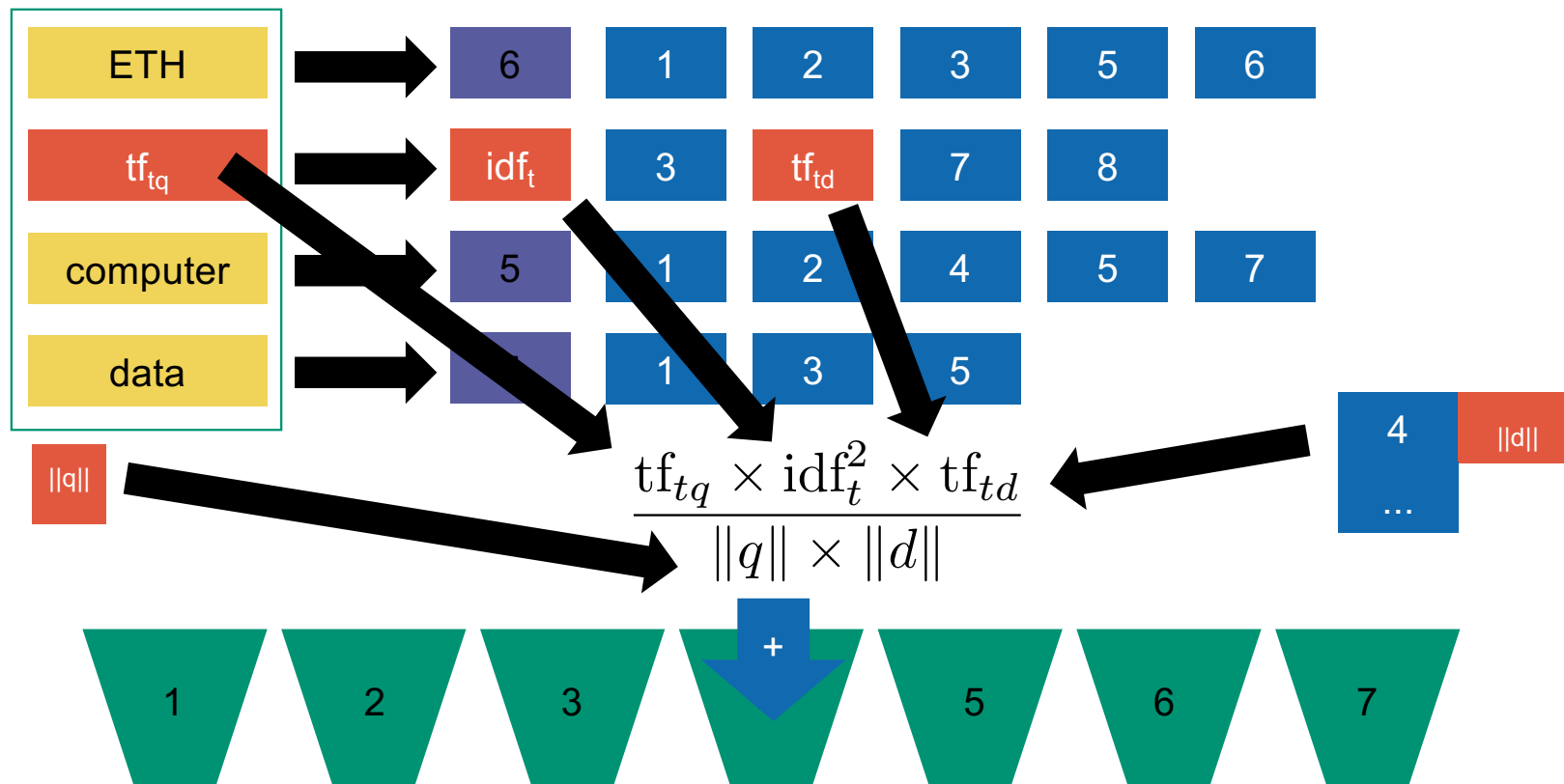
We use one accumulator for each document



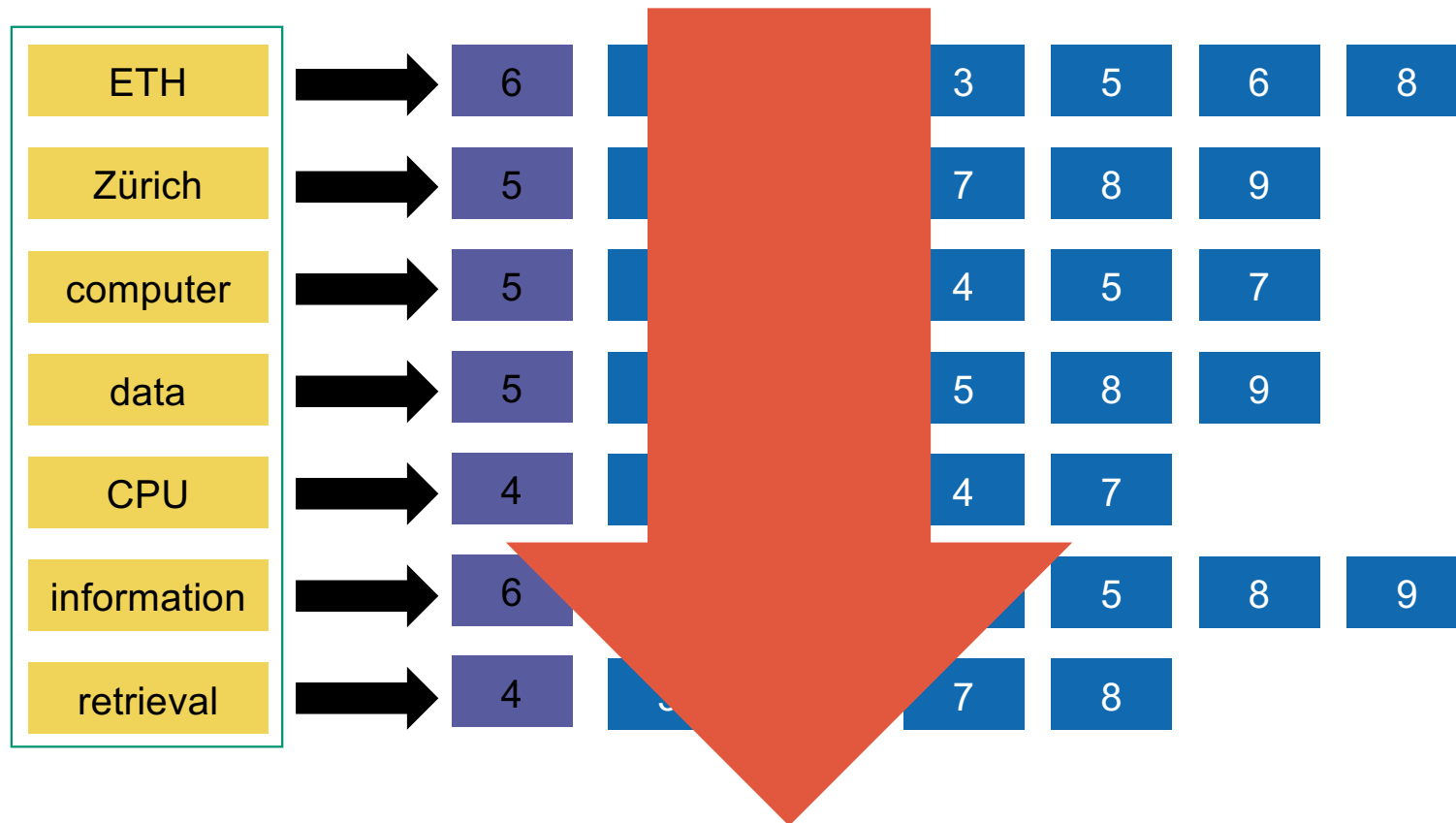
Inverted index adapted to ranked retrieval



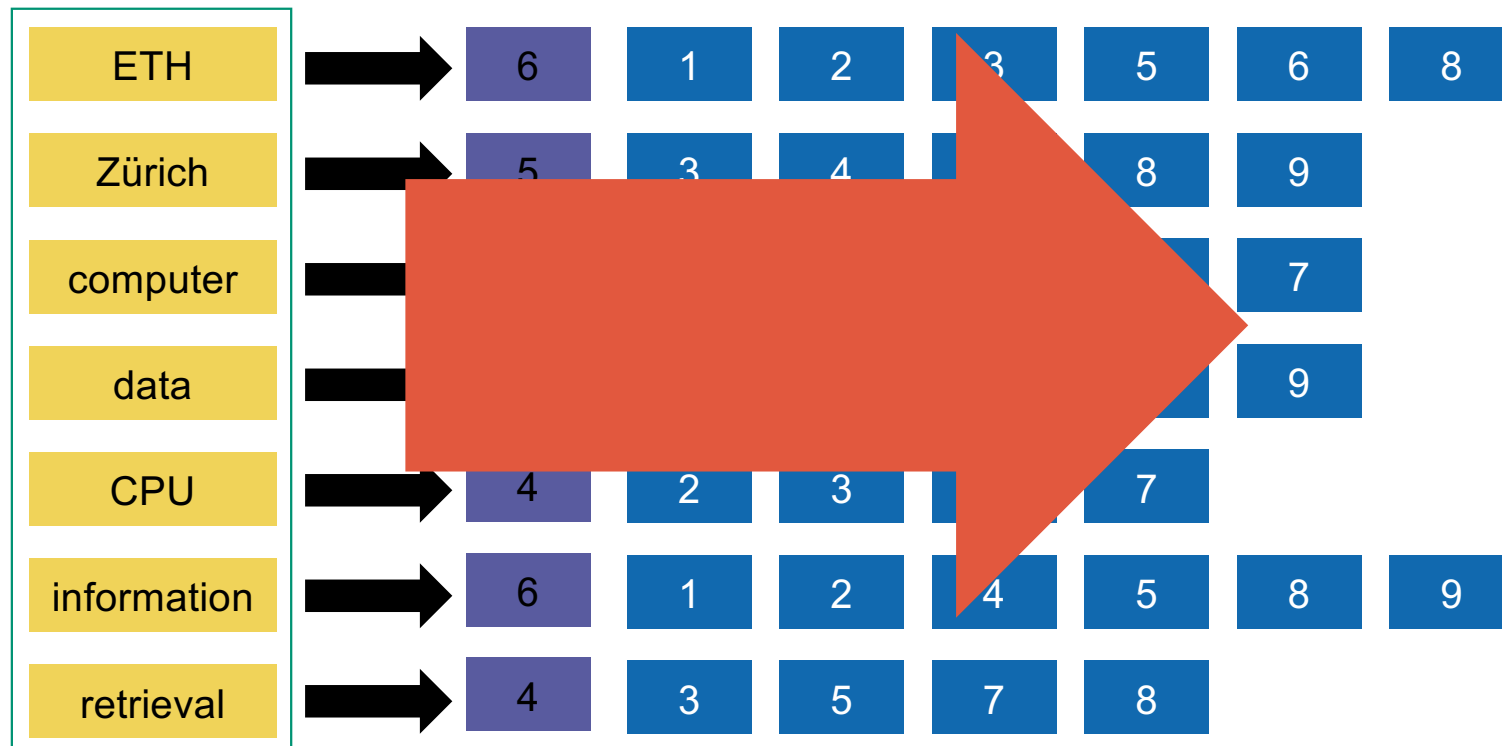
Inverted index adapted to ranked retrieval



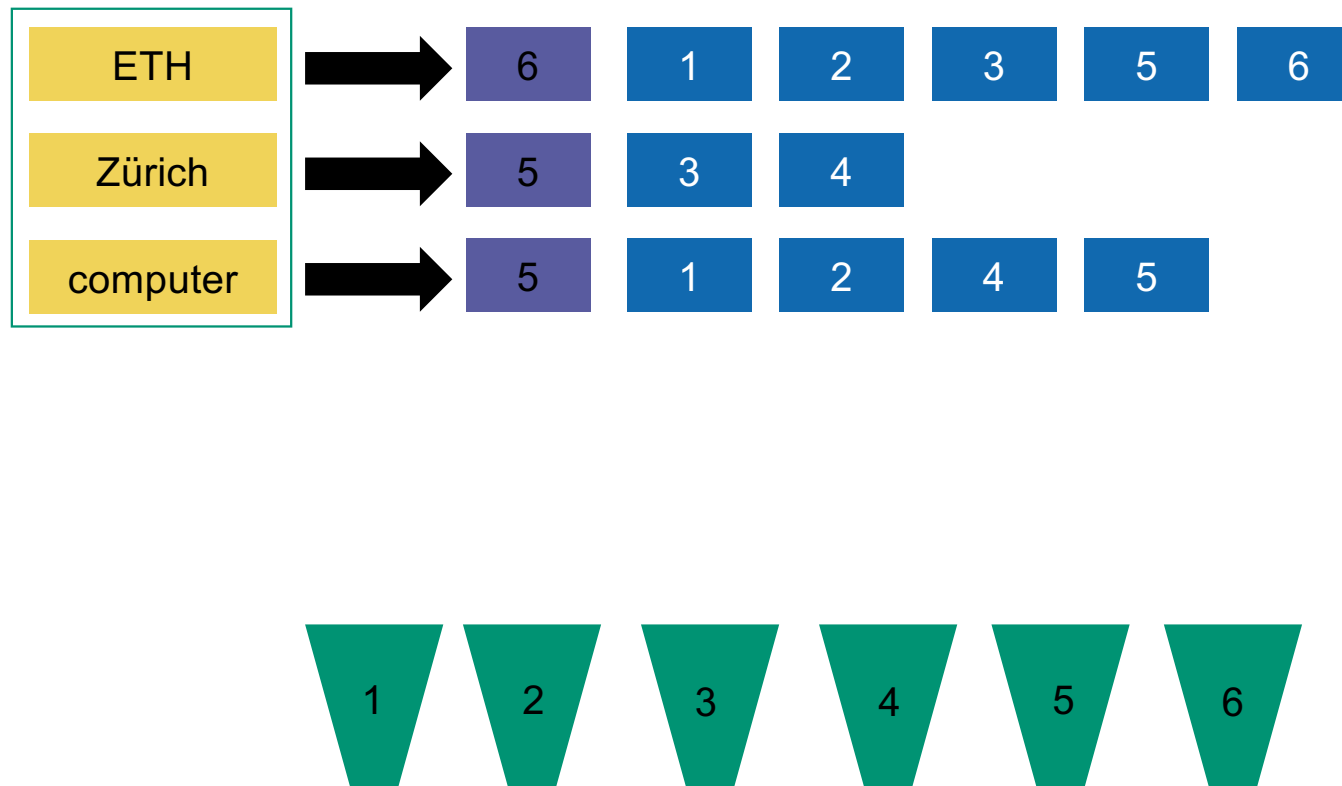
Term-at-a-time



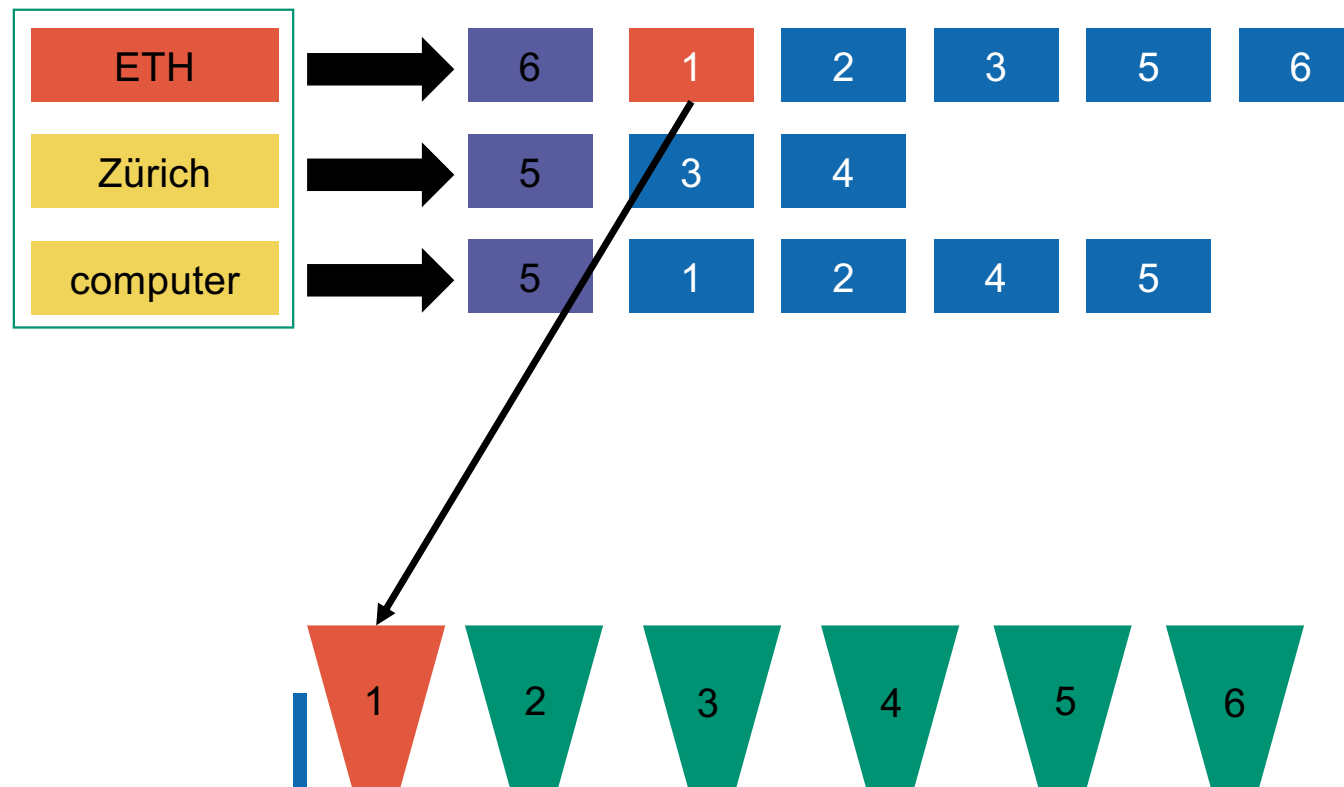
Document-at-a-time



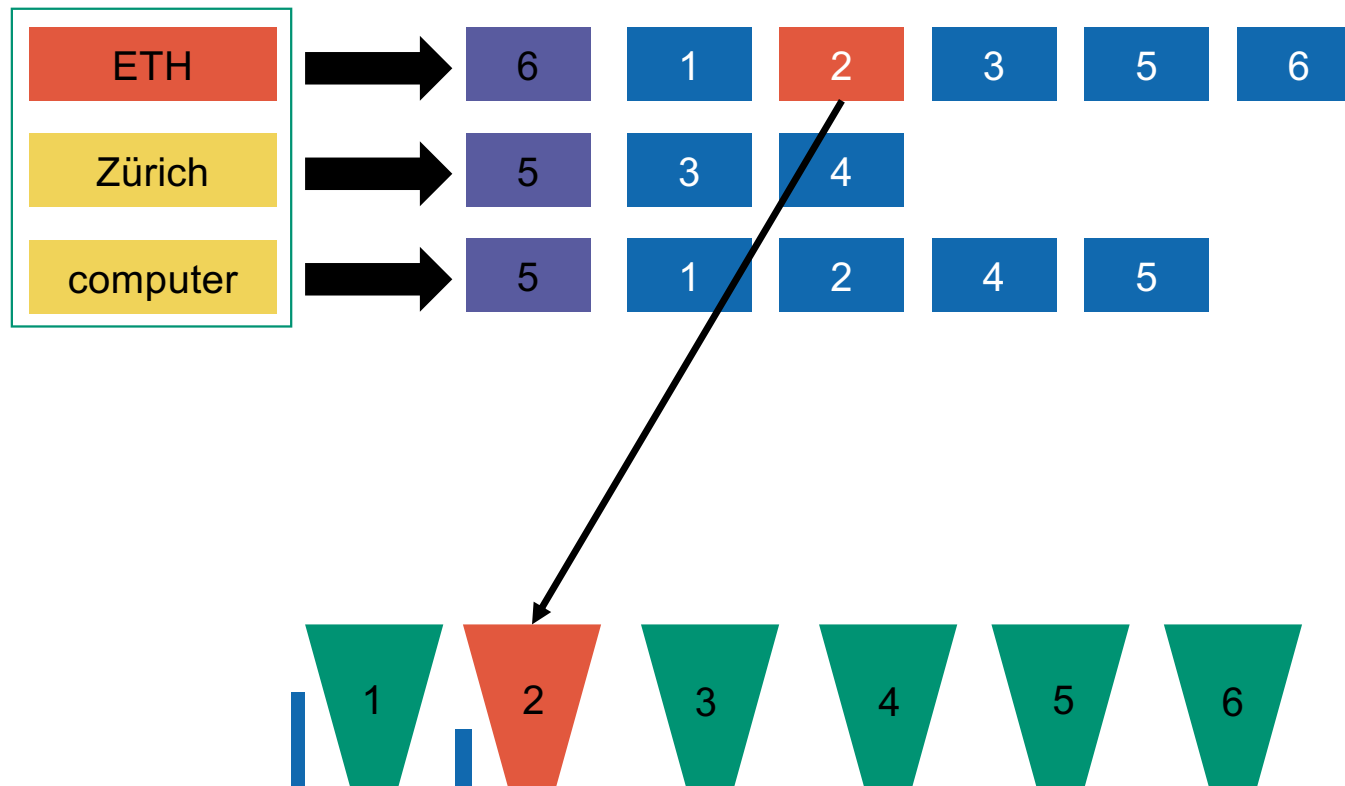
Term-at-a-time



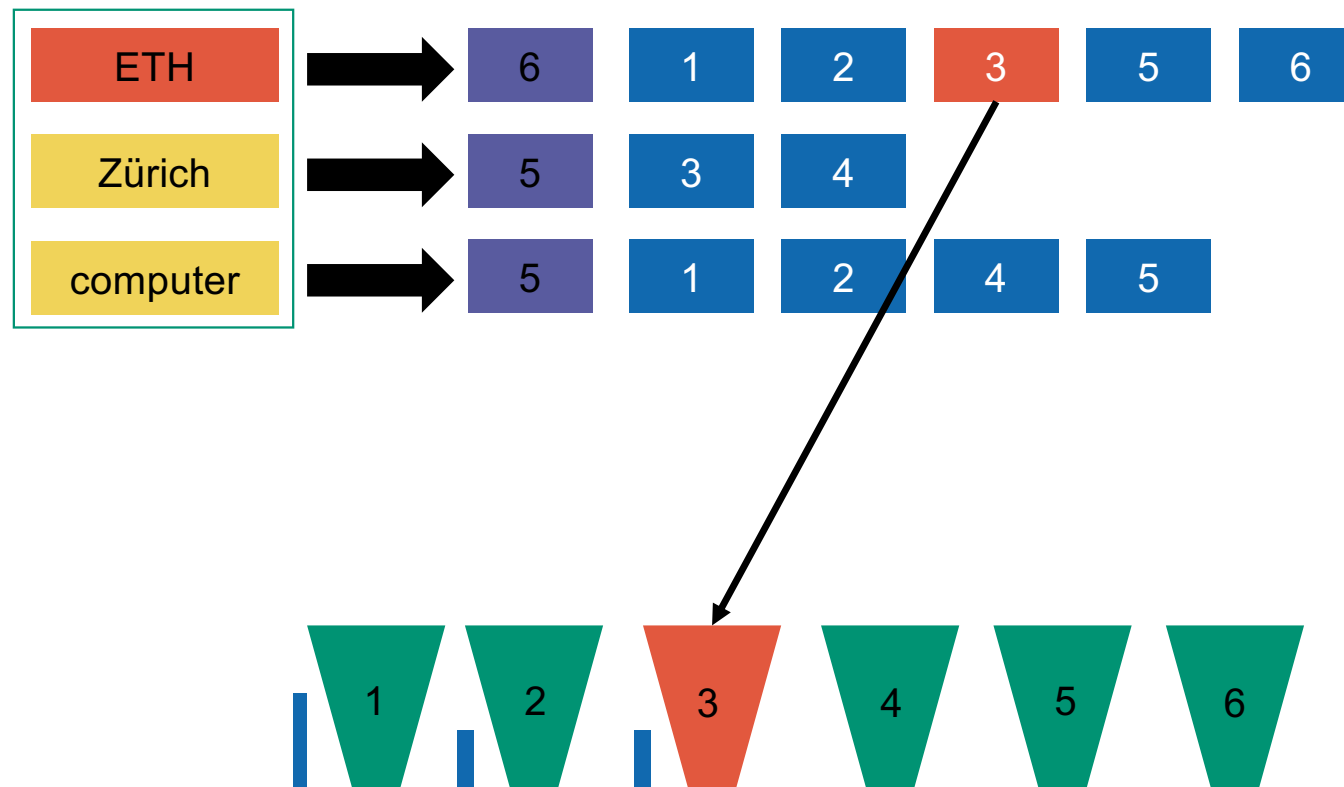
Term-at-a-time



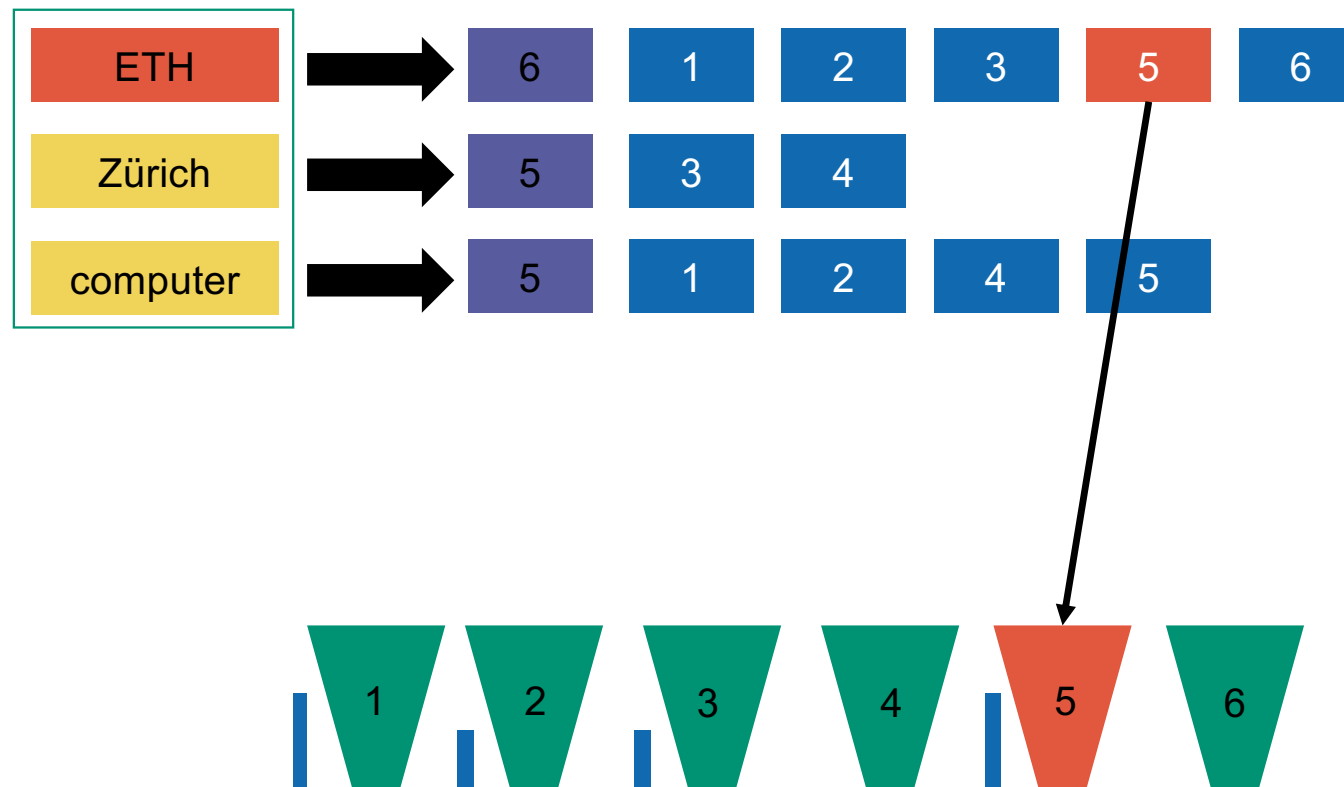
Term-at-a-time



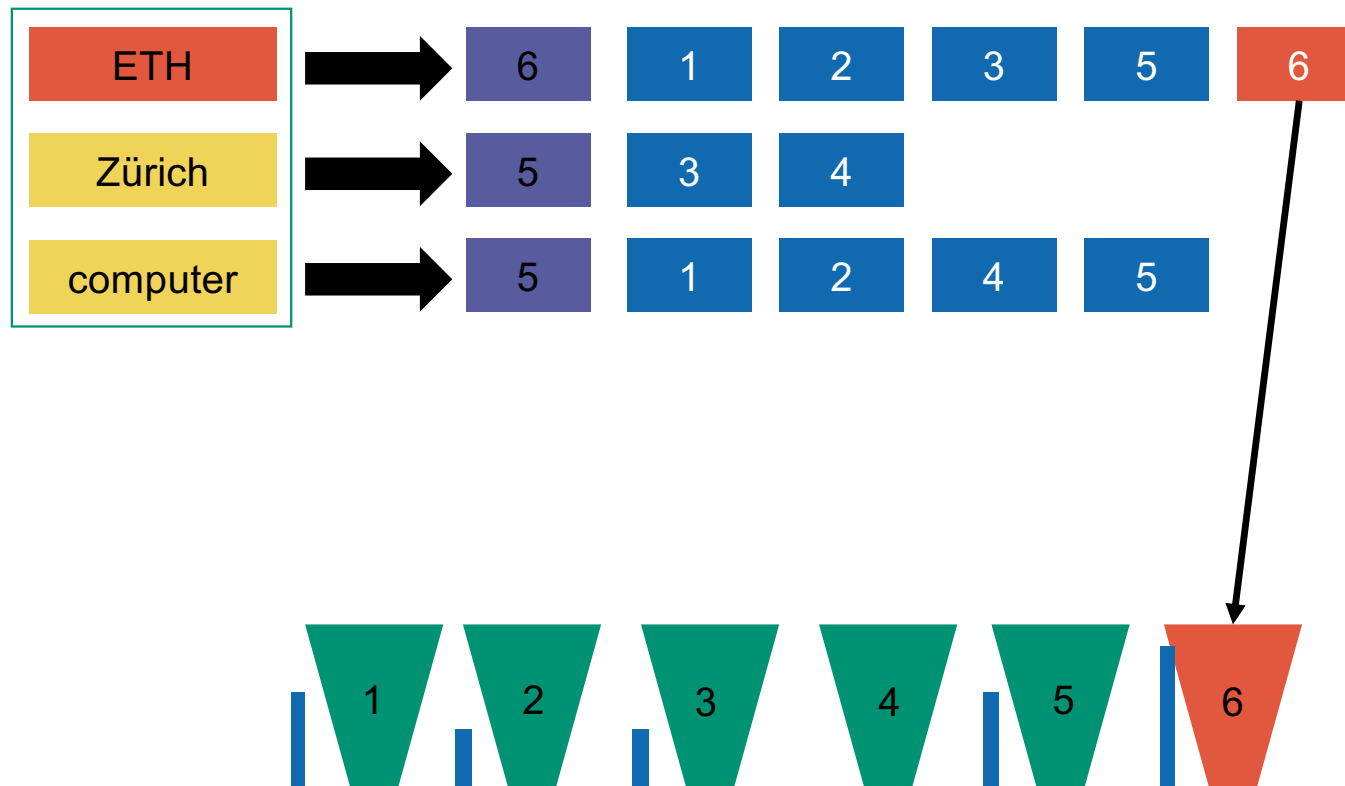
Term-at-a-time



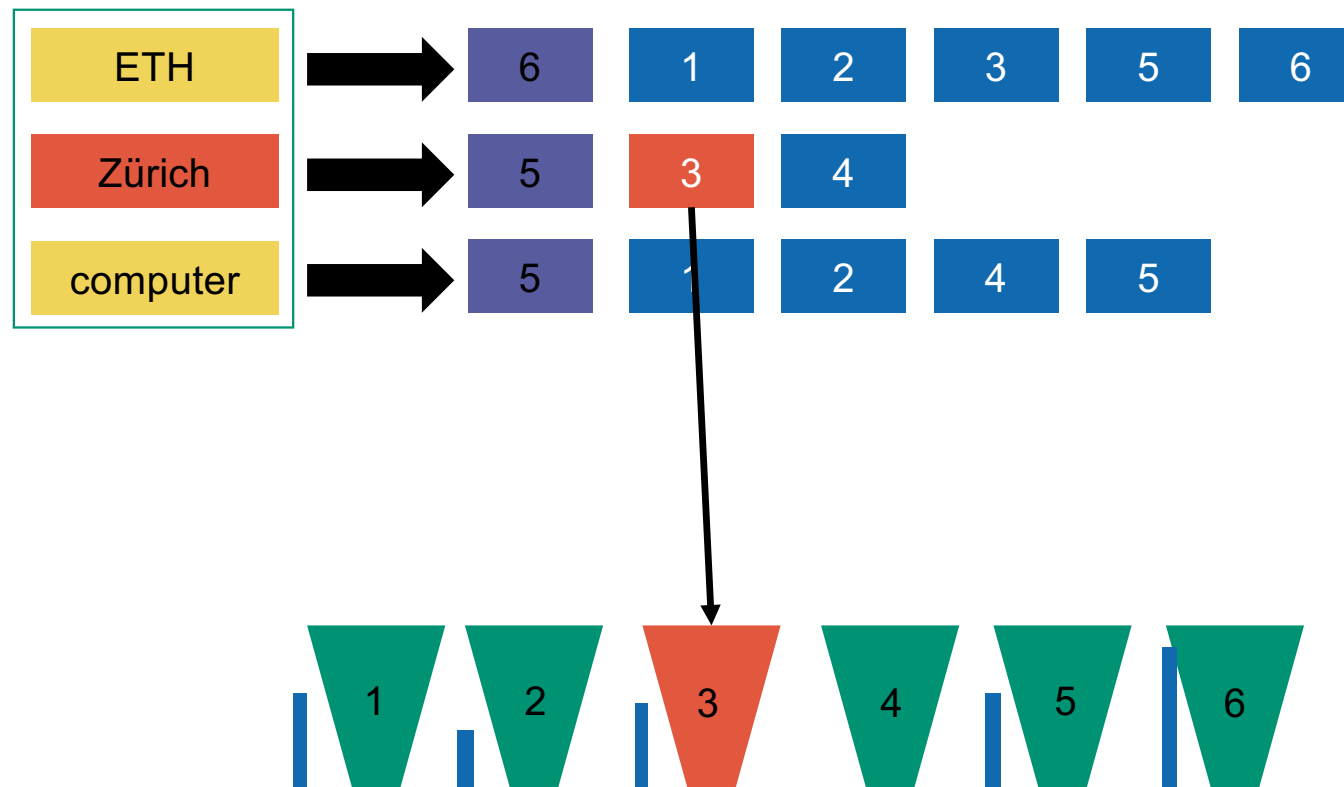
Term-at-a-time



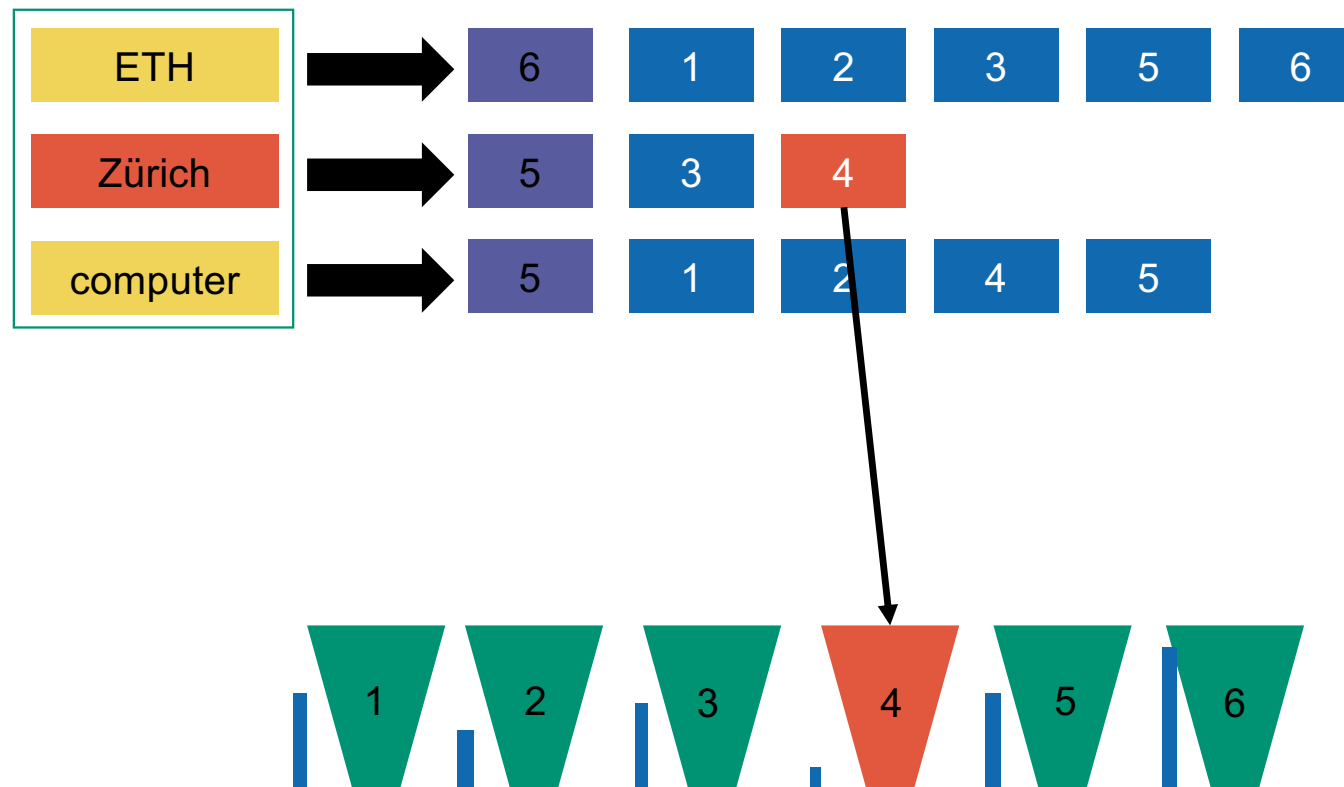
Term-at-a-time



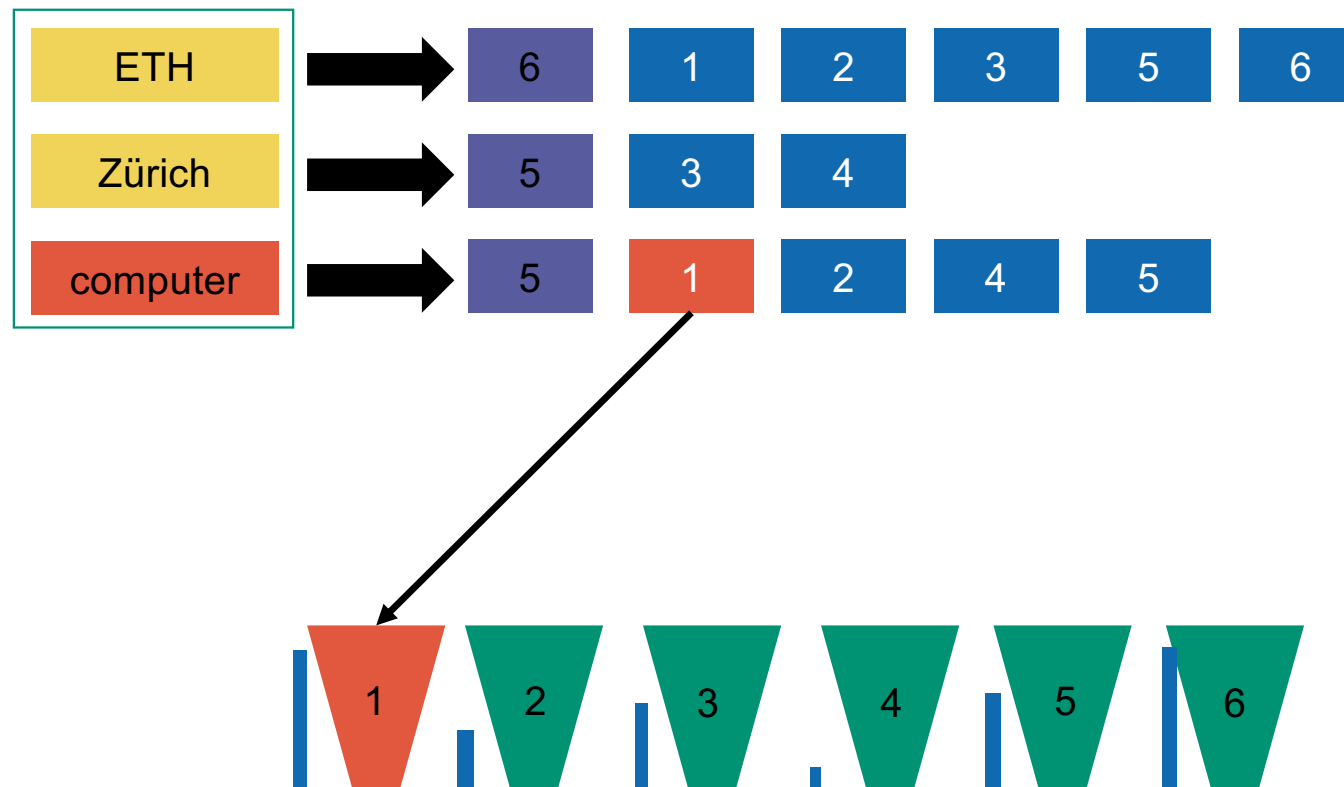
Term-at-a-time



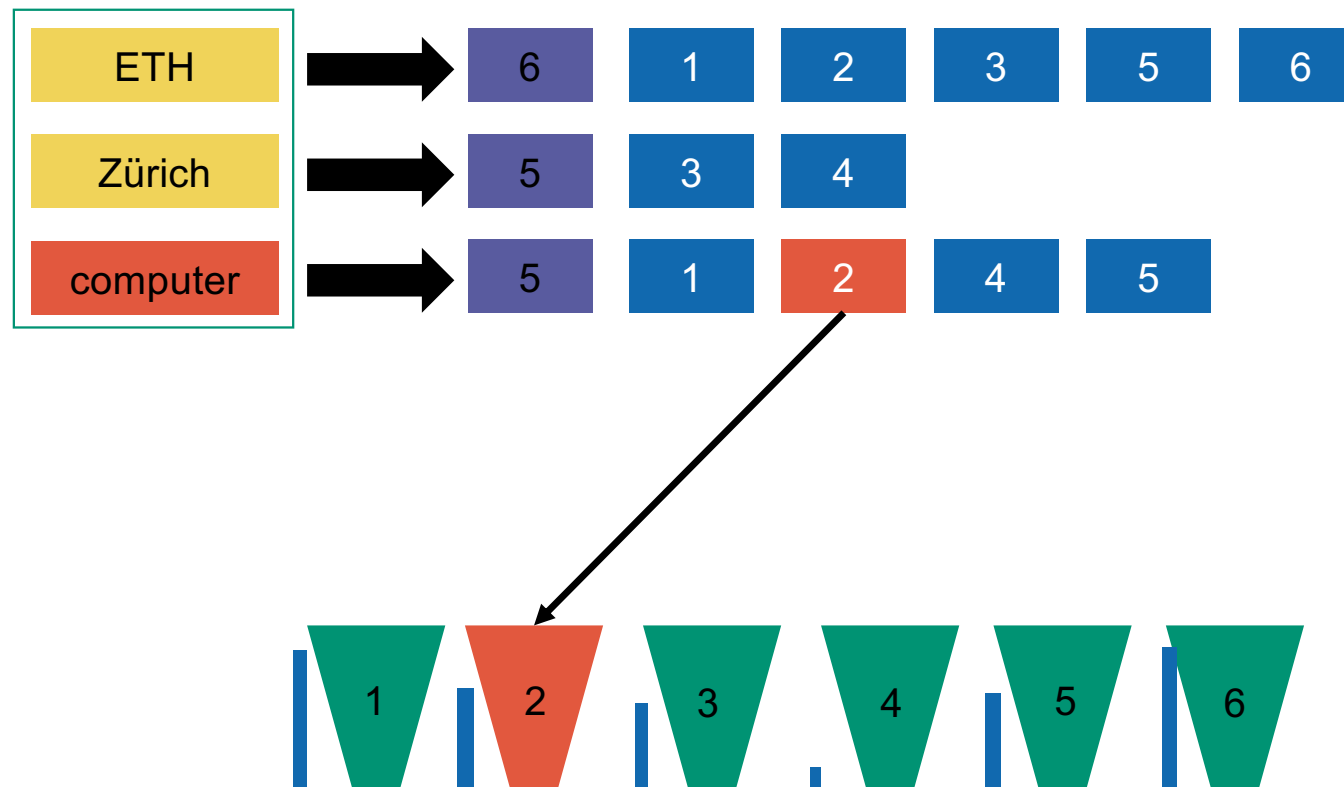
Term-at-a-time



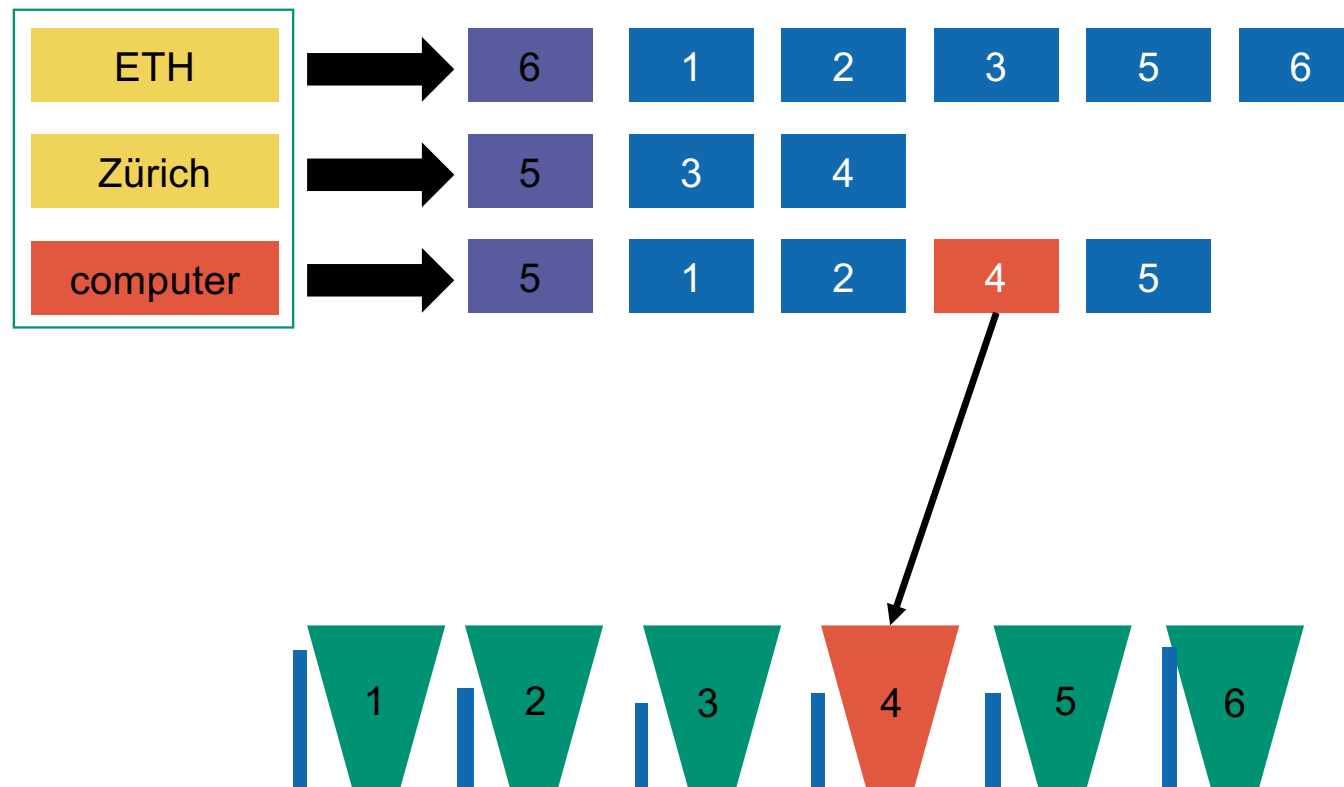
Term-at-a-time



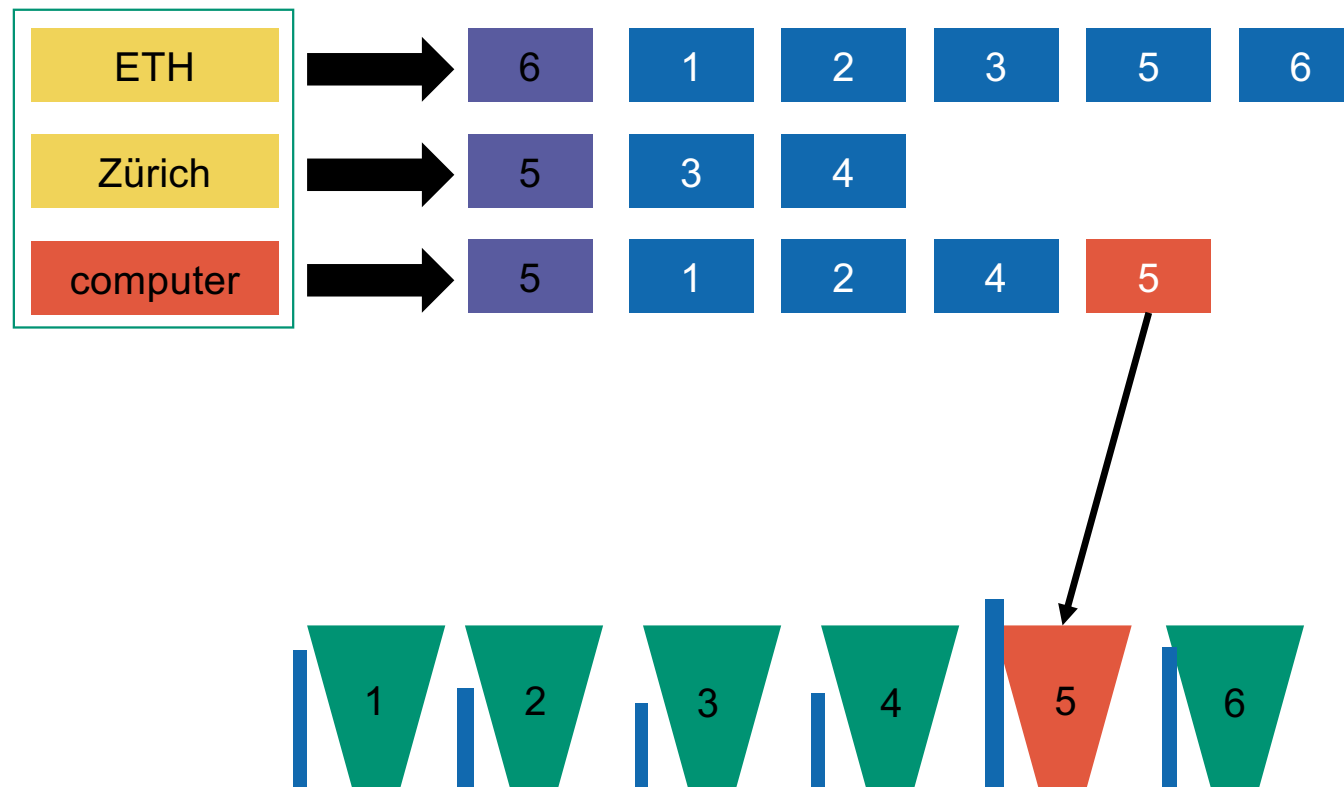
Term-at-a-time



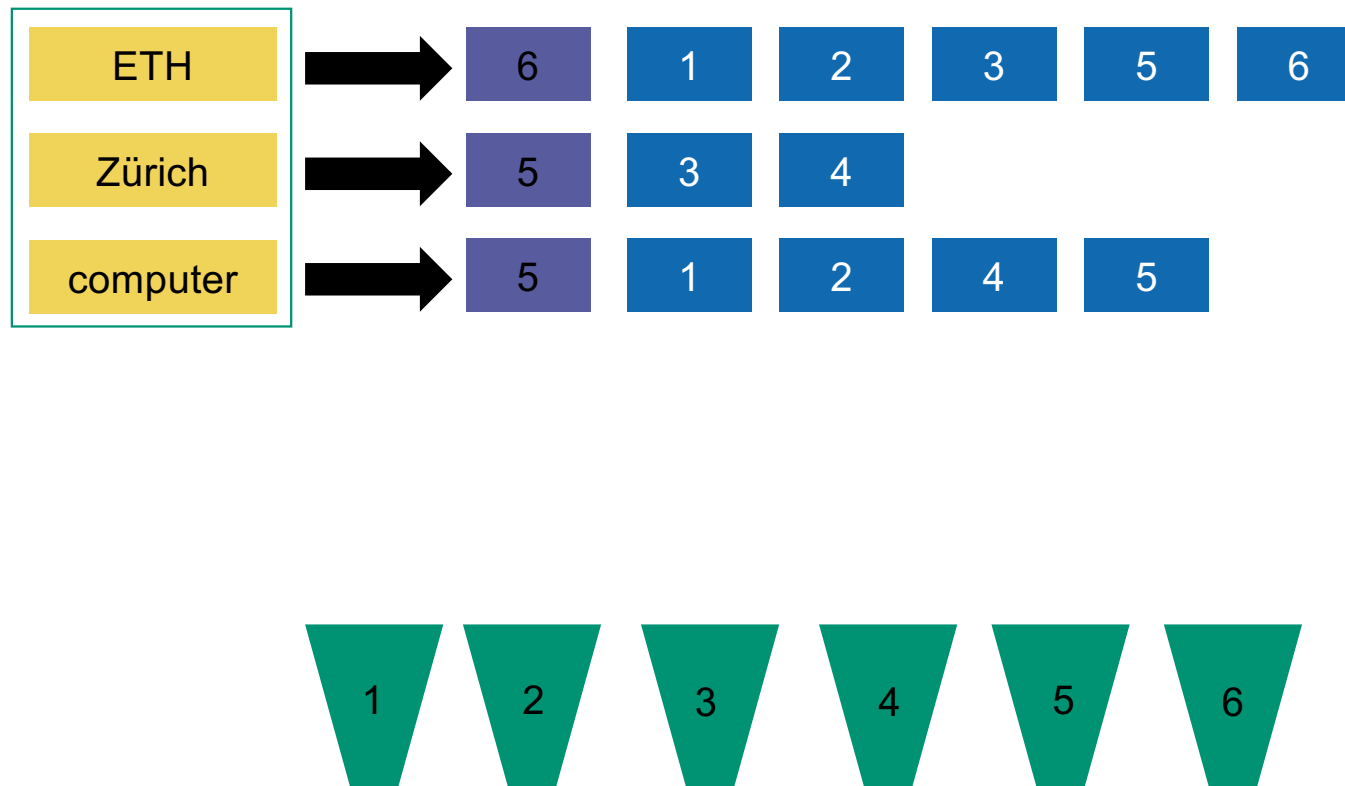
Term-at-a-time



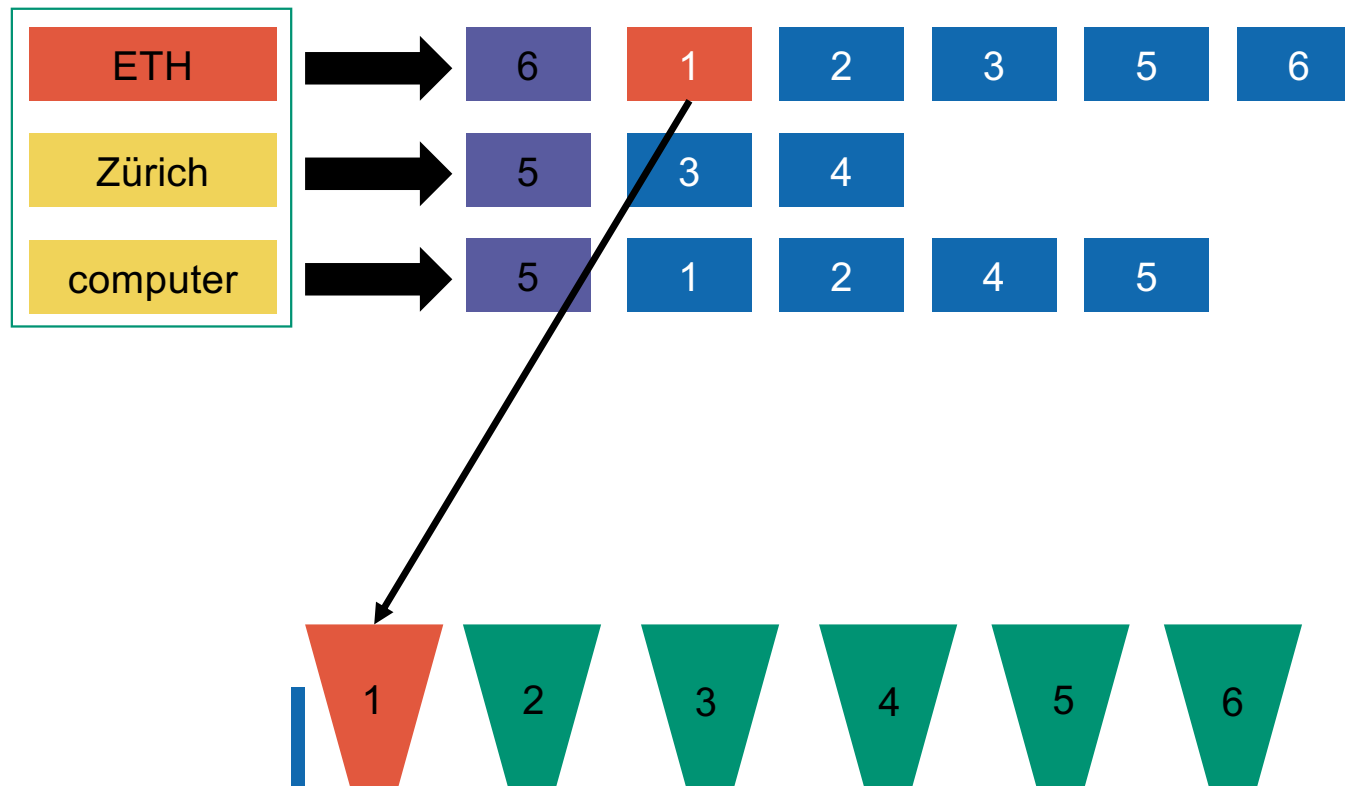
Term-at-a-time



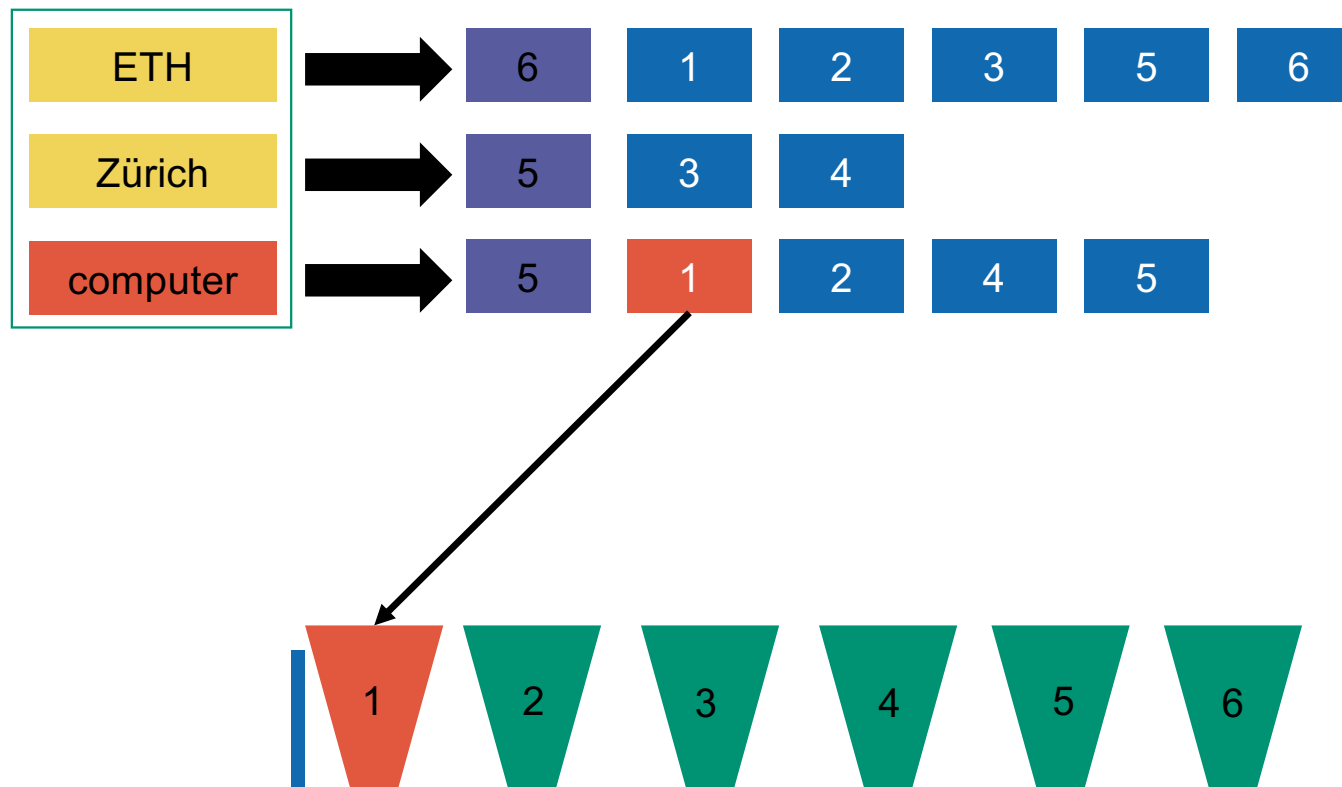
Document-at-a-time



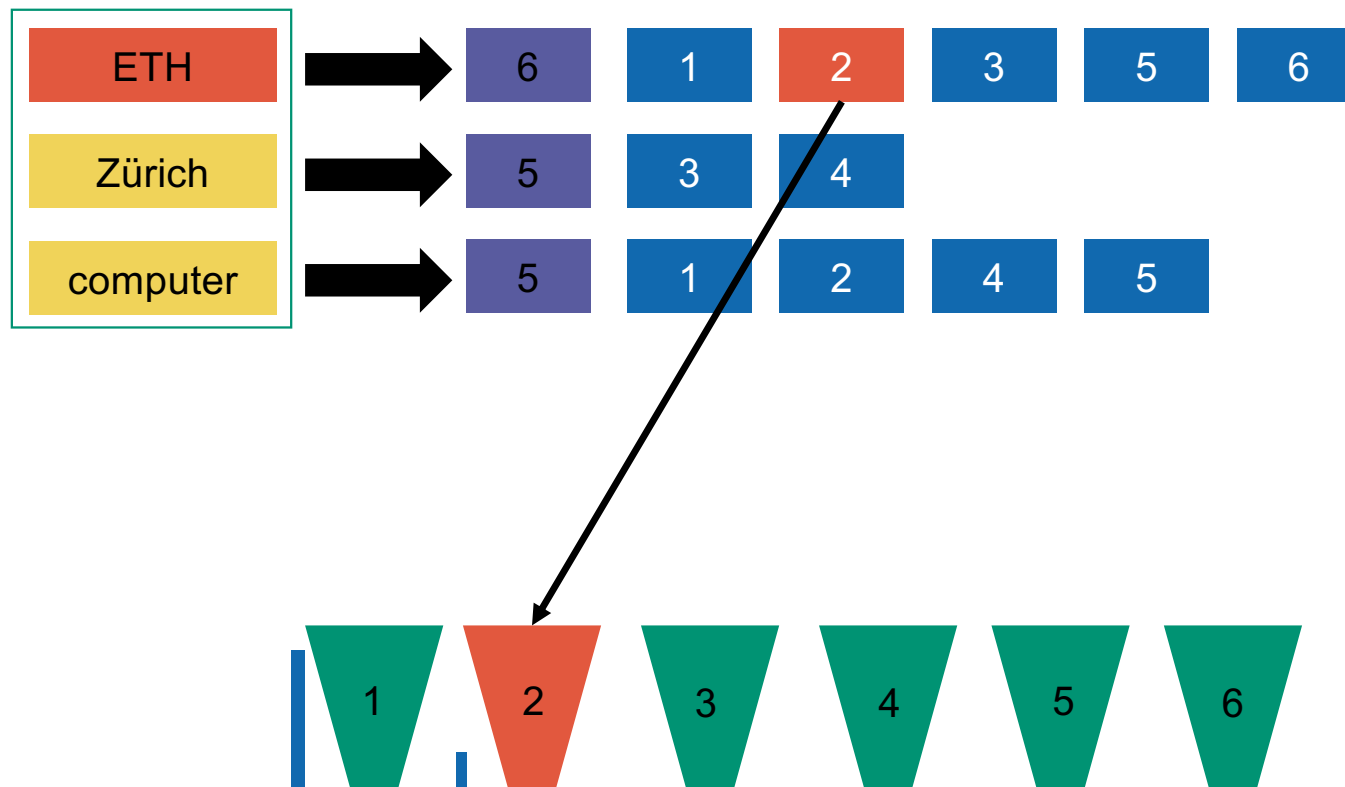
Document-at-a-time



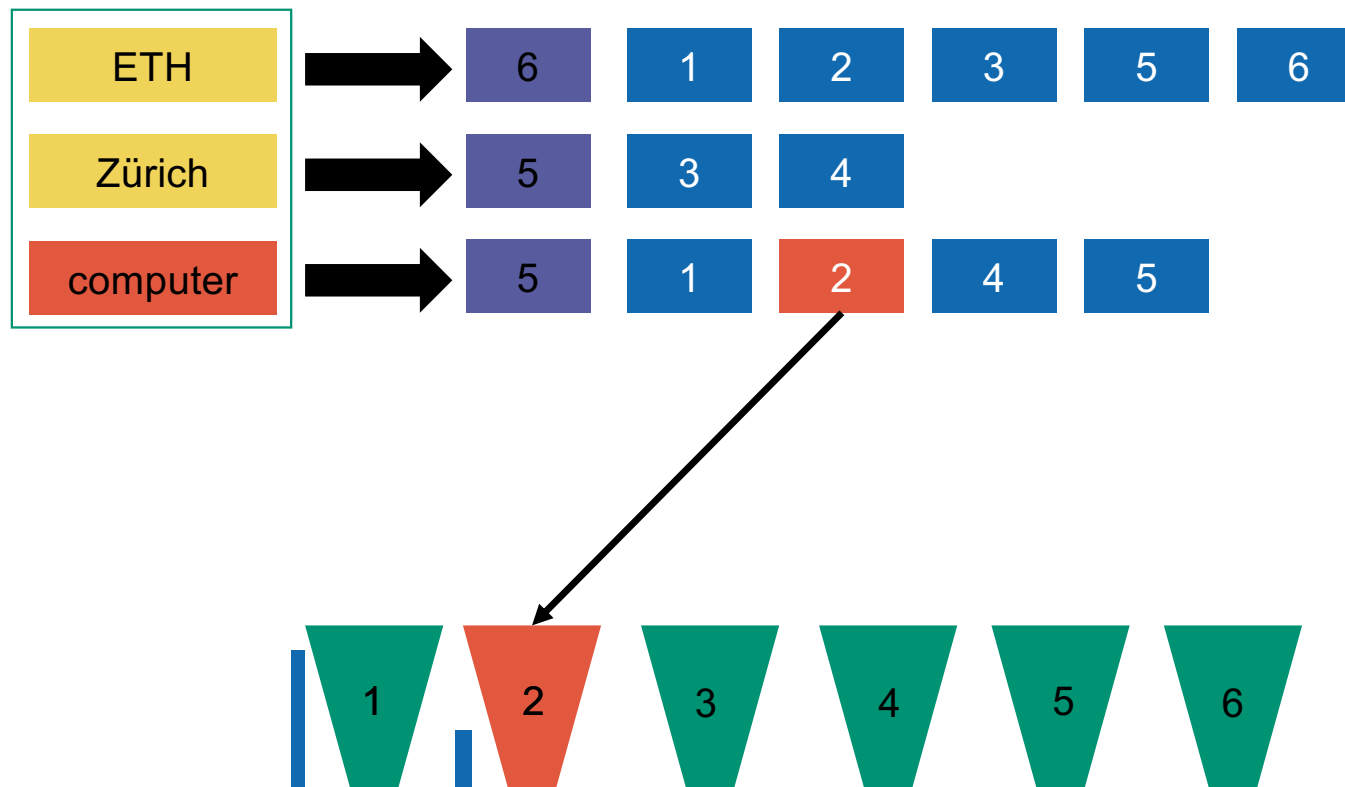
Document-at-a-time



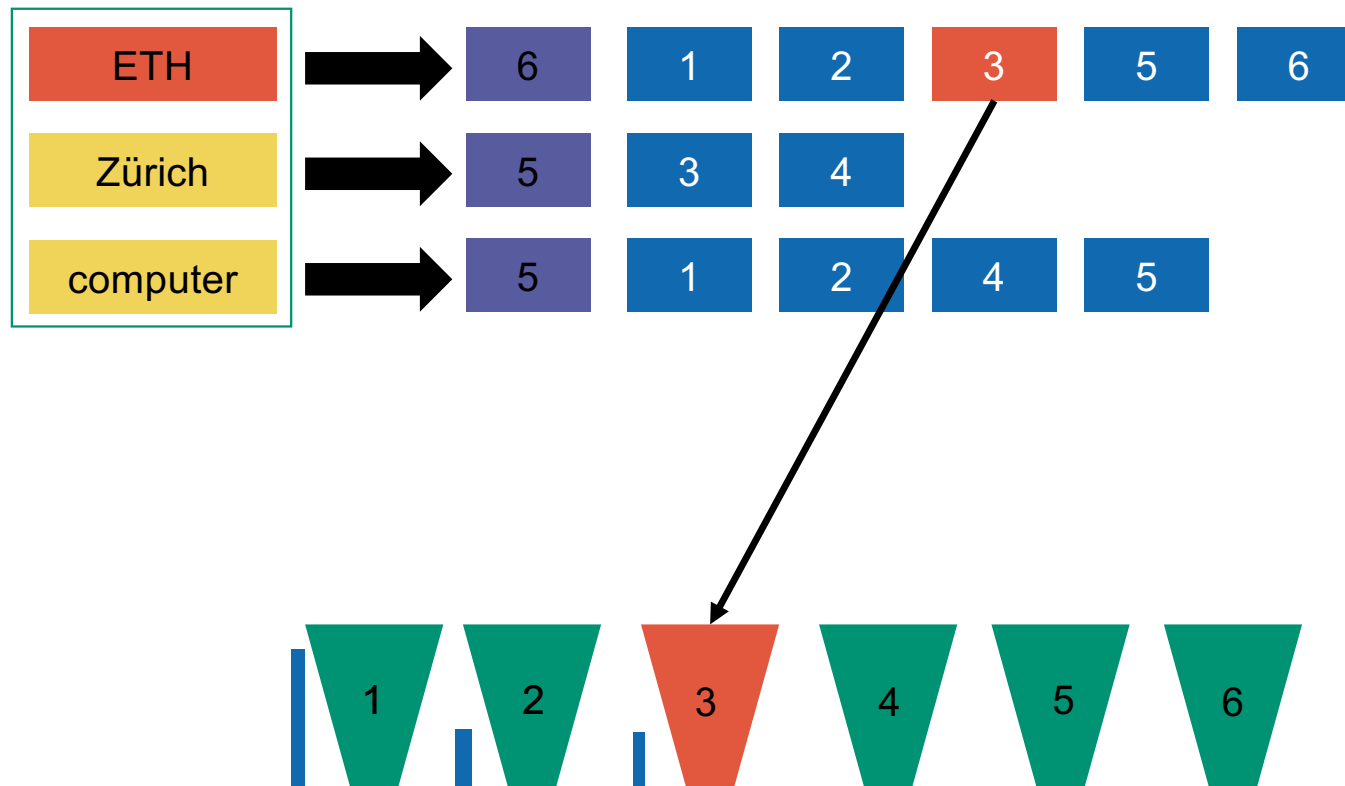
Document-at-a-time



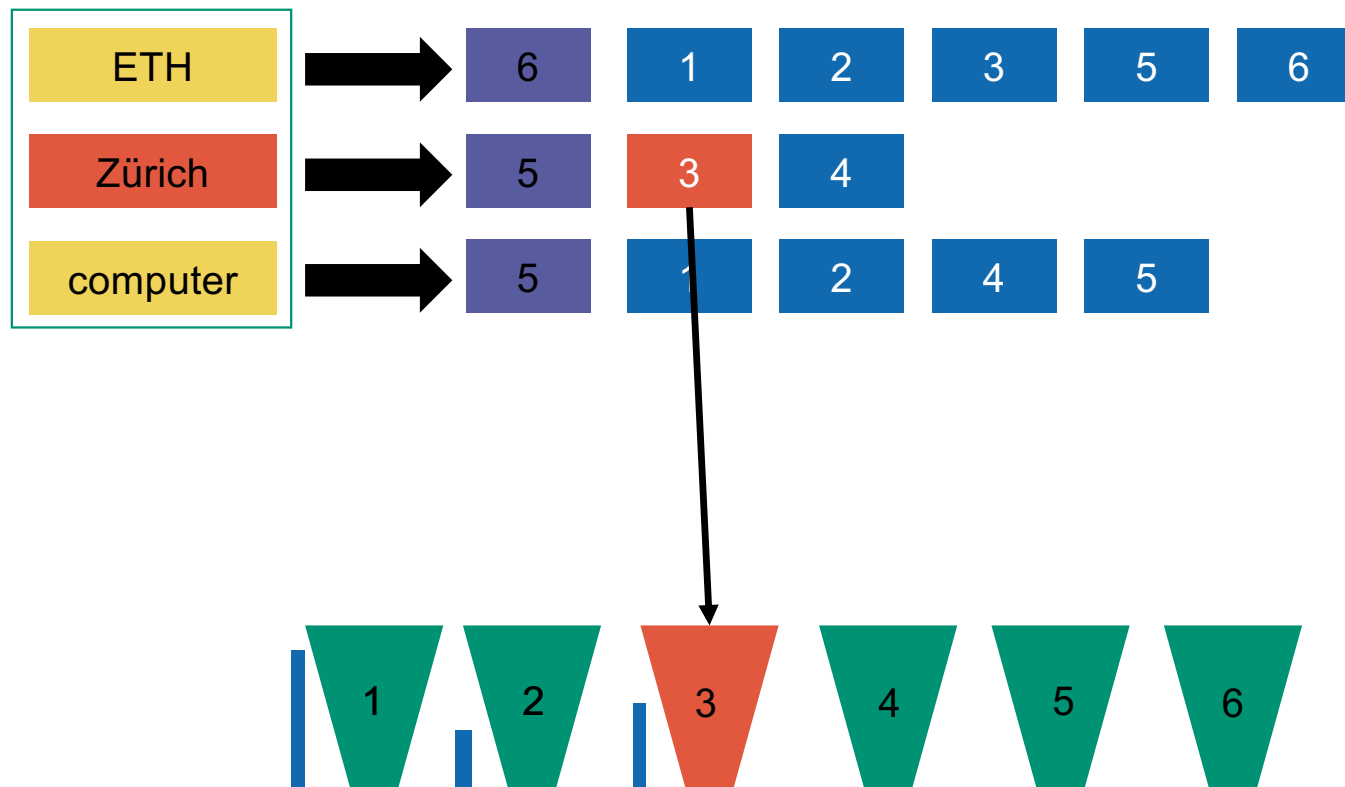
Document-at-a-time



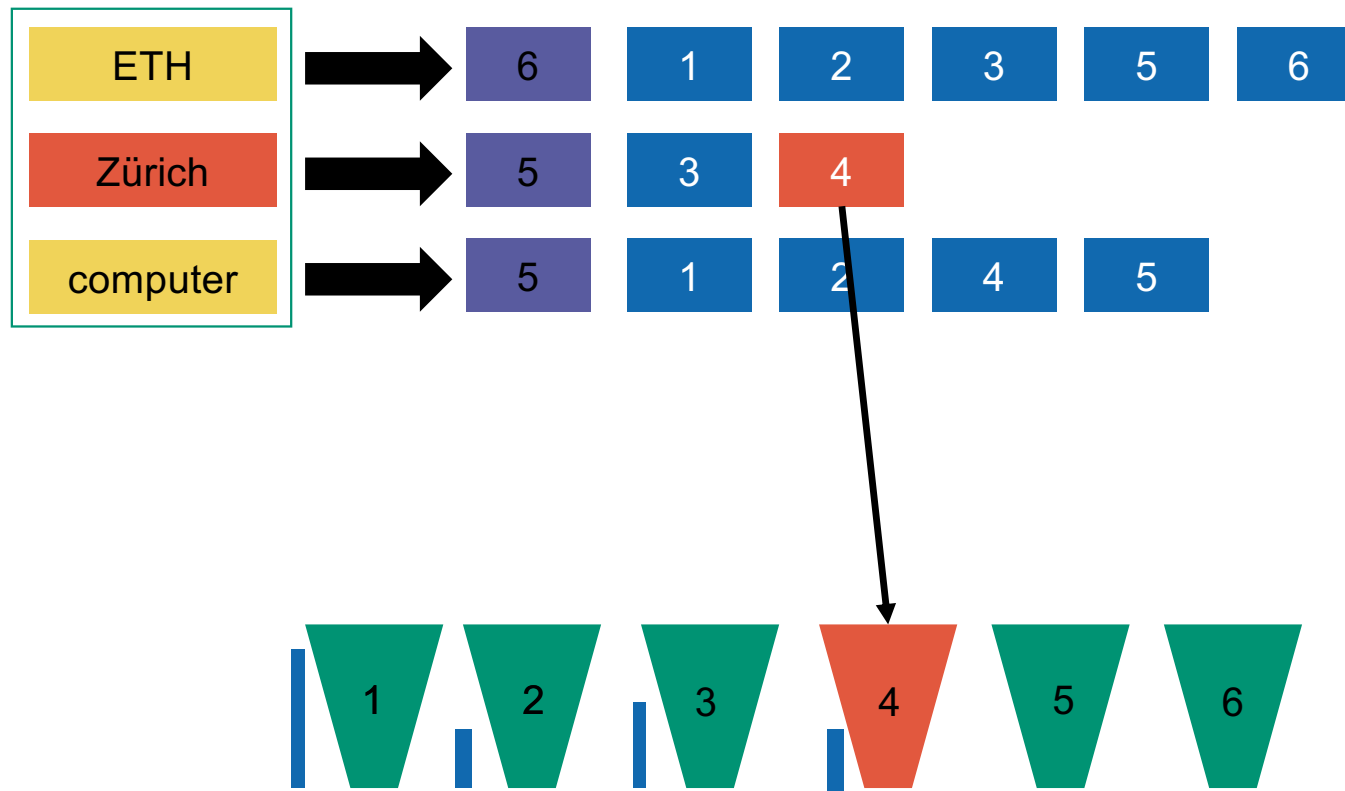
Document-at-a-time



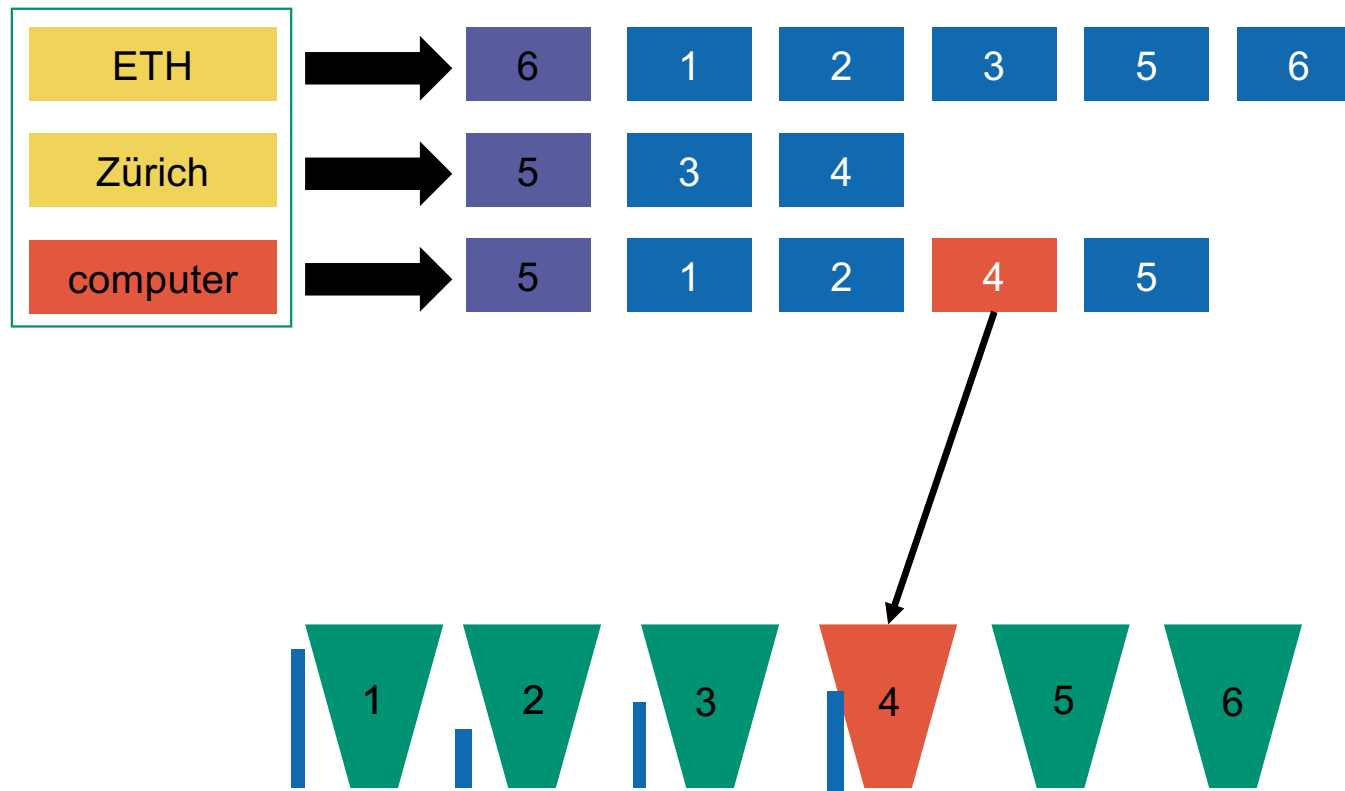
Document-at-a-time



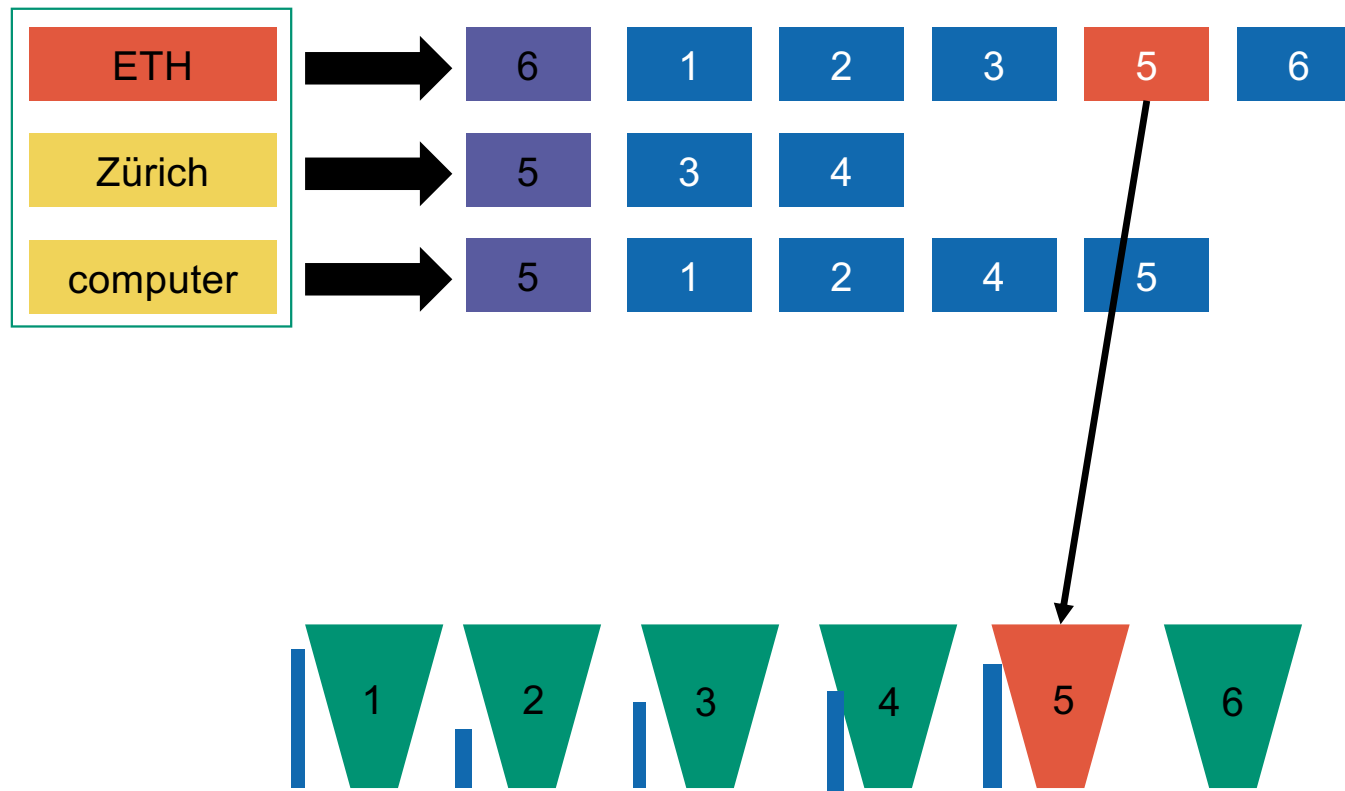
Document-at-a-time



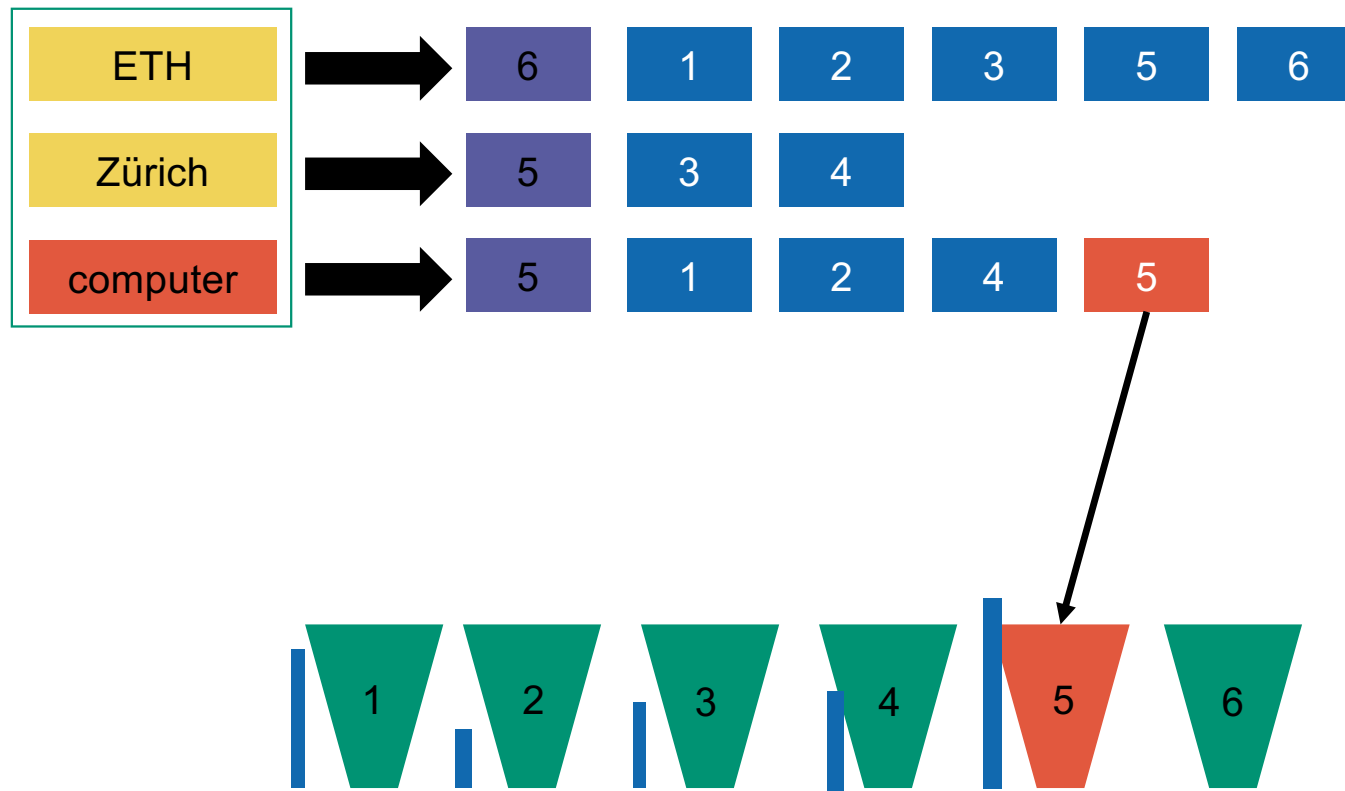
Document-at-a-time



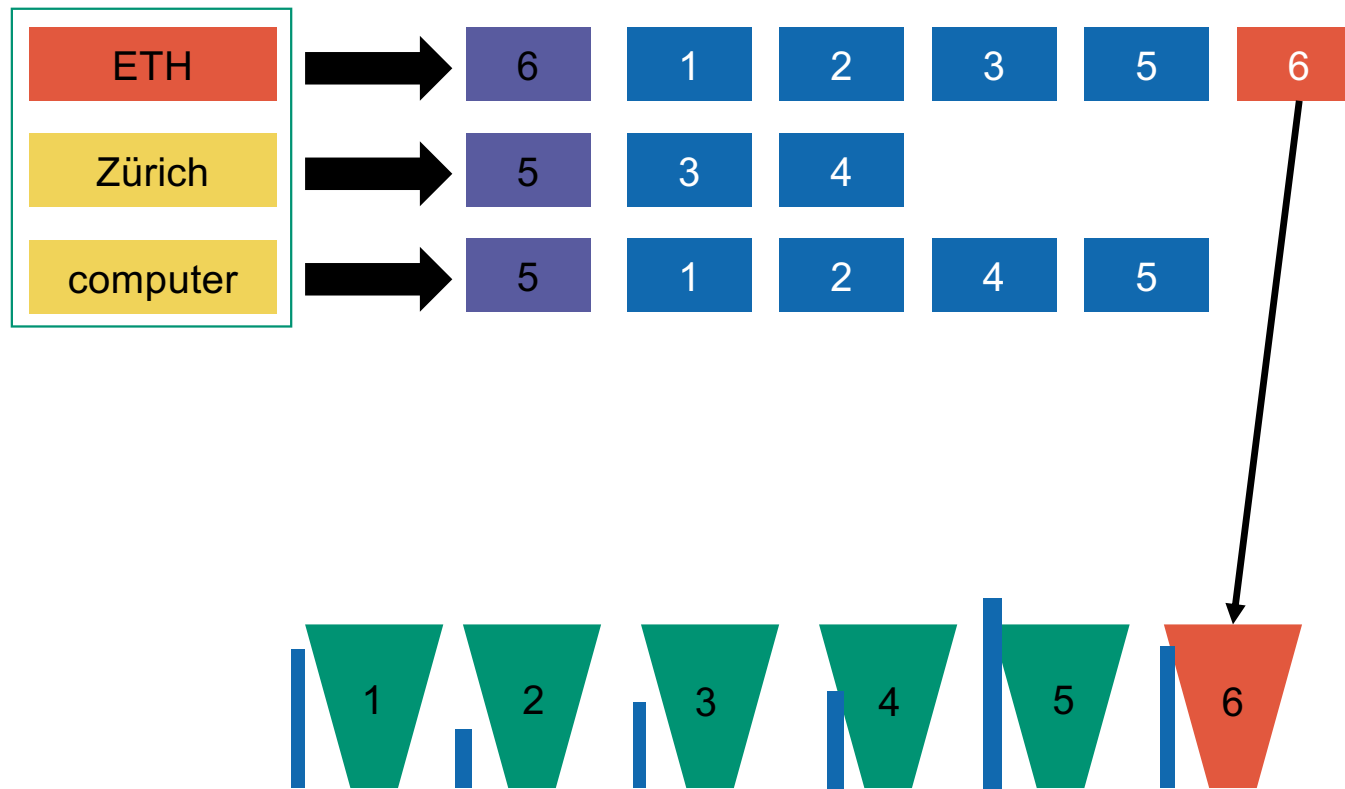
Document-at-a-time



Document-at-a-time



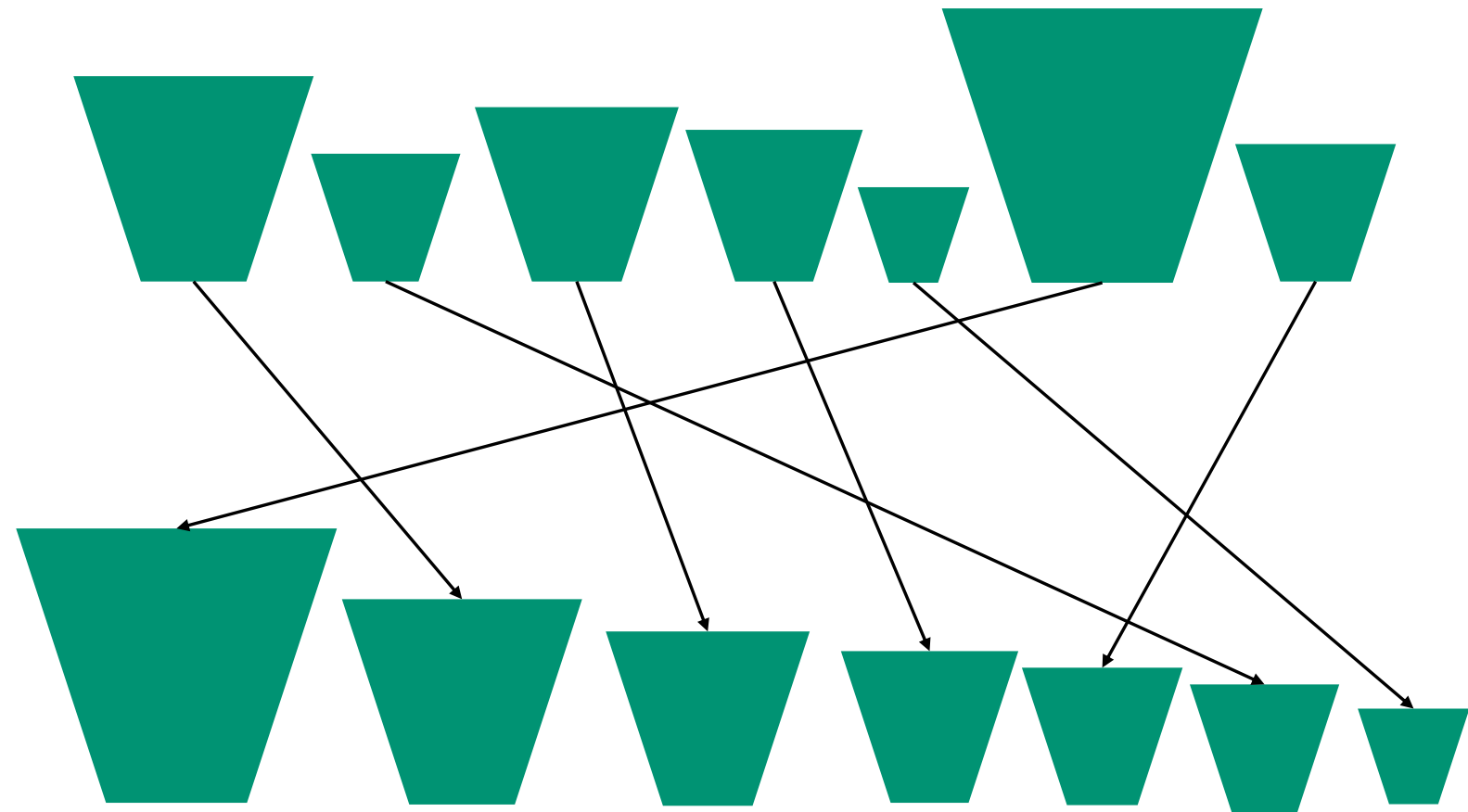
Document-at-a-time



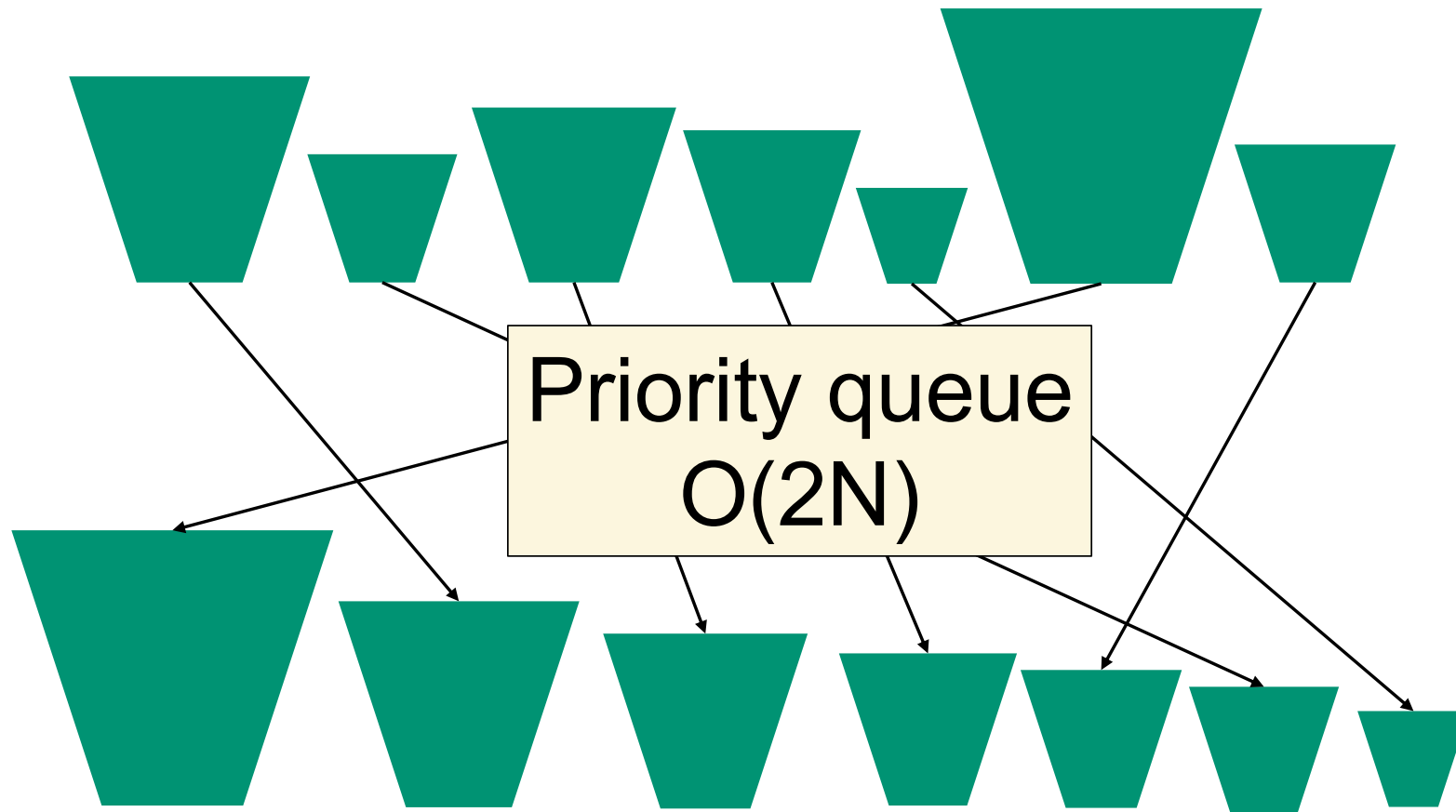
Standard inverted index



Standard inverted index



Standard inverted index



Standard inverted index

Top K



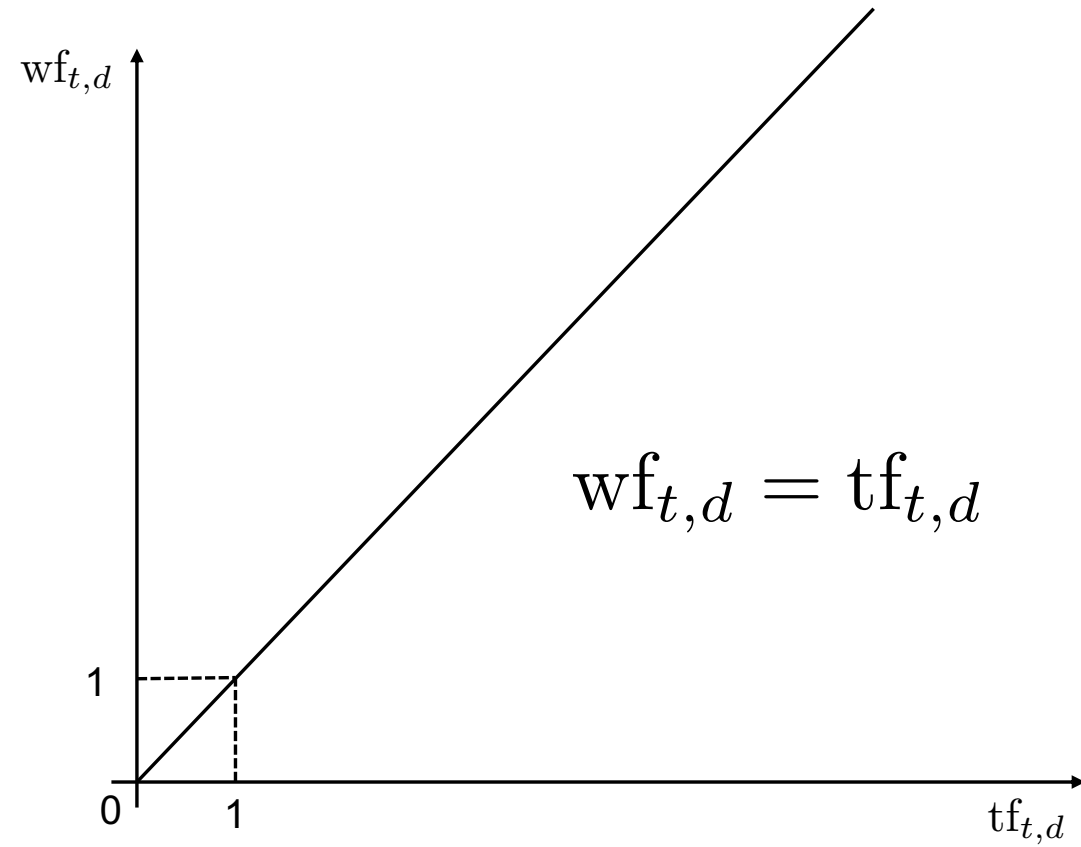
Standard inverted index



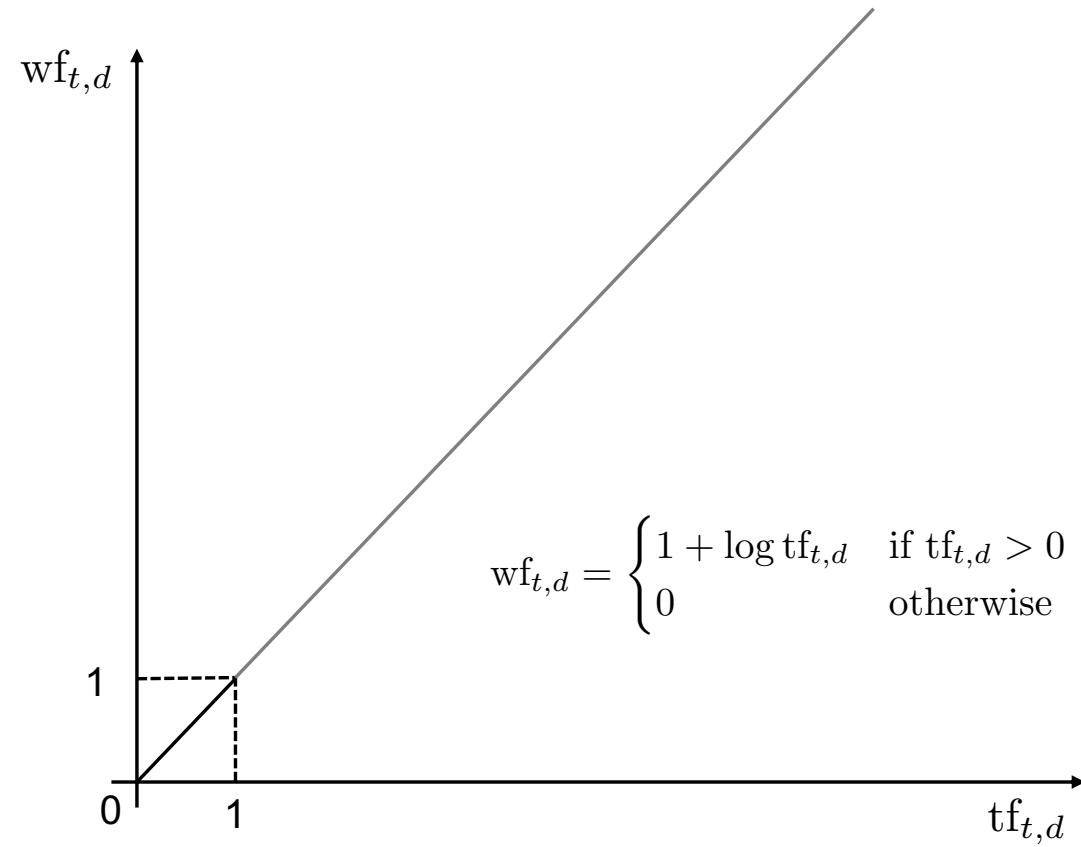
A close-up photograph of a row of sharpened pencils. The pencils are arranged diagonally from the top left towards the bottom right. Most are grey, but one in the foreground is red. A green rectangular box is overlaid on the right side of the image, containing the text 'Alternate weights' in white.

Alternate weights

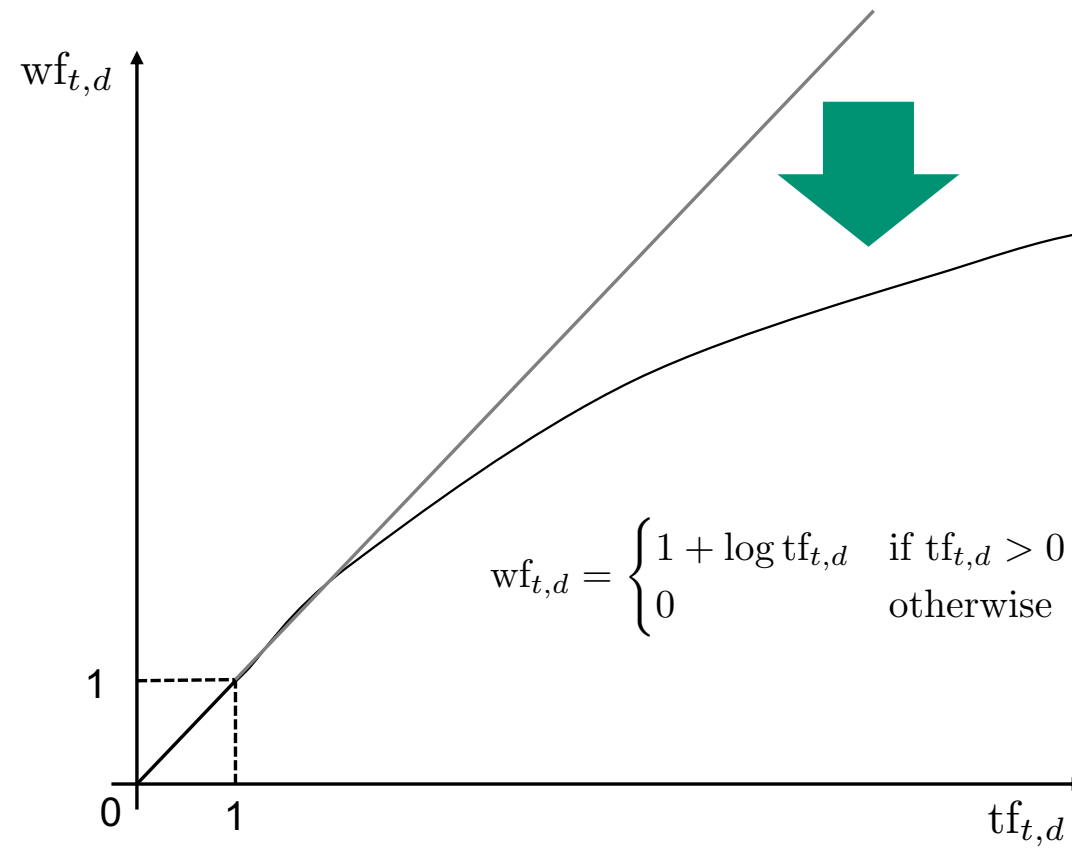
Natural term frequency scaling (n)



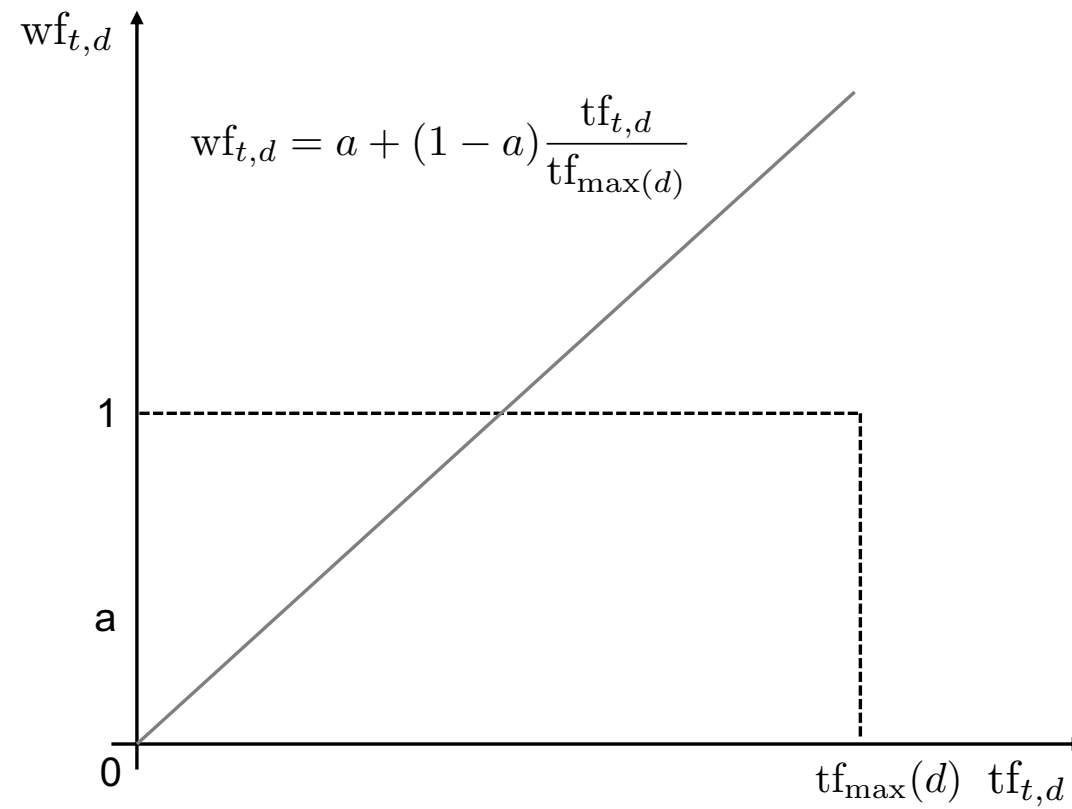
Sublinear term frequency scaling (I)



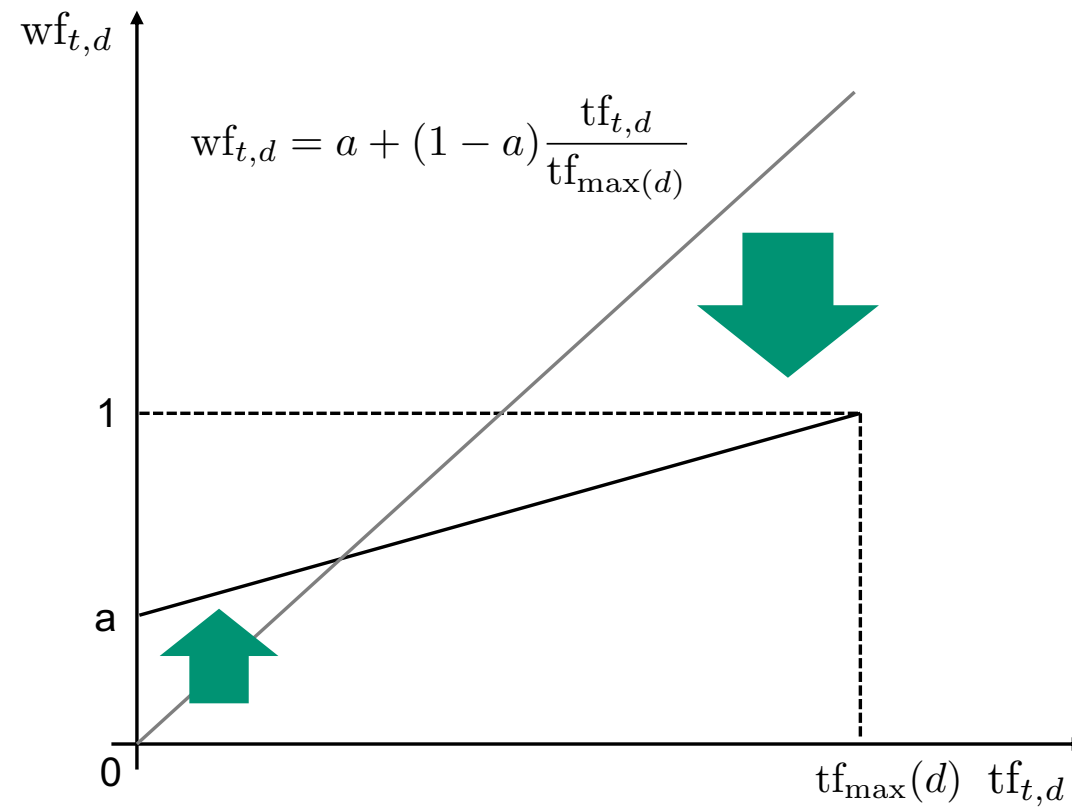
Sublinear term frequency scaling (I)



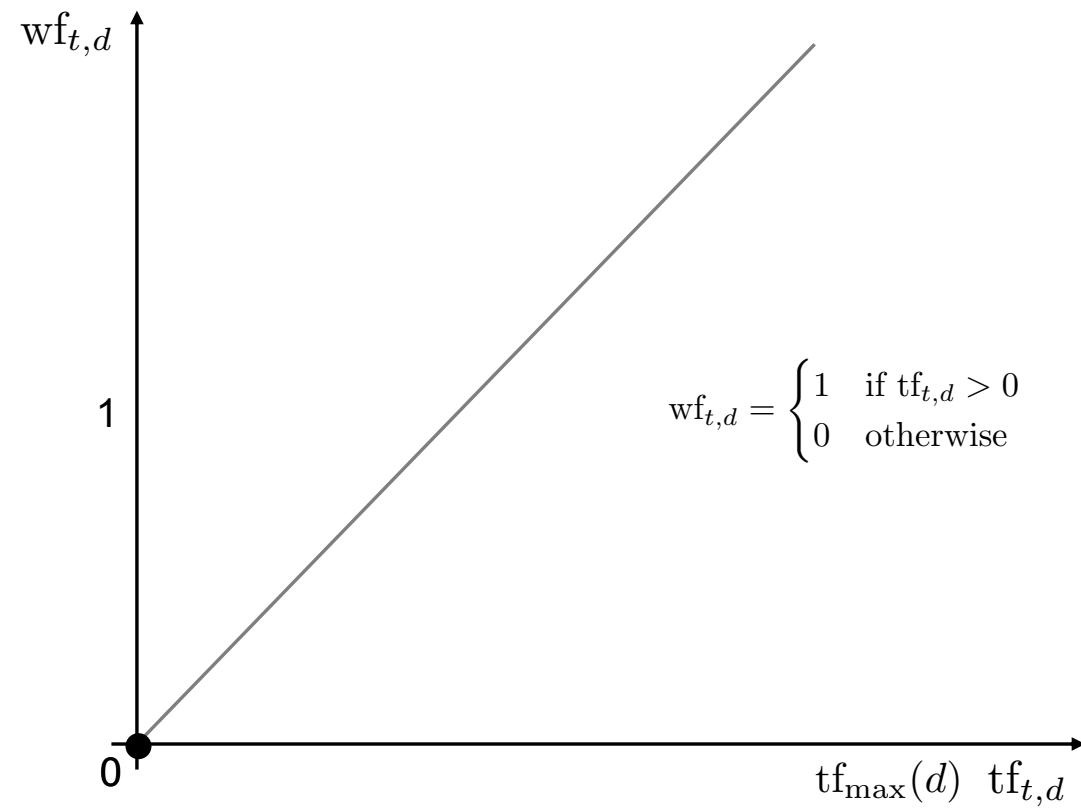
Augmented term frequency scaling (a)



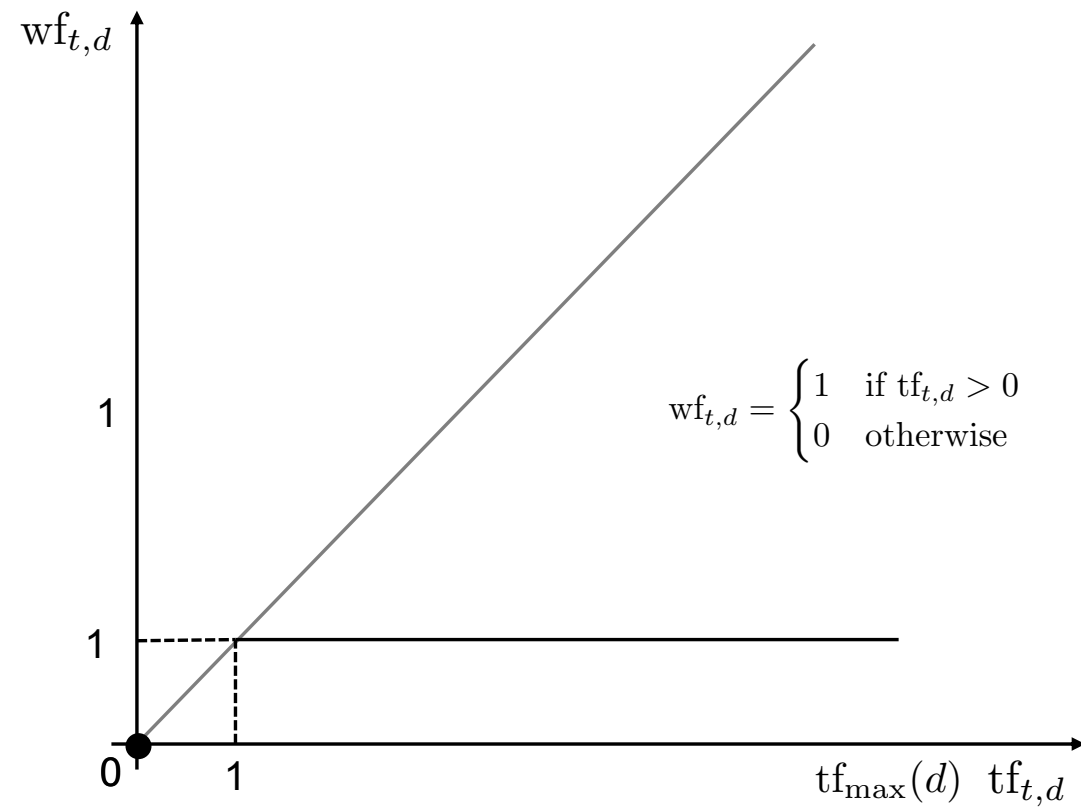
Augmented term frequency scaling (a)



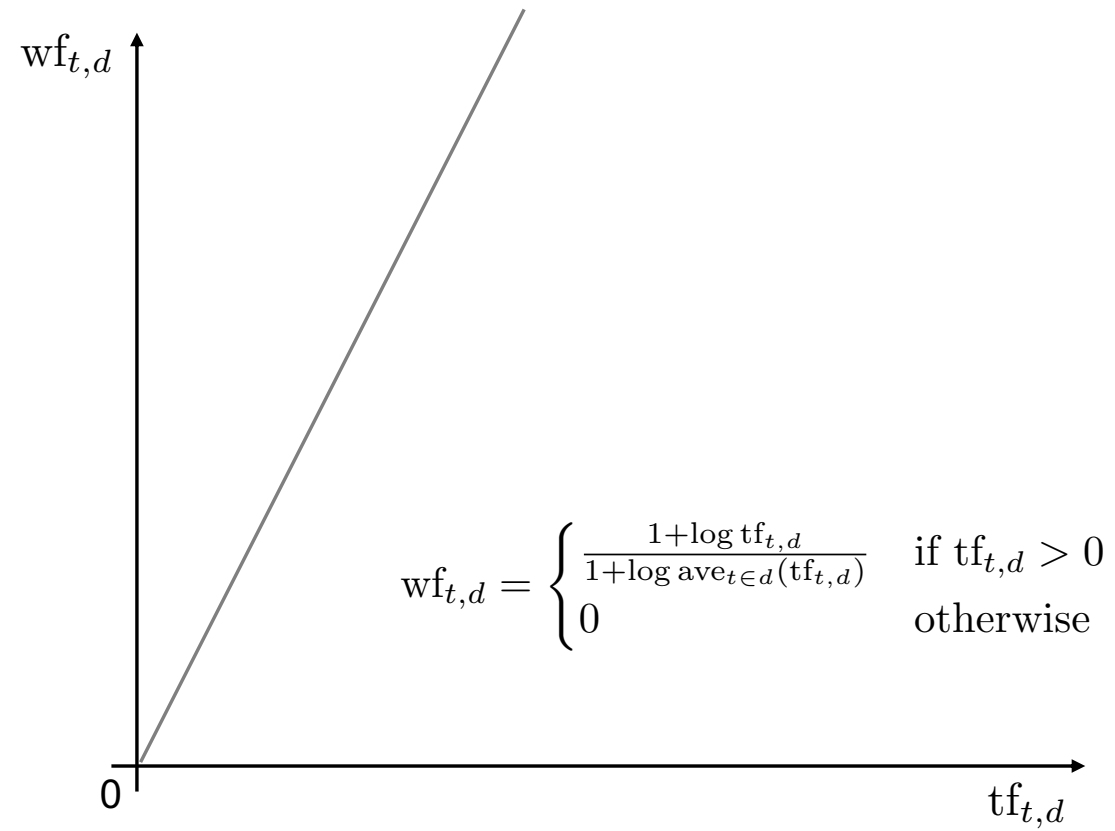
Boolean term frequency scaling (b)



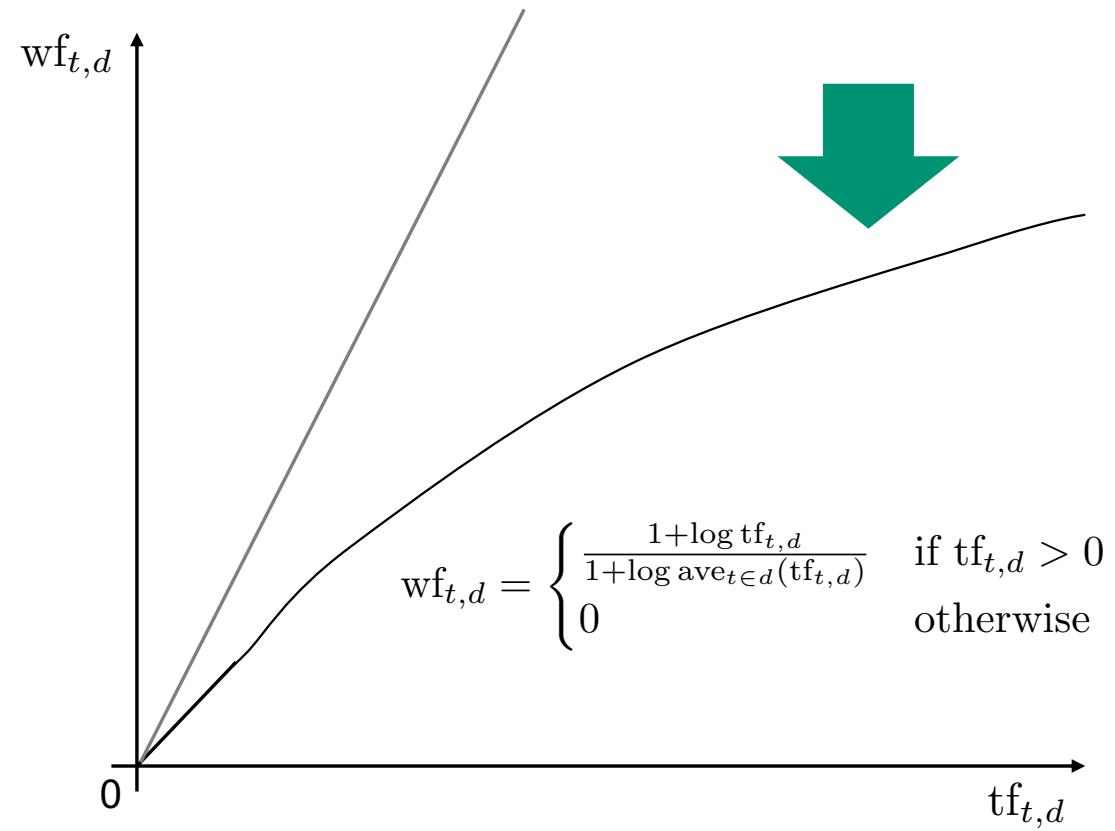
Boolean term frequency scaling (b)



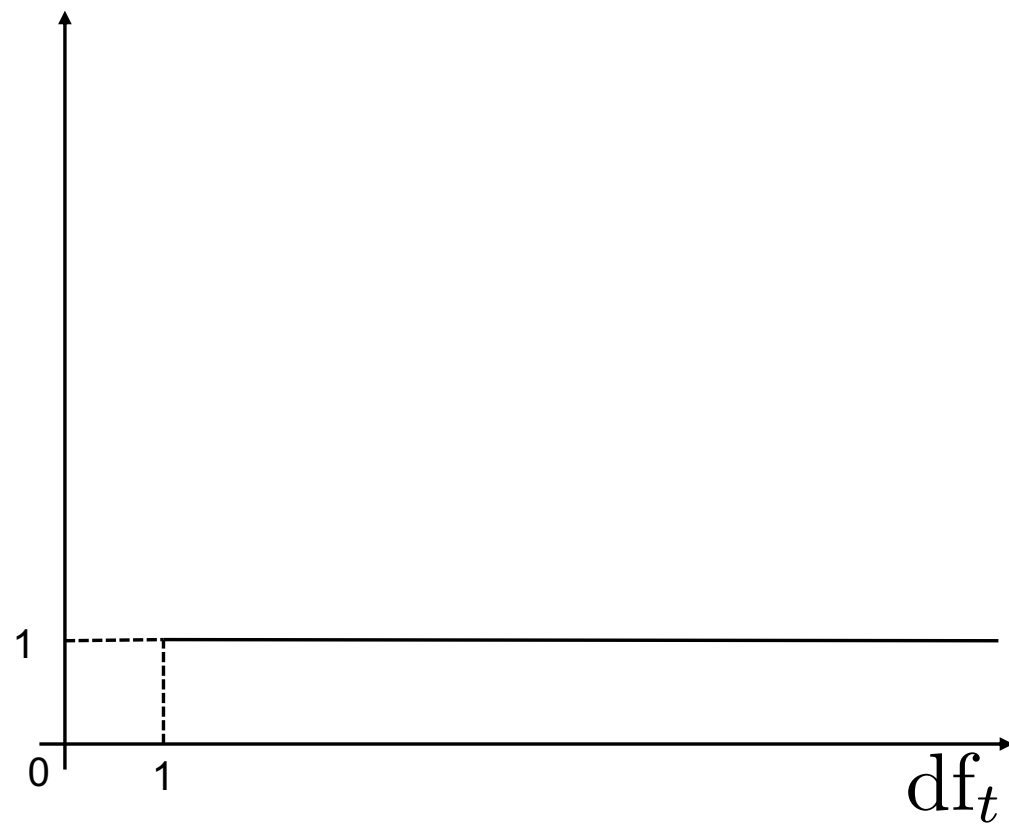
Log-average term frequency scaling (L)



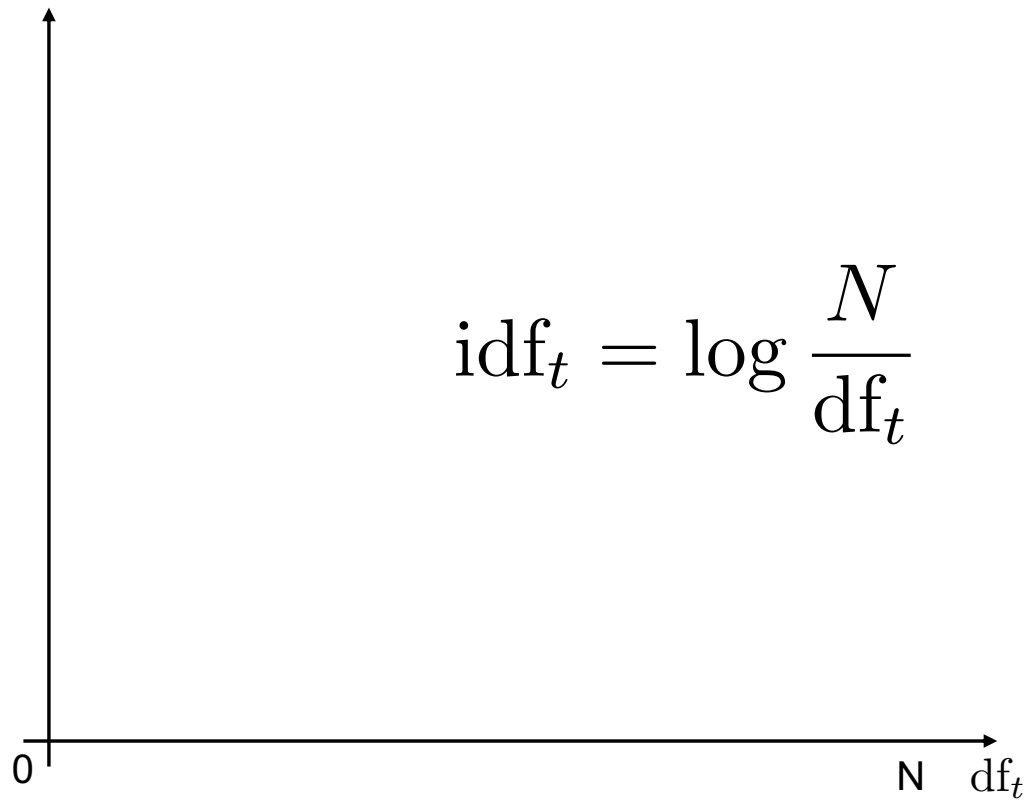
Log-average term frequency scaling (L)



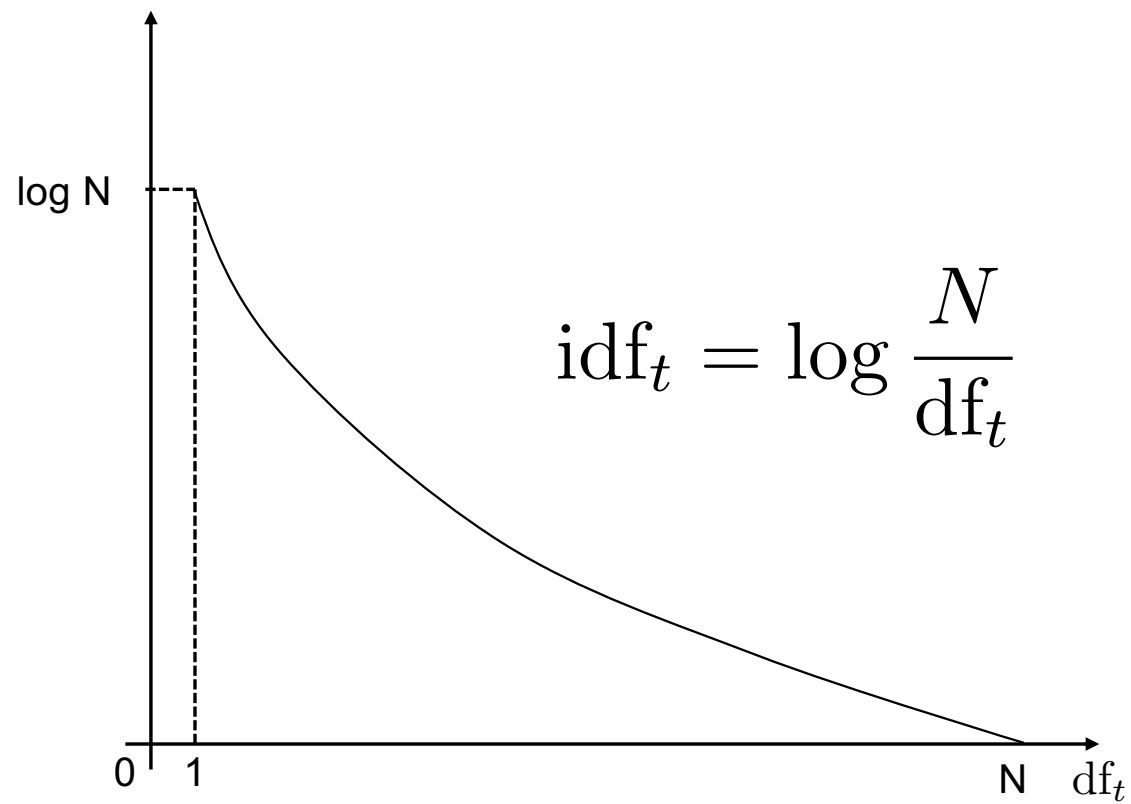
No document frequency scaling (n)



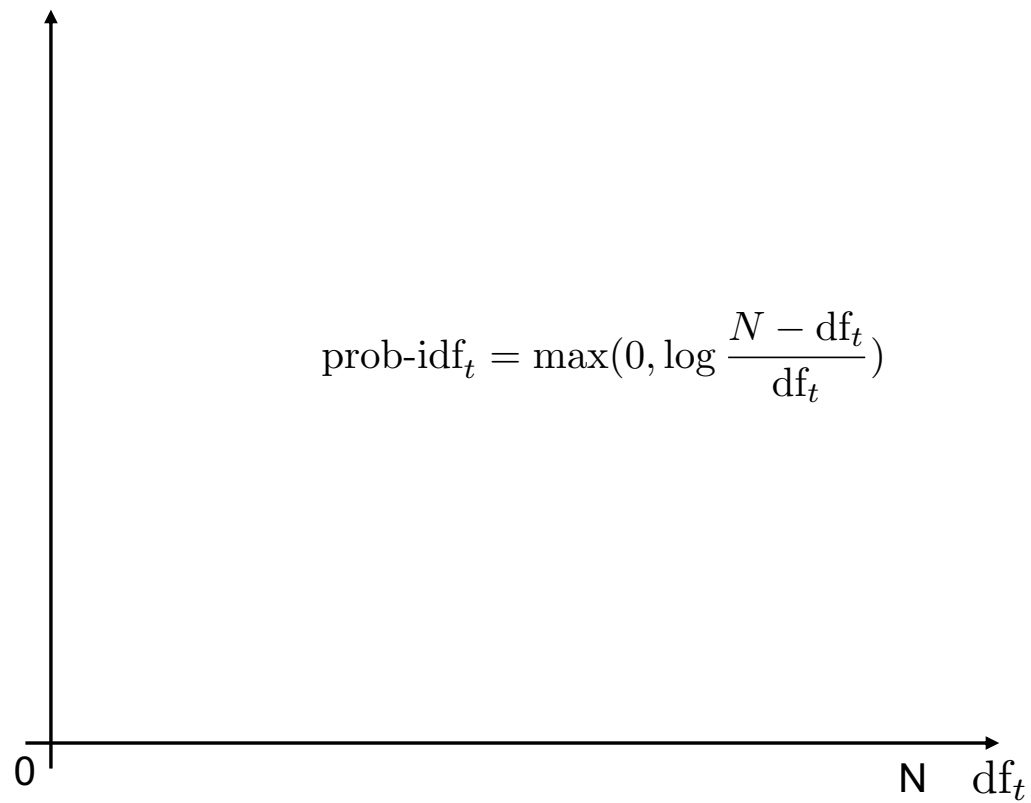
Inverted document frequency scaling (t)


$$\text{idf}_t = \log \frac{N}{\text{df}_t}$$

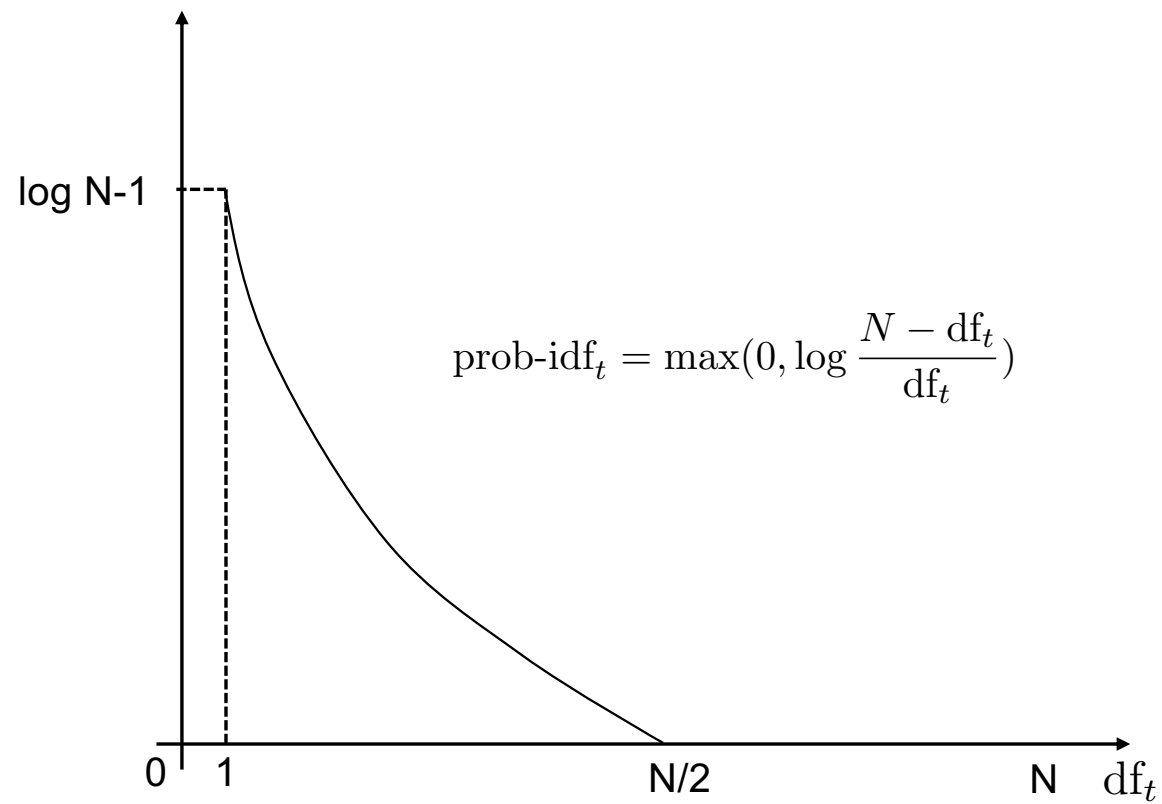
Inverted document frequency scaling (t)



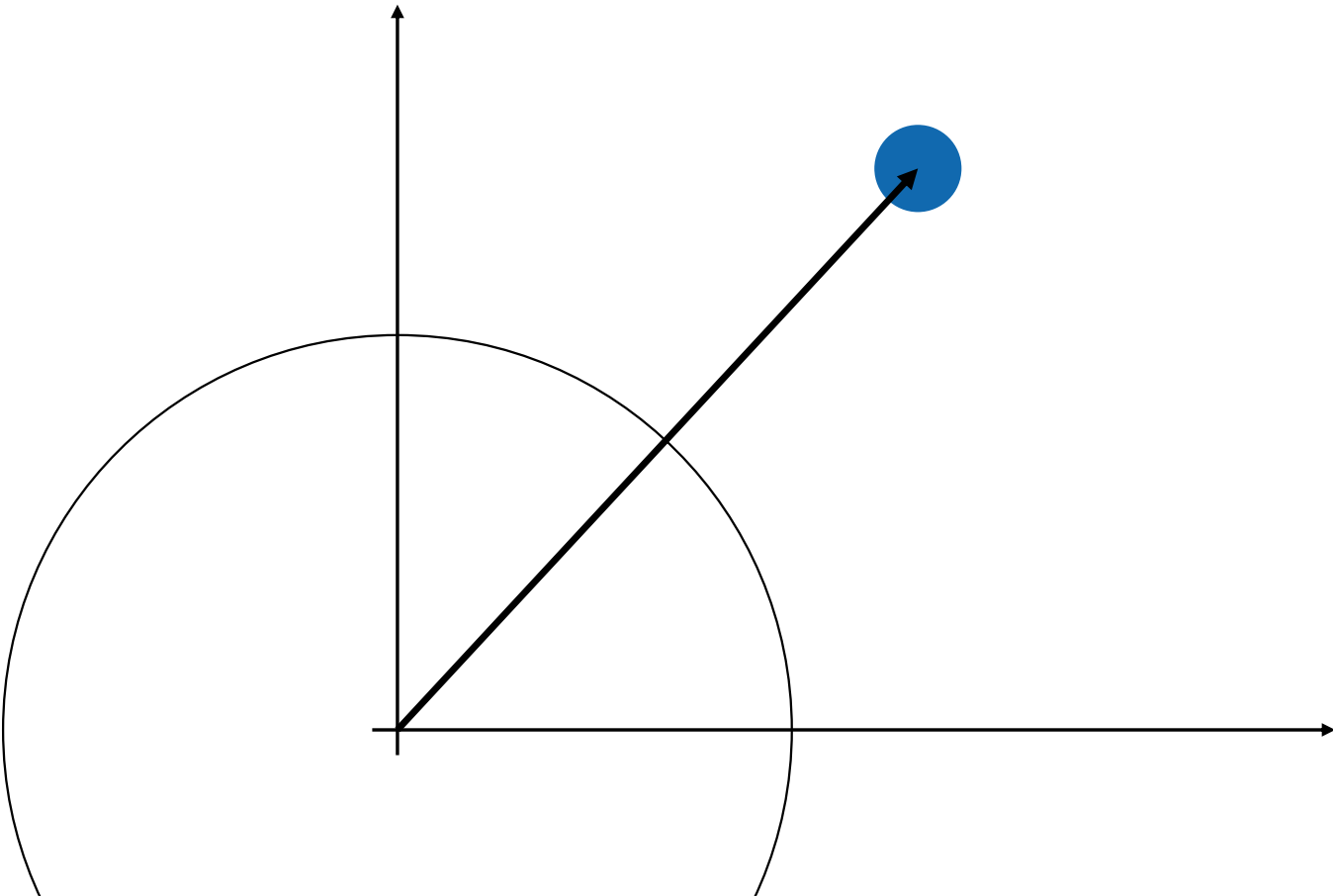
Prob idf document frequency scaling (p)



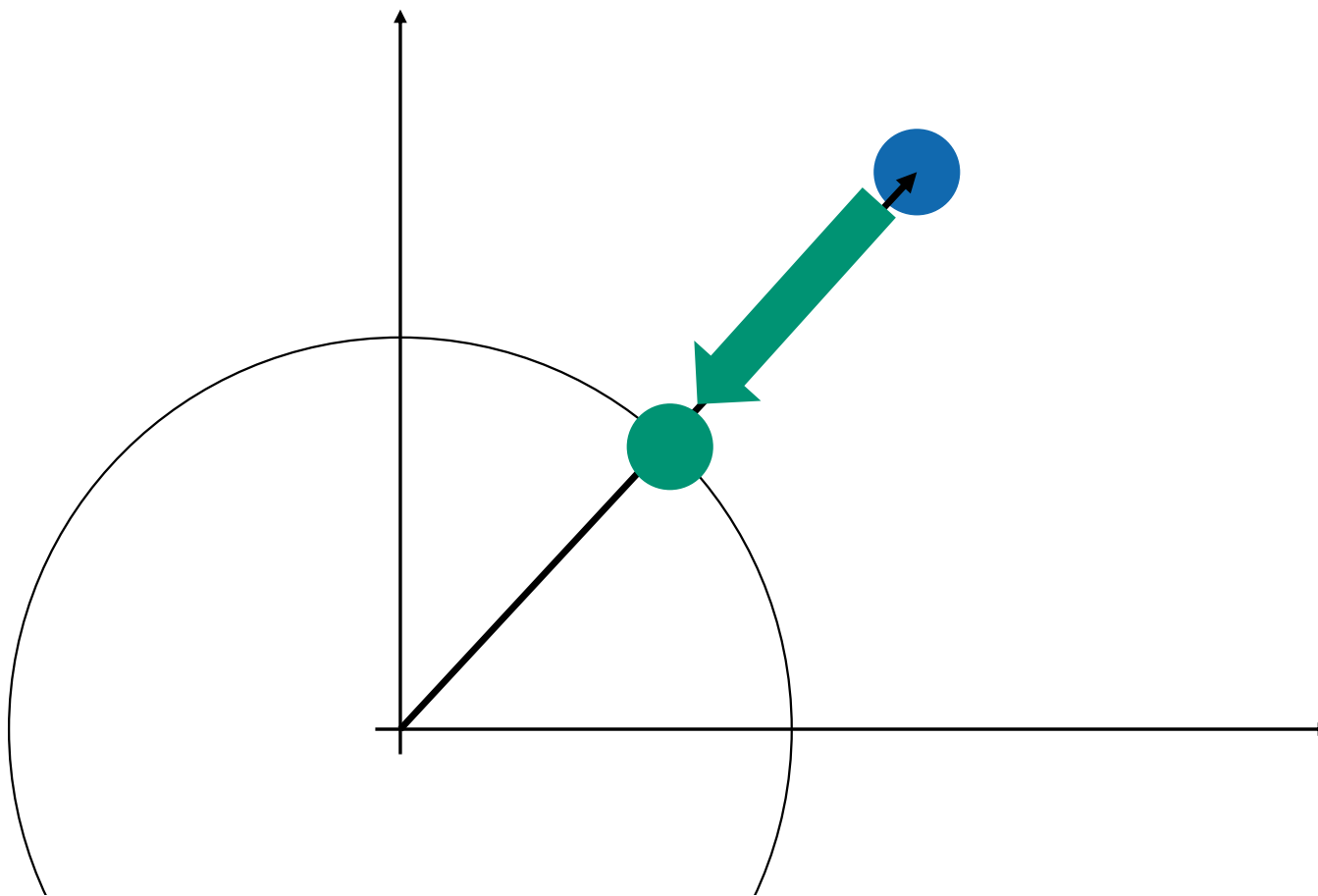
Prob idf document frequency scaling (p)



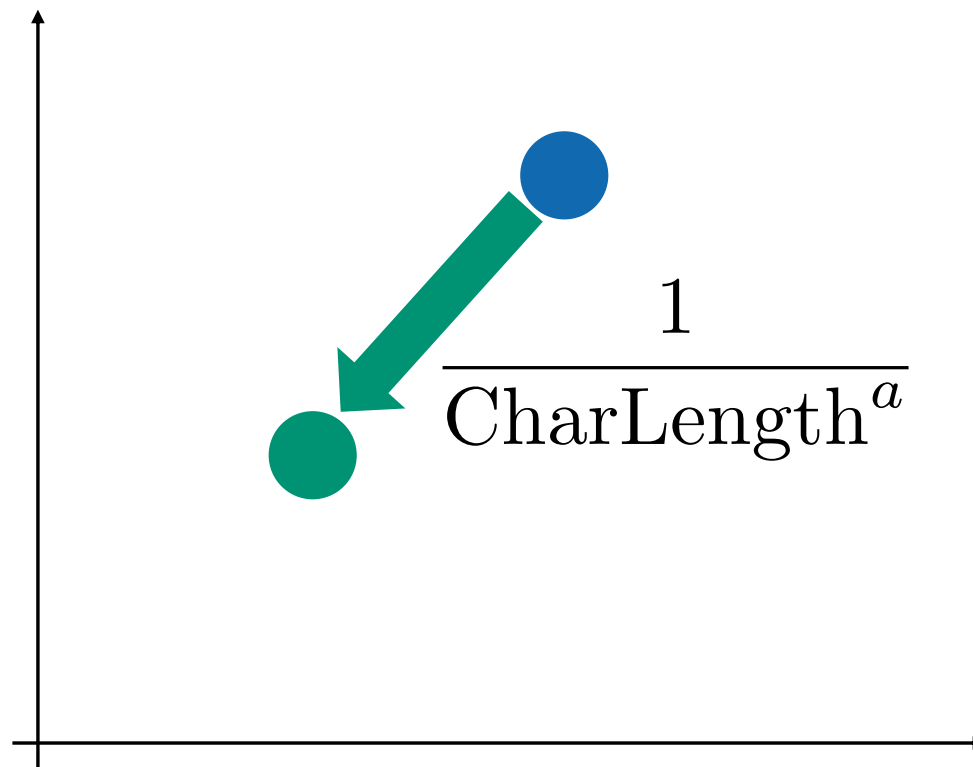
No normalization (n)



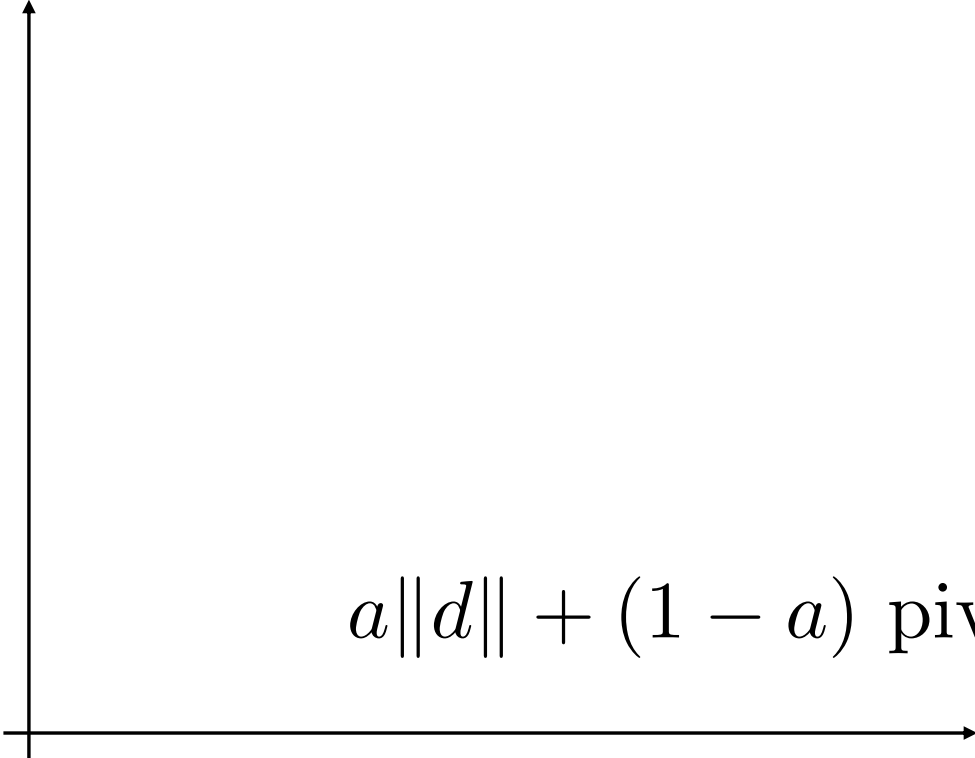
Cosine normalization (c)



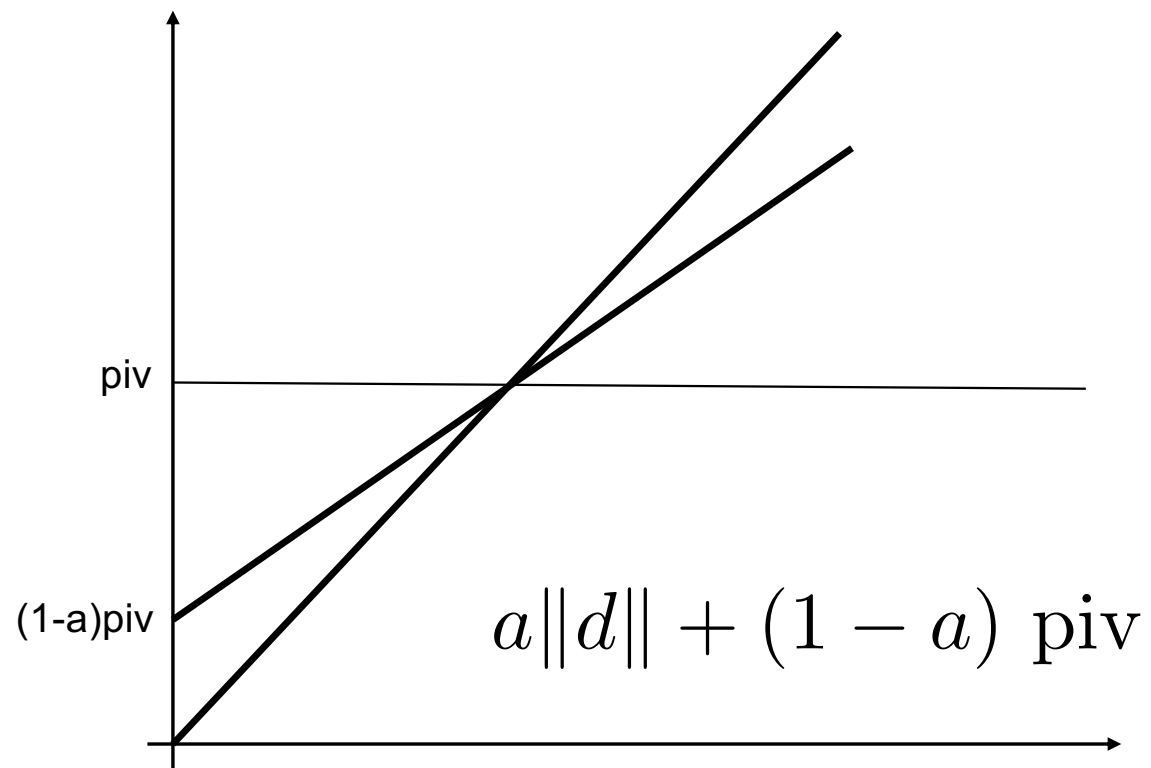
Byte-size normalization (b)



Pivoted normalization (p)


$$a\|d\| + (1 - a) \text{ piv}$$

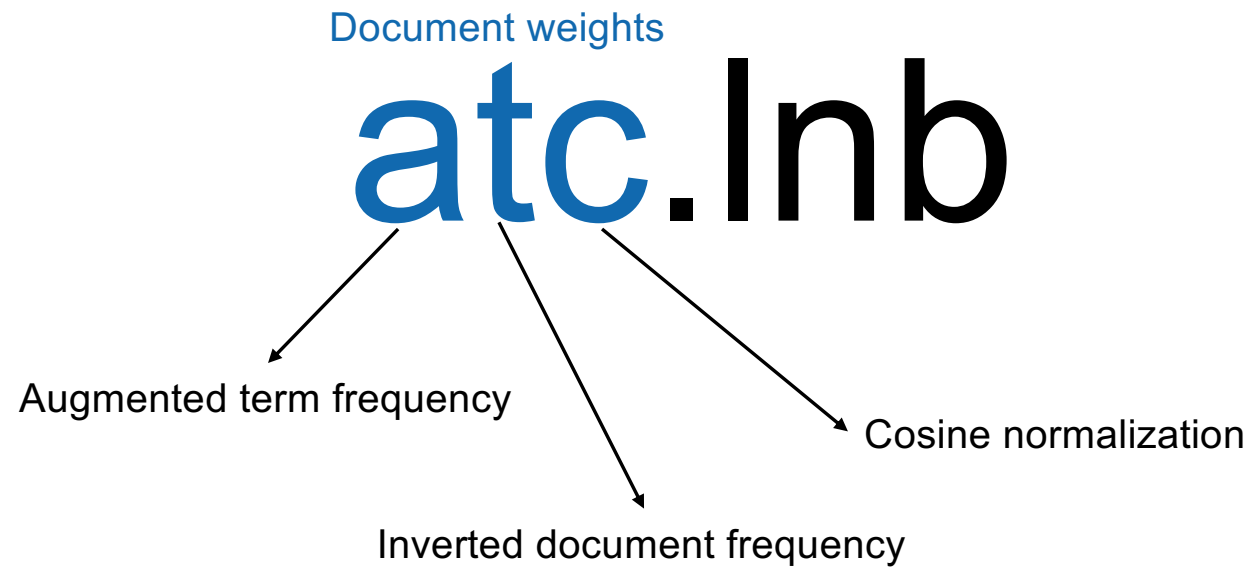
Pivoted normalization (p)



SMART notation

atc.Inb

SMART notation



SMART notation

